

6 Transmitter characteristics

6.1 General

Unless otherwise stated, the transmitter characteristics are specified at the antenna connector of the UE with a single or multiple transmit antenna(s). For UE with integral antenna only, a reference antenna with a gain of 0 dBi is assumed.

Transmitter requirements for UL MIMO operation apply when the UE transmits on 2 ports on the same CDM group. The UE may use higher MPR values outside this limitation.

6.2 Transmitter power

6.2.1 UE maximum output power

The following UE Power Classes define the maximum output power for any transmission bandwidth within the channel bandwidth of NR carrier unless otherwise stated. The period of measurement shall be at least one sub frame (1ms).

Table 6.2.1-1: UE Power Class

NR band	Class 1 (dBm)	Tolerance (dB)	Class 2 (dBm)	Tolerance (dB)	Class 3 (dBm)	Tolerance (dB)
n1					23	±2
n2					23	±2 ³
n3					23	±2 ³
n5					23	±2
n7					23	±2 ³
n8					23	±2 ³
n12					23	±2 ³
n20					23	±2 ³
n25					23	±2 ³
n28					23	+2/-2.5
n34					23	±2
n38					23	±2
n39					23	±2
n40					23	±2
n41			26	+2/-3 ³	23	±2 ³

n50					23	± 2
n51					23	± 2
n66					23	± 2
n70					23	± 2
n71					23	$+2/-2.5$
n74					23	± 2
n77			26	$+2/-3$	23	$+2/-3$
n78			26	$+2/-3$	23	$+2/-3$
n79			26	$+2/-3$	23	$+2/-3$
n80					23	± 2
n81					23	± 2
n82					23	± 2
n83					23	$+2/-2.5$
n84					23	± 2
n86					23	± 2
NOTE 1: $P_{\text{PowerClass}}$ is the maximum UE power specified without taking into account the tolerance						
NOTE 2: Power class 3 is default power class unless otherwise stated						
NOTE 3: Refers to the transmission bandwidths confined within $F_{\text{UL_low}}$ and $F_{\text{UL_low}} + 4$ MHz or $F_{\text{UL_high}} - 4$ MHz and $F_{\text{UL_high}}$, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB						

If a UE supports a different power class than the default UE power class for the band and the supported power class enables the higher maximum output power than that of the default power class:

- if the field of UE capability *maxUplinkDutyCycle-PC2-FR1* is absent and the percentage of uplink symbols transmitted in a certain evaluation period is larger than 50% (The exact evaluation period is no less than one radio frame); or
- if the field of UE capability *maxUplinkDutyCycle-PC2-FR1* is not absent and the percentage of uplink symbols transmitted in a certain evaluation period is larger than *maxUplinkDutyCycle-PC2-FR1* as defined in TS 38.331 (The exact evaluation period is no less than one radio frame); or
- if the IE P-Max as defined in TS 38.331 [7] is provided and set to the maximum output power of the default power class or lower;
- shall apply all requirements for the default power class to the supported power class and set the configured transmitted power as specified in clause 6.2.4;

- else if the IE *P-Max* as defined in TS 38.331 [7] is not provided or set to the higher value than the maximum output power of the default power class and the percentage of uplink symbols transmitted in a certain evaluation period is less than or equal to *maxUplinkDutyCycle-PC2-FR1* as defined in TS 38.331; or
- if the IE *P-Max* as defined in TS 38.331 [7] is not provided or set to the higher value than the maximum output power of the default power class and the percentage of uplink symbols transmitted in a certain evaluation period is less than or equal to 50% when *maxUplinkDutyCycle-PC2-FR1* is absent. (The exact evaluation period is no less than one radio frame);
- shall apply all requirements for the supported power class and set the configured transmitted power as specified in clause 6.2.4.

6.2.2 UE maximum output power reduction

UE is allowed to reduce the maximum output power due to higher order modulations and transmit bandwidth configurations. For UE power class 2 and 3, the allowed maximum power reduction (MPR) is defined in Table 6.2.2-2 and Table 6.2.2-1, respectively for channel bandwidths that meets both following criteria:

Channel bandwidth ≤ 100 MHz.

Relative channel bandwidth ≤ 4 % for TDD bands and ≤ 3 % for FDD bands

Where relative channel bandwidth = $2 \cdot BW_{\text{Channel}} / (F_{\text{UL_low}} + F_{\text{UL_high}})$

The allowed MPR for SRS, PUCCH formats 0, 1, 3 and 4, and PRACH shall be as specified for QPSK modulated DFT-s-OFDM of equivalent RB allocation. The allowed MPR for PUCCH format 2 shall be as specified for QPSK modulated CP-OFDM of equivalent RB allocation.

Table 6.2.2-1 Maximum power reduction (MPR) for power class 3

Modulation		MPR (dB)		
		Edge RB allocations	Outer RB allocations	Inner RB allocations
DFT-s-OFDM	Pi/2 BPSK	$\leq 3.5^1$	$\leq 1.2^1$	$\leq 0.2^1$
		$\leq 0.5^2$	$\leq 0.5^2$	0^2
	QPSK	≤ 1		0
	16 QAM	≤ 2		≤ 1
	64 QAM	≤ 2.5		
	256 QAM	≤ 4.5		
CP-OFDM	QPSK	≤ 3		≤ 1.5
	16 QAM	≤ 3		≤ 2
	64 QAM	≤ 3.5		
	256 QAM	≤ 6.5		

NOTE 1: Applicable for UE operating in TDD mode with Pi/2 BPSK modulation and UE indicates support for UE capability *powerBoosting-pi2BPSK* and if the IE *powerBoostPi2BPSK* is set to 1 and 40 % or less slots in radio frame are used for UL transmission for bands n40, n41, n77, n78 and n79. The reference power of 0 dB MPR is 26 dBm.

NOTE 2: Applicable for conditions where note 1 does not apply.

Table 6.2.2-2 Maximum power reduction (MPR) for power class 2

Modulation		MPR (dB)		
		Edge RB allocations	Outer RB allocations	Inner RB allocations
DFT-s-OFDM	Pi/2 BPSK	≤ 3.5	≤ 0.5	0
	QPSK	≤ 3.5	≤ 1	0
	16 QAM	≤ 3.5	≤ 2	≤ 1
	64 QAM	≤ 3.5	≤ 2.5	
	256 QAM	≤ 4.5		
CP-OFDM	QPSK	≤ 3.5	≤ 3	≤ 1.5
	16 QAM	≤ 3.5	≤ 3	≤ 2
	64 QAM	≤ 3.5		
	256 QAM	≤ 6.5		

Where the following parameters are defined to specify valid RB allocation ranges for Outer and Inner RB allocations:

N_{RB} is the maximum number of RBs for a given Channel bandwidth and sub-carrier spacing defined in Table 5.3.2-1. $RB_{Start,Low} = \max(1, \text{floor}(L_{CRB}/2))$

where $\max()$ indicates the largest value of all arguments and $\text{floor}(x)$ is the greatest integer less than or equal to x .

$$RB_{Start,High} = N_{RB} - RB_{Start,Low} - L_{CRB}$$

The RB allocation is an Inner RB allocation if the following conditions are met

$$RB_{Start,Low} \leq RB_{Start} \leq RB_{Start,High}, \text{ and}$$

$$L_{CRB} \leq \text{ceil}(N_{RB}/2)$$

where $\text{ceil}(x)$ is the smallest integer greater than or equal to x .

An Edge RB allocation is the one for which the RB(s) is (are) allocated at the lowermost or uppermost edge of the channel with $L_{CRB} \leq 2$ RBs.

The RB allocation is an Outer RB allocation for all other allocations which are not an Inner RB allocation or Edge RB allocation.

If CP-OFDM allocation satisfies following conditions, it is considered as almost contiguous allocation

$$N_{RB_gap} / (N_{RB_alloc} + N_{RB_gap}) \leq 0.25$$

and $N_{RB_alloc} + N_{RB_gap}$ is larger than 106, 51 or 24 RBs for 15 kHz, 30 kHz or 60 kHz respectively where N_{RB_gap} is the total number of unallocated RBs between allocated RBs and N_{RB_alloc} is the total number of allocated RBs. The size and location of allocated and unallocated RBs are restricted by RBG parameters specified in clause 6.1.2.2 of TS 38.214 [10]. For UE that indicates support for *almostContiguousCP-OFDM-UL*, the almost contiguous signals in power class 3, the allowed maximum power reduction defined in Table 6.2.2-1 is increased by

$$CEIL\{ 10 \log_{10}(1 + N_{RB_gap} / N_{RB_alloc}), 0.5 \} \text{ dB},$$

where $CEIL\{x, 0.5\}$ means x rounding upwards to closest 0.5dB. The parameter of L_{CRB} which is used to specify valid RB allocation ranges for Outer and Inner RB allocations is replaced by $(N_{RB_alloc} + N_{RB_gap})$ for almost contiguous allocation cases

$$RB_{Start,Low} = \max(1, \text{floor}((N_{RB_alloc} + N_{RB_gap})/2))$$

$$RB_{Start,High} = N_{RB} - RB_{Start,Low} - N_{RB_alloc} - N_{RB_gap}$$

For the UE maximum output power modified by MPR, the power limits specified in clause 6.2.4 apply.

6.2.3 UE additional maximum output power reduction

6.2.3.1 General

Additional emission requirements can be signalled by the network. Each additional emission requirement is associated with a unique network signalling (NS) value indicated in RRC signalling by an NR frequency band number of the applicable operating band and an associated value in the field *additionalSpectrumEmission*. Throughout this specification, the notion of indication or signalling of an NS value refers to the corresponding indication of an NR frequency band number of the applicable operating band, the IE field *freqBandIndicatorNR* and an associated value of *additionalSpectrumEmission* in the relevant RRC information elements [7].

To meet the additional requirements, additional maximum power reduction (A-MPR) is allowed for the maximum output power as specified in Table 6.2.1-1. Unless stated otherwise, the total reduction to UE maximum output power is $\max(\text{MPR}, \text{A-MPR})$ where MPR is defined in clause 6.2.2. Outer and inner allocation notation used in clause 6.2.3 is defined in clause 6.2.2. Unless stated

otherwise, Edge RB allocations get the same AMPR as Outer RB allocations. In absense of modulation and waveform types the A-MPR applies to all modulation and waveform types.

Table 6.2.3.1-1 specifies the additional requirements with their associated network signalling values and the allowed A-MPR and applicable operating band(s) for each NS value. In case of a power class 3 UE, when IE *powerBoostPi2BPSK* is set to 1, power class 2 A-MPR values apply. The mapping of NR frequency band numbers and values of the *additionalSpectrumEmission* to network signalling labels is specified in Table 6.2.3.1-1A.

For almost contiguous allocations in CP-OFDM waveforms in power class 3, the allowed A-MPR defined in clause 6.2.3 is increased by $\text{CEIL}\{10 \log_{10}(1 + N_{\text{RB_gap}} / N_{\text{RB_alloc}}), 0.5\}$ dB, where $\text{CEIL}\{x, 0.5\}$ means x rounding upwards to closest 0.5dB, $N_{\text{RB_gap}}$ is the total number of unallocated RBs between allocated RBs and $N_{\text{RB_alloc}}$ is the total number of allocated RBs, and the parameter L_{CRB} is replaced by $N_{\text{RB_alloc}} + N_{\text{RB_gap}}$ in specifying the RB allocation regions.

Table 6.2.3.1-1: Additional maximum power reduction (A-MPR)

Network signalling label	Requirements (clause)	NR Band	Channel bandwidth (MHz)	Resources blocks (N_{RB})	A-MPR (dB)
NS_01		Table 5.2-1 (NOTE 4)	5, 10, 15, 20, 25, 30, 40, 50, 60, 80, 90, 100	Table 5.3.2-1	N/A
NS_03	6.5.2.3.3	n2, n25, n66, n70, n86			Clause 6.2.3.7
NS_03U	6.5.2.3.3, 6.5.2.4.2	n2, n25, n66, n86 (NOTE 1)			Clause 6.2.3.7
NS_04	6.5.2.3.2, 6.5.3.3.1	n41	10, 15, 20, 40, 50, 60 80, 90, 100		Clause 6.2.3.2
NS_05	6.5.3.3.4	n1, n84	5, 10, 15, 20 (NOTE 2)		Clause 6.2.3.4
NS_05U	6.5.3.3.4, 6.5.2.4.2	n1, n84 (NOTE 1)	5, 10, 15, 20		Clause 6.2.3.4
NS_06	6.5.2.3.4	n12	5, 10, 15		N/A
NS_10		n20, n82	15, 20	Table 6.2.3.3-1	Table 6.2.3.3-1
NS_17	6.5.3.3.2	n28, n83	5,10	Table 5.3.2-1	N/A
NS_18	6.5.3.3.3	n28, n83	5		Table 6.2.3.13-1, A1

			10, 15, 20		Table 6.2.3.13-1, A2
NS_35	6.5.2.3.1	n71	5, 10, 15, 20	Table 5.3.2-1	N/A
NS_37	6.5.3.3.6	n74 (NOTE 3)	10, 15	Table 6.2.3.8-1	Table 6.2.3.8-1
NS_38	6.5.3.3.7	n74	5, 10, 15, 20	Table 6.2.3.9-1	Table 6.2.3.9-1
NS_39	6.5.3.3.8	n74	10, 15, 20	Table 6.2.3.10-1	Table 6.2.3.10-1
NS_40	6.5.3.3.9	n51	5		Table 6.2.3.5-1
NS_41	6.5.3.3.10	n50	5, 10, 15, 20, 40, 50, 60		Table 6.2.3.11-1
NS_42	6.5.3.3.11	n50	5, 10, 15, 20, 40, 50, 60		Table 6.2.3.12-1
NS_43	6.5.3.3.5	n8, n81	5, 10, 15		Clause 6.2.3.6
NS_43U	6.5.3.3.5, 6.5.2.4.2	n8, n81 (NOTE 1)	5, 10, 15		Clause 6.2.3.6
NS_50	6.5.3.3.16	n39	25, 30, 40		Clause 6.2.3.19
NS_100	6.5.2.4.2	n1, n2, n3, n5, n8, n25, n66, n80, n81, n84, n86 (NOTE 1)			Table 6.2.3.1-2

NOTE 1: This NS can be signalled for NR bands that have UTRA services deployed.
NOTE 2: No A-MPR is applied for 5 MHz BW_{Channel} where the upper channel edge is ≥ 1930 MHz, 10 MHz BW_{Channel} where the upper channel edge is ≥ 1950 MHz and 15 MHz BW_{Channel} where the upper channel edge is ≥ 1955 MHz and 20 MHz BW_{Channel} where the upper channel edge is ≥ 1970 MHz.
NOTE 3: Applicable when the NR carrier is within 1447.9 – 1462.9 MHz.
NOTE 4: The NS_01 label with the field *additionalPmax* [7] absent is default for all NR bands.

Table 6.2.3.1-1A: Mapping of network signaling label

NR band	Value of <i>additionalSpectrumEmission</i>							
	0	1	2	3	4	5	6	7
n1	NS_01	NS_100	NS_05	NS_05U				
n2	NS_01	NS_100	NS_03	NS_03U				
n3	NS_01	NS_100						
n5	NS_01	NS_100						

n7	NS_01							
n8	NS_01	NS_100	NS_43	NS_43U				
n12	NS_01	NS_06						
n20	NS_01	Void	NS_10					
n25	NS_01	NS_100	NS_03	NS_03U				
n28	NS_01	NS_17	NS_18					
n34	NS_01							
n38	NS_01							
n39	NS_01	NS_50						
n40	NS_01							
n41	NS_01	NS_04						
n50	NS_01	NS_41	NS_42					
n51	NS_01	NS_40						
n66	NS_01	NS_100	NS_03	NS_03U				
n70	NS_01	NS_03						
n71	NS_01	NS_35						
n74	NS_01	NS_37	NS_38	NS_39				
n77	NS_01							
n78	NS_01							
n79	NS_01							
n80	NS_01	NS_100						
n81	NS_01	NS_100	NS_43	NS_43U				
n82	NS_01	Void	NS_10					
n83	NS_01	NS_17	NS_18					
n84	NS_01	NS_100	NS_05	NS_05U				
n86	NS_01	NS_100	NS_03	NS_03U				
NOTE: <i>additionalSpectrumEmission</i> corresponds to an information element of the same name defined in clause 6.3.2 of TS 38.331 [7].								

Table 6.2.3.1-2: A-MPR for NS_100 (UTRA protection)

Modulation/Waveform	Outer (dB)
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DFT-s-OFDM	Pi/2 BPSK	≤ 2
	QPSK	≤ 2
	16 QAM	≤ 2.5
	64 QAM	≤ 3
	256 QAM	≤ 4.5
CP-OFDM	QPSK	≤ 4
	16 QAM	≤ 4
	64 QAM	≤ 4
	256 QAM	≤ 6.5
NOTE 1: Void		
NOTE 2: Void		

6.2.3.2 A-MPR for NS_04

For NS_04, A-MPR is not added to MPR. Also, when NS_04 is signalled, MPR shall be set to zero in the P_{CMAX} equations to avoid double counting MPR.

Allowed maximum power reduction is defined as A-MPR = max(MPR, A-MPR'),
Note that A-MPR' = 0 dB means only MPR is applied,
where A-MPR' is defined as

if $R_{\text{Bstart}} \leq f_{\text{start,max,IMD3}} / (12\text{SCS})$ and $L_{\text{CRB}} \leq A_{\text{Wmax,IMD3}} / (12\text{SCS})$ and $F_{\text{c}} - B_{\text{Wchannel}}/2 < F_{\text{UL_low}} + \text{offset}_{\text{IMD3}}$,
then
 the A-MPR' is defined according to Table 6.2.3.2-2 PC3_A2 relative to 23 dBm for power class 3 and PC2_A4 relative to 26 dBm for power class 2,
else,
if $R_{\text{Bstart}} \leq L_{\text{CRB}}/2 + F_{\text{044}} / (12\text{SCS})$ and $L_{\text{CRB}} \leq A_{\text{Wmax,regrowth}} / (12\text{SCS})$ and $F_{\text{c}} - B_{\text{Wchannel}}/2 < F_{\text{UL_low}} + \text{offset}_{\text{regrowth}}$,
then
 the A-MPR' is defined according to Table 6.2.3.2-2 PC3_A1 relative to 23 dBm for power class 3 and PC2_A3 relative to 26 dBm for power class 2,
else
 A-MPR' = 0 dB and apply MPR.

With the parameters defined in Table 6.2.3.2-1.

Table 6.2.3.2-1: Parameters for region edges and frequency offsets

Parameter	Symbol	Value		Related condition
		CP-OFDM	DFT-s-OFDM	

Max allocation start in IMD3 region	$f_{\text{start,max,IMD3}}$	0.33 BW _{Channel}		$RB_{\text{start}} \leq f_{\text{start,max,IMD3}} / (12\text{SCS})$
Max allocation BW in IMD3 region	AW _{max,IMD3}	4 MHz		$L_{\text{CRB}} \leq AW_{\text{max,IMD3}} / (12\text{SCS})$
Freq. offset required to avoid A-MPR in IMD3 region	offset _{IMD3}	BW _{Channel} – 6 MHz		$F_{\text{C}} - BW_{\text{Channel}}/2 \geq F_{\text{UL_low}} + \text{offset}_{\text{IMD3}}$
Right edge of regrowth region	$F_{\text{0.45}}^{\text{start}}$	0.08 BW _{Channel}		$RB_{\text{start}} \leq L_{\text{CRB}}/2 + F_{\text{0.45}}^{\text{start}} / (12\text{SCS})$
Max allocation BW in regrowth region	AW _{max,regrowth}	100 MHz		$L_{\text{CRB}} \leq \text{Min}(L_{\text{CRB,Max}}, AW_{\text{max,regrowth}} / (12\text{SCS}))$
Freq. offset required to avoid A-MPR in regrowth region	offset _{regrowth}	Max (10 MHz, 0.25* BW _{Channel} MHz)	Max (10 MHz, 0.45* BW _{Channel} MHz)	$F_{\text{C}} - BW_{\text{Channel}}/2 \geq F_{\text{UL_low}} + \text{offset}_{\text{regrowth}}$

Table 6.2.3.2-2: A-MPR' values Access

Modulation/Waveform		A-MPR' (dB)			
		PC3_A1	PC3_A2	PC2_A3	PC2_A4
DFT-s-OFDM	Pi/2-BPSK	≤ 3.5	≤ 3.5	≤ 3.5	≤ 5.5
	QPSK	≤ 4	≤ 4	≤ 4.5	≤ 6
	16 QAM	≤ 4	≤ 4	≤ 5	≤ 6
	64 QAM	≤ 4	≤ 4.5	≤ 5	≤ 6.5
	256 QAM	≤ 4.5	≤ 6	≤ 6.5	≤ 8
CP-OFDM	QPSK	≤ 5.5	≤ 5.5	≤ 6.5	≤ 7.5
	16 QAM	≤ 5.5	≤ 5.5	≤ 6.5	≤ 7.5
	64 QAM	≤ 5.5	≤ 5.5	≤ 6.5	≤ 7.5
	256 QAM	≤ 6.5	≤ 8	≤ 7.5	≤ 10

6.2.3.3 A-MPR for NS_10

Table 6.2.3.3-1: A-MPR for NS_10

Channel bandwidth (MHz)	Parameters	Region A
15	RB _{start}	0 – 10

	L_{CRB} (RBs)	1 – 20
	A (dB)	≤ 3
	RB_{start}	0 – 15
20	L_{CRB} (RBs)	1 – 20
	A (dB)	≤ 6
NOTE 1: RB_{start} indicates the lowest RB index of transmitted resource blocks		
NOTE 2: L_{CRB} is the length of a contiguous resource block allocation		
NOTE 3: For intra-subframe frequency hopping which intersects Region A, notes 1 and 2 apply on a per slot basis. For intra-slot or intra-subslot frequency hopping which intersects Region A, notes 1 and 2 apply on a $T_{\text{no_hopping}}$ basis.		
NOTE 4: For intra-subframe frequency hopping which intersect Region A, the larger A-MPR value may be applied for both slots in the subframe. For intra-slot frequency hopping which intersects Region A, the larger A-MPR value may be applied for the slot. For intra-subslot frequency hopping which intersects Region A, the larger A-MPR value may be applied for the subslot.		
NOTE 5: The A-MPR for DFT-s-OFDM is the total backoff and is obtained by taking the maximum value of MPR + A-MPR specified in Table 6.2.3-1 and Table 6.2.4-1 in TS 36.101 and A value specified in Table 6.2.3.3-1.		
NOTE 6: The A-MPR for CP-OFDM is the total backoff and is obtained by adding the A value in Table 6.2.3.3-1 to the corresponding MPR specified in Table 6.2.2-1.		

6.2.3.4 A-MPR for NS_05 and NS_05U

Table 6.2.3.4-1: A-MPR regions for NS_05 and NS_05U

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_c (MHz)	Region A			Region B			Region C		
		RB_{start}	L_{CRB}	A-MPR	RB_{start}	L_{CRB}	A-MPR	RB_{start}	L_{CRB}	A-MPR
5	$1922.5 \leq F_c < 1927.5$	$< 1.62 \text{ MHz/12/SCS}$	$> 2.52 \text{ MHz/12/SCS}$	A3						
10	$1925 \leq F_c < 1935$	$\leq 1.62 \text{ MHz/12/SCS}$	> 0	A1	$> 1.62 \text{ MHz/12/SCS}$ $\leq 3.60 \text{ MHz/12/SCS}$	$> 5.4 \text{ MHz/12/SCS}$	A7	$\geq 7.2 \text{ MHz/12/SCS}$	$\leq 1.08 \text{ MHz/12/SCS}$	A2
10	$1935 \leq F_c < 1945$		$> 4.5 \text{ MHz/12/SCS}$	A4						
15	$1927.5 \leq F_c < 1932.5$	$\leq 3.24 \text{ MHz/12/SCS}$	> 0	A1	$> 3.24 \text{ MHz/12/SCS}$ $\leq 5.40 \text{ MHz/12/SCS}$	$> 8.1 \text{ MHz/12/SCS}$	A7	$\geq 10.08 \text{ MHz/12/SCS}$	$\leq 1.08 \text{ MHz/12/SCS}$	A2
15	$1932.5 \leq F_c < 1942.5$	$< 1.62 \text{ MHz/12/SCS}$	> 0	A1				$\geq 12.24 \text{ MHz/12/SCS}$	$\leq 1.08 \text{ MHz/12/SCS}$	A2
15	$1942.5 \leq F_c < 1947.5$		$> 7.2 \text{ MHz/12/SCS}$	A5						
20	$1930 \leq F_c < 1950$	$\leq 4.86 \text{ MHz/12/SCS}$	> 0	A1	$> 4.86 \text{ MHz/12/SCS}$ $\leq 7.20 \text{ MHz/12/SCS}$	$> 9.0 \text{ MHz/12/SCS}$	A7	$\geq 13.68 \text{ MHz/12/SCS}$	$\leq 1.08 \text{ MHz/12/SCS}$	A2
20	$1950 \leq F_c < 1960$		$> 9.0 \text{ MHz/12/SCS}$	A6						

NOTE 1: The A-MPR values are specified in Table 6.2.3.4-2, 6.2.3.4-3 and 6.2.3.4-10.
NOTE 2: Void

Table 6.2.3.4-2: A-MPR for NS_05 and NS_05U

Modulation/Waveform		A1 (dB)	A2 (dB)	A3 (dB)	
		Outer/Inner	Outer/Inner	Outer	
DFT-s-OFDM	Pi/2 BPSK	≤ 10	≤ 5		
	QPSK	≤ 10	≤ 5		
	16 QAM	≤ 10	≤ 5		
	64 QAM	≤ 11	≤ 5		
	256 QAM	≤ 13	≤ 5		
CP-OFDM	QPSK	≤ 10	≤ 5	≤ 2	
	16 QAM	≤ 10	≤ 5		
	64 QAM	≤ 11	≤ 5		
	256 QAM	≤ 13			
NOTE 1: Void					
NOTE 2: Void					

Table 6.2.3.4-3: A-MPR for NS_05

Modulation/Waveform		A4 (dB)		A5 (dB)		A6 (dB)		A7 (dB)		
		Outer	Inner	Outer		Outer	Inner	Outer/Inner		
DFT-s-OFDM	Pi/2 BPSK	≤ 1	N/A	≤ 1		≤ 1	N/A	≤ 6		
	QPSK			≤ 1.5		≤ 1.5		≤ 6		
	16 QAM							≤ 6		
	64 QAM							≤ 6		
	256 QAM							≤ 6		
CP-OFDM	QPSK	≤ 3.5		≤ 3.5		≤ 3.5		≤ 6		
	16 QAM	≤ 3.5		≤ 3.5		≤ 3.5		≤ 6		
	64 QAM							≤ 6		
	256 QAM							≤ 6		
NOTE 1: Void										
NOTE 2: Void										

Table 6.2.3.4-4 - Table 6.2.3.4-9: Void

Table 6.2.3.4-10: A-MPR for modulation and waveform type for NS_05U

Modulation/Waveform		A4 (dB)		A5 (dB)		A6 (dB)		A7 (dB)
		Outer	Inner	Outer	Inner	Outer	Inner	Outer/Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 2	N/A	≤ 2		≤ 2	N/A	≤ 6
	QPSK	≤ 2		≤ 2		≤ 2		≤ 6
	16 QAM	≤ 2.5		≤ 2.5		≤ 2.5		≤ 6
	64 QAM	≤ 3		≤ 3		≤ 3		≤ 6
	256 QAM	≤ 4.5		≤ 4.5		≤ 4.5		≤ 6
CP-OFDM	QPSK	≤ 4		≤ 4		≤ 4		≤ 6
	16 QAM	≤ 4		≤ 4		≤ 4		≤ 6
	64 QAM	≤ 4		≤ 4	≤ 4	≤ 4		≤ 6
	256 QAM	≤ 6.5		≤ 6.5		≤ 6.5		≤ 6.5

NOTE 1: Void
NOTE 2: Void

6.2.3.5 A-MPR for NS_40

Table 6.2.3.5-1: A-MPR for NS_40

Modulation/ Waveform		A (dB)	
		Channel bandwidth: 5 MHz	
		Outer	Inner
DFT-s-OFDM	QPSK	≤ 15.5	≤ 12
	16 QAM	≤ 14.5	≤ 11
	64 QAM	≤ 14.5	≤ 10
	256 QAM	≤ 12.5	≤ 7.5
CP-OFDM	QPSK	≤ 14.5	≤ 10
	16 QAM	≤ 14.5	≤ 10
	64 QAM	≤ 14	≤ 8
	256 QAM	≤ 11	≤ 5.5
NOTE 1: The A-MPR for NS_40 is the total backoff and is obtained by taking the maximum value of MPR + A-MPR specified in Table 6.2.3-1 and Table 6.2.4-30a in TS 36.101 and MPR + A specified in Table 6.2.2-1 and Table 6.2.3.5-1.			

6.2.3.6 A-MPR for NS_43 and NS_43U

Table 6.2.3.6-1: A-MPR regions for NS_43

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_c (MHz)	Region A			Region B		
		RB_{start}	L_{CRB}	A-MPR	RB_{start}	L_{CRB}	A-MPR
5 MHz	$902.526 \leq F_c < 912.5$		> 15	A1			
10 MHz	$F_c = 910$		> 40	A2		$> 5.4 \text{ MHz}/12/ \text{SCS}$	A4
			> 45	A3		$> 7.2 \text{ MHz}/12/ \text{SCS}$	A5
15 MHz	$F_c = 907.5$	$< 1.8 \text{ MHz}/12/ \text{SCS}$	> 0	A6	$> 1.8 \text{ MHz}/12/ \text{SCS}$	$\geq 7.2 \text{ MHz}/12/ \text{SCS}$	A6

					< 6.12 MHz/12/SCS		
		> 12.24 MHz/12/SCS	> 0	A6			
NOTE 1: The A-MPR values are specified in Table 6.2.3.6-2.							
NOTE 2: 15 kHz SCS unless otherwise stated							
NOTE 3: Void							

Table 6.2.3.6-2: A-MPR for NS_43

Modulation/ Waveform		A1 (dB)		A2 (dB)		A3 (dB)		A4 (dB)		A5 (dB)		A6 (dB)
		Outer	Inner	Outer	Inner	Outer	Inner	Outer	Inner	Outer	Inner	Outer / Inner
DFT-s-OFDM	Pi/2 BPSK		N/A	≤ 1.5	N/A		N/A		N/A		N/A	≤ 9
	QPSK	≤ 2						≤ 2.5				≤ 9
	16 QAM									≤ 2.5		≤ 9
	64 QAM					≤ 2.5						≤ 9
	256 QAM											≤ 9
CP-OFDM	QPSK	≤ 3.5								≤ 4		≤ 9
	16 QAM	≤ 3.5								≤ 4		≤ 9
	64 QAM					≤ 4						≤ 9
	256 QAM											≤ 9

Table 6.2.3.6-3: Void

When NS_43U is signalled for 5 and 10 MHz channel bandwidths A-MPR is defined in Table 6.2.3.1-2 except for DFT-s-OFDM QPSK when $L_{CRB} > 5.4$ MHz/12/SCS the A-MPR is 2.5 dB. For 15 MHz channel bandwidth Table 6.2.3.6-4 applies.

Table 6.2.3.6-4: A-MPR for for NS_43U

Modulation/Waveform	15 MHz
	Outer/ Inner (dB)

DFT-s-OFDM	Pi/2 BPSK	≤ 9
	QPSK	≤ 9
	16 QAM	≤ 9
	64 QAM	≤ 9
	256 QAM	≤ 9
CP-OFDM	QPSK	≤ 9
	16 QAM	≤ 9
	64 QAM	≤ 9
	256 QAM	≤ 9

6.2.3.7 A-MPR for NS_03 and NS_03U

Table 6.2.3.7-1 A-MPR for NS_03

Modulation/Waveform		Outer (dB)	Inner (dB)
DFT-s-OFDM	PI/2 BPSK	≤ 1.5	N/A
	QPSK	≤ 2	
	16 QAM	≤ 3	
	64 QAM	≤ 3.5	
	256 QAM	≤ 5.5	
CP-OFDM	QPSK	≤ 4	
	16 QAM	≤ 4	
	64 QAM	≤ 4.5	
	256 QAM	≤ 7.5	
NOTE 1: Void			
NOTE 2: Void			

In case UE operates in a band where NS_03U applies and it receives *additionalSpectrumEmission* value of 3 then A-MPR values specified in Table 6.2.3.7-1 apply with an exception that DFT-s-OFDM Pi/2 BPSK A-MPR is 2 dB.

6.2.3.8 A-MPR for NS_37

Table 6.2.3.8-1: A-MPR regions for B11/B21 protection (NS_37) (1447.9 - 1462.9 MHz)

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_c (MHz)	Region A (Outer/Inner)			Region B (Outer/Inner)			Region C (Outer/Inner)		
		RB_{start}	L_{CRB}	A-MPR	RB_{start}	L_{CRB}	A-MPR	RB_{start}	L_{CRB}	A-MPR
10	$1452.9 < F_c \leq 1457.9$	≥ 0	$> 7.2 \text{ MHz}/12/\text{SCS}$	$\leq A1$	N/A	N/A	N/A	N/A	N/A	N/A
15	$F_c = 1455.4$	≥ 0	$> 9.9 \text{ MHz}/12/\text{SCS}$	$\leq A1$	$< 0.54 \text{ MHz}/12/\text{SCS}$	$< 1.08 \text{ MHz}/12/\text{SCS}$	$\leq A2$	$> 13.86 \text{ MHz}/12/\text{SCS}$	$< 1.08 \text{ MHz}/12/\text{SCS}$	$\leq A2$
NOTE 1: The A-MPR values are specified in Table 6.2.3.8-2 NOTE 2: Void NOTE 3: Void NOTE 4: No A-MPR for SCS = 60 kHz for region B and C only.										

Table 6.2.3.8-2: A-MPR for NS_37

Modulation/Waveform		A1 (dB)		A2 (dB)
		Outer	Inner	Outer/Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 1	N/A	≤ 3
	QPSK	≤ 1.5		≤ 3
	16 QAM	≤ 2.5		≤ 3
	64 QAM	≤ 3		≤ 3
	256 QAM			
CP-OFDM	QPSK	≤ 3.5		≤ 3
	16 QAM	≤ 3.5		≤ 3
	64 QAM			
	256 QAM			
NOTE 1: Void				
NOTE 2: Void				

6.2.3.9 A-MPR for NS_38

Table 6.2.3.9-1: A-MPR for EESS (NS_38) Protection (1430 – 1470 MHz)

Channel	Carrier Centre	Region A	Region B
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Bandwidth (MHz)	Frequency, F _c (MHz)	Outer/Inner			Outer/Inner	
		RB _{start}	L _{CRB}	A-MPR (dB)	RB _{start} +L _{CRB}	A-MPR (dB)
5	1432.5 ≤ F _c < 1434.55	≤ -1.8 MHz/12/SCS + L _{CRB} /2	> 3.6 MHz/12/ SCS	≤ 7	≤ 2.16 MHz/12/SCS	≤ 5.5
10	1435 ≤ F _c < 1442	≤ -1.8 MHz/12/SCS + L _{CRB} /2	> 3.6 MHz/12/ SCS	≤ 12	≤ 2.16 MHz/12/SCS	≤ 9
15	1437.5 ≤ F _c < 1447.5	≤ -1.8 MHz/12/SCS + L _{CRB} /2	> 3.6 MHz/12/ SCS	≤ 13	≤ 3.6 MHz/12/SCS	≤ 10
20	1440 ≤ F _c < 1450	≤ -1.8 MHz/12/SCS + L _{CRB} /2	> 3.6 MHz/12/ SCS	≤ 13	≤ 5.4 MHz/12/SCS	≤ 10
NOTE 1 - 4: Void						

6.2.3.10 A-MPR for NS_39

Table 6.2.3.10-1: A-MPR for own RX (NS_39) Protection (1440 – 1470 MHz)

Channel Bandwidth, MHz	Carrier Centre Frequency, F _c , MHz	Region A (Outer/Inner)	
		RB _{start} +L _{CRB}	A-MPR (dB)
10	1460 < F _c ≤ 1465	> 7.9 MHz/12/SCS	≤ 6
15	1452.5 < F _c ≤ 1462.5	> 11.2 MHz/12/SCS	≤ 6
20	1450 < F _c ≤ 1460	> 14.4 MHz/12/SCS	≤ 6
NOTE 1 - 4: Void			

6.2.3.11 A-MPR for NS_41

Table 6.2.3.11-1: A-MPR for NS_41

Channel Bandwidth (MHz)	Carrier Centre Frequency, F _c (MHz)	Region A Outer/Inner			Region B Outer/Inner	
		RB _{start}	L _{CRB}	A-MPR (dB)	RB _{start} +L _{CRB}	A-MPR (dB)

5	-	-	-	-	-	-
10	$1437 \leq F_C < 1442$	$\leq -4.5 \text{ MHz}/12/\text{SCS} + L_{\text{CRB}}$	$> 4.5 \text{ MHz}/12/\text{SCS}$	≤ 9	$< 1.8 \text{ MHz}/12/\text{SCS}$	≤ 9
15	$1439.5 \leq F_C < 1447.5$	$\leq -5.4 \text{ MHz}/12/\text{SCS} + L_{\text{CRB}}$	$> 5.4 \text{ MHz}/12/\text{SCS}$	≤ 11	$< 3.42 \text{ MHz}/12/\text{SCS}$	≤ 9
20	$1442 \leq F_C < 1450$	$\leq -5.4 \text{ MHz}/12/\text{SCS} + L_{\text{CRB}}$	$> 5.4 \text{ MHz}/12/\text{SCS}$	≤ 12	$< 5.04 \text{ MHz}/12/\text{SCS}$	≤ 9
40	$1452 \leq F_C < 1497$	$\leq -7.2 \text{ MHz}/12/\text{SCS} + L_{\text{CRB}}$	$> 7.2 \text{ MHz}/12/\text{SCS}$	≤ 13.5	$< 11.7 \text{ MHz}/12/\text{SCS}$	≤ 13.5
50	$1457 \leq F_C < 1492$	$\leq -7.2 \text{ MHz}/12/\text{SCS} + L_{\text{CRB}}$	$> 7.2 \text{ MHz}/12/\text{SCS}$	≤ 13.5	$< 15.12 \text{ MHz}/12/\text{SCS}$	≤ 13.5
60	$1462 \leq F_C < 1487$	$\leq -7.2 \text{ MHz}/12/\text{SCS} + L_{\text{CRB}}$	$> 7.2 \text{ MHz}/12/\text{SCS}$	≤ 13.5	$< 18.72 \text{ MHz}/12/\text{SCS}$	≤ 13.5
NOTE 1 - 4: Void						

6.2.3.12 A-MPR for NS_42

Table 6.2.3.12-1: A-MPR for NS_42

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_C (MHz)	Region A		Region B			
		$RB_{\text{start}} + L_{\text{CRB}}$	A-MPR Outer/Inner (dB)	RB_{start}	$RB_{\text{start}} + L_{\text{CRB}}$	A-MPR Inner (dB)	A-MPR Outer (dB)
5	$1512 \leq F_C \leq 1514.5$	$> 3.1 \text{ MHz} / 12 / \text{SCS}$	≤ 7	$< 0.90 \text{ MHz} / 12 / \text{SCS}$	$\leq 3.1 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 4
10	$1497 \leq F_C \leq 1512$	$> 6.2 \text{ MHz} / 12 / \text{SCS}$	≤ 8	$< 0.90 \text{ MHz} / 12 / \text{SCS}$	$\leq 6.2 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 5
15	$1502 \leq F_C \leq 1509.5$	$> 9.3 \text{ MHz} / 12 / \text{SCS}$	≤ 8	$< 3.06 \text{ MHz} / 12 / \text{SCS}$	$\leq 9.3 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 5
20	$1497 \leq F_C \leq 1507$	$> 12.4 \text{ MHz} / 12 / \text{SCS}$	≤ 8	$< 4.50 \text{ MHz} / 12 / \text{SCS}$	$\leq 12.4 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 5

40	$1477 \leq F_C \leq 1497$	$> 24.8 \text{ MHz} / 12 / \text{SCS}$	≤ 8	$< 5.40 \text{ MHz} / 12 / \text{SCS}$	$\leq 24.8 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 5
50	$1467 \leq F_C \leq 1492$	$> 31 \text{ MHz} / 12 / \text{SCS}$	≤ 8	$< 7.20 \text{ MHz} / 12 / \text{SCS}$	$\leq 31 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 5
60	$1462 \leq F_C \leq 1487$	$> 37.2 \text{ MHz} / 12 / \text{SCS}$	≤ 8	$< 7.20 \text{ MHz} / 12 / \text{SCS}$	$\leq 37.2 \text{ MHz} / 12 / \text{SCS}$	≤ 1.5	≤ 5
NOTE 1 - 5: Void							

6.2.3.13 A-MPR for NS_18

Table 6.2.3.13-1: A-MPR for NS_18

Modulation/Waveform		A1 (dB)		A2 (dB)
		Outer	Inner	Inner/Outer
DFT-s-OFDM	Pi/2 BPSK	≤ 2	N/A	≤ 5
	QPSK	≤ 2		≤ 5
	16 QAM	≤3		≤ 6
	64 QAM	≤ 4		≤ 7
	256 QAM	≤ 6		≤ 9
CP-OFDM	QPSK	≤ 5		≤ 6.5
	16 QAM	≤ 5		≤ 7
	64 QAM	≤ 5.5		≤ 8.5
	256 QAM	≤ 8.5		≤ 11.5
NOTE 1: Void				
NOTE 2: Void				

6.2.3.14 Void

6.2.3.15 Void

6.2.3.16 Void

- 6.2.3.17 Void
- 6.2.3.18 Void
- 6.2.3.19 A-MPR for NS_50

Table 6.2.3.19-1: A-MPR regions for NS_50

Channel Bandwidth (MHz)	$RB_{start} * 12 * SCS$ (MHz)	$L_{CRB} * 12 * SCS$ (MHz)	A-MPR
25 MHz	$\leq L_{CRB} * 12 * SCS - 5$	> 5	A7
	≤ 6.48	≤ 1.44	A8
		≤ 3.6	A9
30 MHz	$\leq L_{CRB} * 12 * SCS - 5$	> 5	A7
	≤ 8.64	≤ 1.44	A8
		≤ 3.6	A9
40 MHz	≤ 4.32	> 0	A1
	$> 4.32, \leq 10.44$	≤ 10.8	A3
	$> 4.32, \leq 18$	> 10.8	A2
	$> 18, \leq 31.68$	$> \max(31.68 - RB_{start} * 12 * SCS, 0)$	A6
	> 31.68	> 0	A5
NOTE 1: The A-MPR values are specified in Table 6.2.3.19-2.			

Table 6.2.3.19-2: A-MPR for NS_50

Modulation/ Waveform		A1 (dB)	A2 (dB)	A3 (dB)	A5 (dB)	A6 (dB)	A7 (dB)	A8 (dB)	A9 (dB)
		Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 11	≤ 7	≤ 3	≤ 5	≤ 2	≤ 4	≤ 2	
	QPSK	≤ 11	≤ 7	≤ 3	≤ 5	≤ 2	≤ 5	≤ 2	
	16 QAM	≤ 11	≤ 7	≤ 3	≤ 5	≤ 2	≤ 5	≤ 2.5	
	64 QAM	≤ 11	≤ 7	≤ 3	≤ 5		≤ 5		
	256 QAM	≤ 11	≤ 7		≤ 5		≤ 5		
CP-OFDM	QPSK	≤ 12	≤ 8	≤ 4.5	≤ 5	≤ 3.5	≤ 6.5		≤ 3.0

	16 QAM	≤ 12	≤ 8	≤ 4.5	≤ 5	≤ 3.5	≤ 6.5		≤ 3.0
	64 QAM	≤ 12	≤ 8	≤ 4.5	≤ 5		≤ 6.5		
	256 QAM	≤ 12	≤ 8				≤ 6.5		

6.2.4 Configured transmitted power

The UE is allowed to set its configured maximum output power $P_{\text{CMAX},f,c}$ for carrier f of serving cell c in each slot. The configured maximum output power $P_{\text{CMAX},f,c}$ is set within the following bounds:

$$P_{\text{CMAX}_L,f,c} \leq P_{\text{CMAX},f,c} \leq P_{\text{CMAX}_H,f,c} \text{ with}$$

$$P_{\text{CMAX}_L,f,c} = \text{MIN} \{ P_{\text{EMAX},c} - \Delta T_{C,c}, (P_{\text{PowerClass}} - \Delta P_{\text{PowerClass}}) - \text{MAX}(\text{MAX}(\text{MPR}_c, A\text{-MPR}_c) + \Delta T_{\text{IB},c} + \Delta T_{C,c} + \Delta T_{\text{RxSRS}}, P\text{-MPR}_c) \}$$

$$P_{\text{CMAX}_H,f,c} = \text{MIN} \{ P_{\text{EMAX},c}, P_{\text{PowerClass}} - \Delta P_{\text{PowerClass}} \}$$

where

$P_{\text{EMAX},c}$ is the value given by either the *p-Max* IE or the field *additionalPmax* of the *NR-NS-PmaxList* IE, whichever is applicable according to TS 38.331[7];

$P_{\text{PowerClass}}$ is the maximum UE power specified in Table 6.2.1-1 without taking into account the tolerance specified in the Table 6.2.1-1;

When the IE *powerBoostPi2BPSK* is set to 1, $P_{\text{EMAX},c}$ is increased by +3 dB for a power class 3 UE operating in TDD bands n40, n41, n77, n78, and n79 with PI/2 BPSK modulation and UE indicates support for UE capability *powerBoosting-pi2BPSK* and 40% or less symbols in certain evaluation period are used for UL transmission when $P_{\text{EMAX},c} \geq 20$ dBm (The exact evaluation period is no less than one radio frame).

When the IE *powerBoostPi2BPSK* is set to 1, $\Delta P_{\text{PowerClass}} = -3$ dB for a power class 3 UE operating in TDD bands n40, n41, n77, n78, and n79 with Pi/2 BPSK modulation and UE indicates support for UE capability *powerBoosting-pi2BPSK* and 40% or less slots in radio frame are used for UL transmission.

$\Delta P_{\text{PowerClass}} = 3$ dB for a power class 2 UE when P-max of 23 dBm or lower is indicated; or when the field of UE capability *maxUplinkDutyCycle-PC2-FR1* is absent and the percentage of uplink symbols transmitted in a certain evaluation period is larger than 50%; or when the field of UE capability *maxUplinkDutyCycle-PC2-FR1* is not absent and the percentage of uplink symbols transmitted in a certain evaluation period is larger

than $\text{maxUplinkDutyCycle-PC2-FR1}$ as defined in TS 38.331 (The exact evaluation period is no less than one radio frame); otherwise $\Delta P_{\text{PowerClass}} = 0$ dB;

$\Delta T_{\text{IB},c}$ is the additional tolerance for serving cell c as specified in clause 6.2A.4.2 for NR CA, clause 6.2C.2 for SUL, or TS 38.101-3 clause 6.2B.4.2 for EN-DC; $\Delta T_{\text{IB},c} = 0$ dB otherwise; In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then

- a) When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta T_{\text{IB},c}$ shall be the average value for all band combinations defined in clause 6.2A.4.2, 6.2C.2 in this specification and 6.2B.4.2 in TS 38.101-3 [3], truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta T_{\text{IB},c}$ among the different supported band combinations involving such band shall be applied
- b) When the operating band frequency range is > 1 GHz, the applicable additional $\Delta T_{\text{IB},c}$ shall be the maximum value for all band combinations defined in clause 6.2A.4.2, 6.2C.2 in this specification and 6.2B.4.2 in TS 38.101-3 [3] for the applicable operating bands.

$\Delta T_{\text{C},c} = 1.5\text{dB}$ when NOTE 3 in Table 6.2.1-1 in 38.101-1 applies for a serving cell c , otherwise $\Delta T_{\text{C},c} = 0$ dB ;

MPR_c and A-MPR_c for serving cell c are specified in clause 6.2.2 and clause 6.2.3, respectively;

ΔT_{RxSRS} is applied during SRS transmission occasions with *usage* in *SRS-ResourceSet* set as 'antennaSwitching' when

- a) UE transmits SRS on the second SRS resource in every configured SRS resource set when the *SRS-TxSwitch* capability is indicated as 't1r2'
- b) UE transmits SRS on the second, third and fourth SRS resources of the total 4 SRS resources from all configured SRS resource set(s) consisting of one SRS port when the *SRS-TxSwitch* capability is indicated as 't1r4' or 't1r4-t2r4' but in 't1r4' mode.
- c) UE transmits SRS from the SRS port pair on the second SRS resource in every configured SRS resource set consisting of two SRS ports when the *SRS-TxSwitch* capability is indicated as 't2r4' or 't1r4-t2r4' but in 't2r4' mode, or
- d) UE transmits SRS to a DL-only carrier.

The value of ΔT_{RxSRS} is 4.5dB for n79 and 3 dB for bands whose $F_{\text{UL_high}}$ is lower than the $F_{\text{UL_low}}$ of n79 when the device is power class 3 in the band, or when the device is power class 2 in the band and $\Delta P_{\text{PowerClass}} = 3$ dB. The value of ΔT_{RxSRS} is 7.5dB for n79 and 6 dB for bands whose $F_{\text{UL_high}}$ is lower than the $F_{\text{UL_low}}$ of n79 when the device is power class 2 in the band and $\Delta P_{\text{PowerClass}} = 0$ dB.

For other SRS transmissions ΔT_{RxSRS} is zero;

P-MPR_c is the power management maximum power reduction for

- a) ensuring compliance with applicable electromagnetic energy absorption requirements and addressing unwanted emissions / self desense requirements in case of simultaneous transmissions on multiple RAT(s) for scenarios not in scope of 3GPP RAN specifications;
- b) ensuring compliance with applicable electromagnetic energy absorption requirements in case of proximity detection is used to address such requirements that require a lower maximum output power.

The UE shall apply $P\text{-MPR}_c$ for serving cell c only for the above cases. For UE conducted conformance testing $P\text{-MPR}_c$ shall be 0 dB

NOTE 1: $P\text{-MPR}_c$ was introduced in the $P_{\text{CMAX},f,c}$ equation such that the UE can report to the gNB the available maximum output transmit power. This information can be used by the gNB for scheduling decisions.

NOTE 2: $P\text{-MPR}_c$ may impact the maximum uplink performance for the selected UL transmission path.

T_{REF} and T_{eval} are specified in Table 6.2.4-1. For each T_{REF} , the $P_{\text{CMAX},L,c}$ for serving cell c are evaluated per T_{eval} and given by the minimum value taken over the transmission(s) within the T_{eval} ; the minimum $P_{\text{CMAX},L,f,c}$ over one or more T_{eval} is then applied for the entire T_{REF}

Table 6.2.4-1: Evaluation and reference periods for P_{cm}

T_{REF}	T_{eval}	T_{eval} with frequency hopping
Physical channel length	Physical channel length	$\text{Min}(T_{\text{no_hopping}}, \text{Physical Channel Length})$

The measured configured maximum output power $P_{\text{UMAX},f,c}$ shall be within the following bounds:

$$P_{\text{CMAX},L,f,c} - \text{MAX}\{T_{L,c}, T(P_{\text{CMAX},L,f,c})\} \leq P_{\text{UMAX},f,c} \leq P_{\text{CMAX},H,f,c} + T(P_{\text{CMAX},H,f,c}).$$

where the tolerance $T(P_{\text{CMAX},f,c})$ for applicable values of $P_{\text{CMAX},f,c}$ is specified in Table 6.2.4-1. The tolerance $T_{L,c}$ is the absolute value of the lower tolerance for the applicable operating band as specified in Table 6.2.1-1.

Table 6.2.4-1: P_{CMAX} tolerance

$P_{\text{CMAX},f,c}$ (dBm)	Tolerance $T(P_{\text{CMAX},f,c})$ (dB)
$23 < P_{\text{CMAX},c} \leq 33$	2.0
$21 \leq P_{\text{CMAX},c} \leq 23$	2.0
$20 \leq P_{\text{CMAX},c} < 21$	2.5
$19 \leq P_{\text{CMAX},c} < 20$	3.5

$18 \leq P_{\text{CMAX},c} < 19$	4.0
$13 \leq P_{\text{CMAX},c} < 18$	5.0
$8 \leq P_{\text{CMAX},c} < 13$	6.0
$-40 \leq P_{\text{CMAX},c} < 8$	7.0

6.2A Transmitter power for CA

6.2A.1 UE maximum output power for CA

6.2A.1.1 Void

6.2A.1.2 Void

6.2A.1.3 UE maximum output power for Inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the transmitter power requirements in clause 6.2 apply.

For inter-band carrier aggregation with uplink assigned to two NR bands, UE maximum output power shall be measured over all component carriers from different bands. If each band has separate antenna connectors, maximum output power is measured as the sum of maximum output power at each UE antenna connector. The period of measurement shall be at least one sub frame (1 ms). The maximum output power is specified in Table 6.2A.1.3-1.

Table 6.2A.1.3-1 UE Power Class for uplink inter-band CA (two bands)

NR CA Configuration	Class 1 (dBm)	Tolerance (dB)	Class 2 (dBm)	Tolerance (dB)	Class 3 (dBm)	Tolerance (dB)	Class 5 (dBm)	Tolerance (dB)
CA_n3A-n78A					23	+2/-3		
CA_n8A-n78A					23	+2/-3		

NOTE 1: Void

NOTE 2: For an NR CA configuration with one uplink carrier assigned to one NR band, if the uplink NR band has NOTE 3 in Table 6.2.1-1, the band is allowed to reduce the lower tolerance limit by 1.5 dB when the transmission bandwidths of the band(s) is confined within $F_{\text{UL_low}}$ and $F_{\text{UL_low}} + 4$ MHz or $F_{\text{UL_high}} - 4$ MHz and $F_{\text{UL_high}}$.

An uplink NR CA configuration in which at least one of the bands has NOTE 3 in Table 6.2.1-1 is allowed to reduce the lower tolerance limit by 1.5 dB when the transmission bandwidths of at least one of the bands is confined within F_{UL_low} and $F_{UL_low} + 4$ MHz or $F_{UL_high} - 4$ MHz and F_{UL_high} .

NOTE 3: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance

NOTE 4: For inter-band carrier aggregation the maximum power requirement should apply to the total transmitted power over all component carriers (per UE).

NOTE 5: Power class 3 is the default power class unless otherwise stated

6.2A.2 UE maximum output power reduction for CA

6.2A.2.1 Void

6.2A.2.2 Void

6.2A.2.3 UE maximum output power reduction for Inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the requirements in subclause 6.2.2 apply. For inter-band carrier aggregation with uplink assigned to two NR bands, the requirements in clause 6.2.2 apply for each uplink component carrier.

6.2A.3 UE additional maximum output power reduction for CA

6.2A.3.1.1 Void

6.2A.3.1.2 Void

6.2A.3.1.3 UE additional maximum output power reduction for Inter-band CA

Unless otherwise stated, for inter-band carrier aggregation with one uplink carrier assigned to one NR band, the requirements in subclause 6.2.3 apply.

Unless specified in Table 6.2A.3.1.3-1, for inter-band carrier aggregation with uplink assigned to two NR bands, the requirements in clause 6.2.3 apply only to the indicated carrier. The requirements in Table 6.2A.3.1.3-1 are specified in terms of an additional spectrum emission requirement with their associated network signalling values and the allowed A-MPR. Unless otherwise stated, the combined requirements and allowed A-MPR are applicable on both bands when both component carriers are active.. Additional spurious emission requirements are signalled by the network to indicate that the UE shall meet the additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

To meet the additional requirements, additional maximum power reduction (A-MPR) is allowed for the maximum output power as specified in Table 6.2.1-1. Unless stated otherwise, the total reduction to UE maximum output power is $\max(\text{MPR}, \text{A-MPR})$ where MPR is defined in clause 6.2.2. In case of a power class 3 UE, when IE powerBoostP2BPSK is set to 1, power class 2 A-MPR values apply.

For almost contiguous allocations in CP-OFDM waveforms in power class 3, the allowed A-MPR defined in clause 6.2.3 is increased by $\text{CEIL}\{10 \log_{10}(1 + \text{NRB_gap} / \text{NRB_alloc}), 0.5\}$ dB, where NRB_gap is the total number of unallocated RBs between allocated RBs and NRB_alloc is the total number of allocated RBs, and the parameter LCRB is replaced by $\text{NRB_alloc} + \text{NRB_gap}$ in specifying the RB allocation regions.

Unless otherwise specified, $\pi/2$ BPSK in following A-MPR tables refers to both variants of $\pi/2$ BPSK referenced in 6.2.2 tables 6.2.2-1.

The emission requirements specified in Table 6.2A.3.1.3-1 also apply for the frequency ranges that are less than FOOB (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.2A.3.1.3-1: Additional Requirements for uplink inter-band carrier aggregation (two-bands)

NR CA combination	Band	Applied NS	Requirements (clause)	A-MPR (table/clause)	Note
CA_n3-n78	n3	100	6.5.2.4.2	Table 6.2.3.1-2	1
CA_n8-n78	n8	43	6.5.3.3.5	Clause 6.2.3.6	1
		43U	6.5.3.3.5, 6.5.2.4.2	Clause 6.2.3.6	
NOTE 1: NS_05U, NS_43U and NS_100 can be signalled for NR bands that have UTRA services deployed and the requirements in clause 6.5.2.4.2 are only applicable to the signalling carrier.					

6.2A.4 Configured output power for CA

6.2A.4.1 Configured transmitted power level

6.2A.4.1.1 Void

6.2A.4.1.2 Void

6.2A.4.1.3 Configured transmitted power for Inter-band CA

For uplink carrier aggregation the UE is allowed to set its configured maximum output power $P_{\text{CMAX},c}$ for serving cell c and its total configured maximum output power P_{CMAX} .

The configured maximum output power $P_{\text{CMAX},c}$ on serving cell c shall be set as specified in clause 6.2.4.

For uplink inter-band carrier aggregation, MPR_c and A-MPR_c apply per serving cell c and are specified in clause 6.2.2 and clause 6.2.3, respectively. P-MPR_c accounts for power management for serving cell c . $P_{\text{CMAX},c}$ is calculated under the assumption that the transmit power is increased independently on all component carriers.

The total configured maximum output power P_{CMAX} shall be set within the following bounds:

$$P_{\text{CMAX_L}} \leq P_{\text{CMAX}} \leq P_{\text{CMAX_H}}$$

For uplink inter-band carrier aggregation with one serving cell c per operating band when same slot symbol pattern is used in all aggregated serving cells,

$$P_{\text{CMAX_L}} = \text{MIN} \{ 10 \log_{10} \sum \text{MIN} [p_{\text{EMAX},c} / \left(\frac{F_0}{44} \right)_{C,c}, p_{\text{PowerClass},c} / (\text{MAX}(\text{mpr}_c, \text{a-mpr}_c) \left(\frac{F_0}{44} \right)_{C,c} \left(\frac{F_0}{44} \right)_{\text{IB},c} \left(\frac{F_0}{44} \right)_{\text{RxSRS},c}), p_{\text{PowerClass},c} / \text{pmpr}_c], P_{\text{EMAX,CA}}, P_{\text{PowerClass,CA}} \}$$

$$P_{\text{CMAX_H}} = \text{MIN} \{ 10 \log_{10} \sum p_{\text{EMAX},c}, P_{\text{EMAX,CA}}, P_{\text{PowerClass,CA}} \}$$

where

- $p_{\text{EMAX},c}$ is the linear value of $P_{\text{EMAX},c}$ which is given by IE *P-Max* for serving cell c in [7];
- $P_{\text{PowerClass,CA}}$ is the maximum UE power specified in Table 6.2A.1.3-1 without taking into account the tolerance specified in the Table 6.2A.1.3-1;
- $p_{\text{PowerClass},c}$ is the linear value of the maximum UE power for serving cell c specified in Table 6.2.1-1 without taking into account the tolerance;
- mpr_c and a-mpr_c are the linear values of MPR_c and A-MPR_c as specified in clause 6.2.2 and clause 6.2.3, respectively;
- pmpr_c is the linear value of P-MPR_c ;
- $\Delta t_{\text{RxSRS},c}$ is the linear value of $\Delta T_{\text{RxSRS},c}$;
- $\left(\frac{F_0}{44} \right)_{C,c}$ is the linear value of $\left(\frac{F_0}{44} \right)_{C,c} \left(\frac{F_0}{44} \right)_{\text{IB},c} \left(\frac{F_0}{44} \right)_{\text{RxSRS},c} = 1.41$ when NOTE 2 in Table 6.2A.1.3-1 applies for a serving cell c , otherwise $\left(\frac{F_0}{44} \right)_{C,c} = 1$;

- $\Delta T_{IB,c}$ is the linear value of the inter-band relaxation term $\Delta T_{IB,c}$ of the serving cell c as specified in clause 6.2A.4.2 for NR CA, clause 6.2C.2 for SUL, or TS 38.101-3 clause 6.2B.4.2 for EN-DC; otherwise $\Delta T_{IB,c}$ is the maximum value of $\Delta T_{IB,c}$ among the different supported band combinations involving such band shall be applied. In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then
 - a) When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta T_{IB,c}$ shall be the average value for all band combinations defined in clause 6.2A.4.2, 6.2C.2 in this specification and 6.2B.4.2 in TS 38.101-3 [3], truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta T_{IB,c}$ among the different supported band combinations involving such band shall be applied
 - b) When the operating band frequency range is > 1 GHz, the applicable additional $\Delta T_{IB,c}$ shall be the maximum value for all band combinations defined in clause 6.2A.4.2, 6.2C.2 in this specification and 6.2B.4.2 in TS 38.101-3 [3] for the applicable operating bands.
- $P_{EMAX,CA}$ is the value indicated by $p\text{-NR-FR1}$ or by $p\text{-UE-FR1}$ whichever is the smallest if both are present. For uplink inter-band carrier aggregation with one serving cell c per operating band when at least one different numerology/slot pattern is used in aggregated cells, the UE is allowed to set its configured maximum output power $P_{CMAX,c(i),i}$ for serving cell $c(i)$ of slot numerology type i , and its total configured maximum output power P_{CMAX} . The configured maximum output power $P_{CMAX,c(i),i}(p)$ in slot p of serving cell $c(i)$ on slot numerology type i shall be set within the following bounds:

$$P_{CMAX_L,f,c(i),i}(p) \leq P_{CMAX,f,c(i),i}(p) \leq P_{CMAX_H,f,c(i),i}(p)$$

where $P_{CMAX_L,f,c(i),i}(p)$ and $P_{CMAX_H,f,c(i),i}(p)$ are the limits for a serving cell $c(i)$ of slot numerology type i as specified in clause 6.2.4. The total UE configured maximum output power $P_{CMAX}(p,q)$ in a slot p of slot numerology or symbol pattern i , and a slot q of slot numerology or symbol pattern j that overlap in time shall be set within the following bounds unless stated otherwise:

$$P_{CMAX_L}(p,q) \leq P_{CMAX}(p,q) \leq P_{CMAX_H}(p,q)$$

When slots p and q have different transmissions lengths and belong to different cells on different bands:

$$P_{CMAX_L}(p,q) = \text{MIN} \{ 10 \log_{10} [P_{CMAX_L,f,c(i),i}(p) + P_{CMAX_L,f,c(i),j}(q)], P_{PowerClass,CA}, P_{EMAX,CA} \}$$

$$P_{CMAX_H}(p,q) = \text{MIN} \{ 10 \log_{10} [P_{CMAX_H,f,c(i),i}(p) + P_{CMAX_H,f,c(i),j}(q)], P_{PowerClass,CA}, P_{EMAX,CA} \}$$

where $P_{CMAX_L,f,c(i),i}$ and $P_{CMAX_H,f,c(i),i}$ are the respective limits $P_{CMAX_L,f,c(i),i}$ and $P_{CMAX_H,f,c(i),i}$ expressed in linear scale.

T_{REF} and T_{eval} are specified in Table 6.2A.4.1.3-0 when same and different slot patterns are used in aggregated carriers. For each T_{REF} , the P_{CMAX_L} is evaluated per T_{eval} and given by the minimum value taken over the transmission(s) within the T_{eval} ; the minimum P_{CMAX_L} over the one or more T_{eval} is then applied for the entire T_{REF} . The lesser of $P_{PowerClass,CA}$ and $P_{EMAX,CA}$ shall not be exceeded by the UE during any period of time.

Table 6.2A.4.1.3-0: P_{CMAX} evaluation window for different slot and channel durations

T_{REF}	T_{eval}	T_{eval} with frequency hopping
T_{REF} of largest slot duration over both UL CCs	Physical channel length	$\text{Min}(T_{no_hopping}, \text{Physical Channel Length})$

If the UE is configured with multiple TAGs and transmissions of the UE on slot i for any serving cell in one TAG overlap some portion of the first symbol of the transmission on slot $i + 1$ for a different serving cell in another TAG, the UE minimum of P_{CMAX_L} for slots i and $i + 1$ applies for any overlapping portion of slots i and $i + 1$. The lesser of $P_{PowerClass,CA}$ and $P_{EMAX,CA}$ shall not be exceeded by the UE during any period of time.

The measured maximum output power P_{UMAX} over all serving cells with same slot pattern shall be within the following range:

$$P_{CMAX_L} - \text{MAX}\{T_L, T_{LOW}(P_{CMAX_L})\} \leq P_{UMAX} \leq P_{CMAX_H} + T_{HIGH}(P_{CMAX_H})$$

$$P_{UMAX} = 10 \log_{10} \sum p_{UMAX,c}$$

where $p_{UMAX,c}$ denotes the measured maximum output power for serving cell c expressed in linear scale. The tolerances $T_{LOW}(P_{CMAX})$ and $T_{HIGH}(P_{CMAX})$ for applicable values of P_{CMAX} are specified in Table 6.2A.4.1.3-1. The tolerance T_L is the absolute value of the lower tolerance for applicable NR CA configuration as specified in Table 6.2A.1.3-1 for inter-band carrier aggregation. The measured maximum output power P_{UMAX} over all serving cells, when at least one slot has a different transmission numerology or symbol pattern, shall be within the following range:

$$P'_{CMAX_L} - \text{MAX}\{T_L, T_{LOW}(P'_{CMAX_L})\} \leq P'_{UMAX} \leq P'_{CMAX_H} + T_{HIGH}(P'_{CMAX_H})$$

$$P'_{UMAX} = 10 \log_{10} \sum p'_{UMAX,c}$$

where $p'_{\text{UMAX},c}$ denotes the average measured maximum output power for serving cell c expressed in linear scale over T_{REF} . The tolerances $T_{\text{LOW}}(P'_{\text{CMAX}})$ and $T_{\text{HIGH}}(P'_{\text{CMAX}})$ for applicable values of P'_{CMAX} are specified in Table 6.2A.4.1.3-1 for inter-band carrier aggregation. The tolerance T_L is the absolute value of the lower tolerance for applicable NR CA configuration as specified in Table 6.2A.1.3-1 for inter-band carrier aggregation.

where:

$$P'_{\text{CMAX_L}} = \text{MIN}\{ \text{MIN}\{10\log_{10}\sum(p_{\text{CMAX_L},f,c(i),i}), P_{\text{PowerClass,CA}}\} \text{ over all overlapping slots in } T_{\text{REF}}\}$$

$$P'_{\text{CMAX_H}} = \text{MAX}\{ \text{MIN}\{10\log_{10}\sum(p_{\text{EMAX},c}), P_{\text{PowerClass,CA}}\} \text{ over all overlapping slots in } T_{\text{REF}}\}$$

Table 6.2A.4.1.3-1: P_{CMAX} tolerance for uplink inter-band CA (two bands)

P_{CMAX} (dBm)	Tolerance $T_{\text{LOW}}(P_{\text{CMAX}})$ (dB)	Tolerance $T_{\text{HIGH}}(P_{\text{CMAX}})$ (dB)
$P_{\text{CMAX}} = 23$	3.0	2.0
$22 \leq P_{\text{CMAX}} < 23$	5.0	2.0
$21 \leq P_{\text{CMAX}} < 22$	5.0	3.0
$20 \leq P_{\text{CMAX}} < 21$	5.0	4.0
$16 \leq P_{\text{CMAX}} < 20$	5.0	
$11 \leq P_{\text{CMAX}} < 16$	6.0	
$-40 \leq P_{\text{CMAX}} < 11$	7.0	

6.2A.4.2 $\Delta T_{\text{IB},c}$ for CA

For the UE which supports inter-band NR CA configuration, $\Delta T_{\text{IB},c}$ in tables below applies. Unless otherwise stated, $\Delta T_{\text{IB},c}$ is set to zero.

6.2A.4.2.1 Void

6.2A.4.2.2 Void

6.2A.4.2.3 $\Delta T_{\text{IB},c}$ for Inter-band CA (two bands)

Table 6.2A.4.2.3-1: $\Delta T_{\text{IB},c}$ due to NR CA (two bands)

Inter-band CA combination	NR Band	$\Delta T_{IB,c}$ (dB)
CA_n3-n77	n3	0.6
	n77	0.8
CA_n3-n78	n3	0.6
	n78	0.8
CA_n3-n79	n3	0.3
	n79	0.8
CA_n8-n75	n8	0.3
CA_n8-n78	n8	0.6
	n78	0.8
CA_n8-n79	n8	0.3
	n79	0.8
CA_n28-n75	n28	0.3
CA_n28-n78	n28	0.5
	n78	0.8
CA_n41-n78 ¹	n41	0.3
	n78	0.8
CA_n75-n78	n78	0.8
CA_n76-n78	n78	0.8
CA_n77-n79	n77	0.5
	n79	0.5
CA_n78-n79	n78	0.5
		1.5 ²
	n79	0.5
		1.5 ²
NOTE: The requirements only apply when the sub-frame and Tx-Rx timings are synchronized between the component carriers. In the absence of synchronization, the requirements are not within scope of these specifications.		
NOTE 2: The requirements only apply for UE supporting inter-band carrier aggregation with simultaneous Rx/Tx capability, and NR UL carrier frequencies are confined to 3700 MHz-3800MHz for n78 and 4400 MHz-4500MHz for n79. Simultaneous Rx/Tx capability does not apply for UEs supporting band n78 with a n77 implementation.		

6.2B Void

6.2C Transmitter power for SUL

6.2C.1 Configured transmitted power for SUL

When a UE is configured with both NR UL and NR SUL carriers in a serving cell with active transmission either on the UL carrier or SUL carrier, the configured transmit power requirements specified in clause 6.2.4 are applicable for the UL carrier and the SUL carrier, respectively.

6.2C.2 $\Delta T_{IB,c}$

For the UE which supports SUL band combination, $\Delta T_{IB,c}$ in Tables below applies. Unless otherwise stated, $\Delta T_{IB,c}$ is set to zero.

Table 6.2C.2-1: $\Delta T_{IB,c}$ due to SUL

Band combination for SUL	NR Band	$\Delta T_{IB,c}$ (dB)
SUL_n78-n80	n78	0.8
	n80	0.6
SUL_n78-n81	n78	0.8
	n81	0.6
SUL_n78-n82	n78	0.8
	n82	0.6
SUL_n78-n83	n78	0.8
	n83	0.5
SUL_n78-n84	n78	0.8
	n84	0.3
SUL_n78-n86	n78	0.8
	n86	0.6

6.2D Transmitter power for UL MIMO

6.2D.1 UE maximum output power for UL MIMO

For power class 2 UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the maximum output power for any transmission bandwidth within the channel bandwidth is specified in Table 6.2D.1-1. The requirements shall be met with the UL MIMO configurations specified in Table 6.2D.1-2. For UE supporting UL MIMO, the maximum output power is defined as the sum of the maximum output power from both UE antenna connectors. The period of measurement shall be at least one sub frame (1 ms).

The requirements shall be met with the UL MIMO configurations of using 2-layer UL MIMO transmission with codebook of $\begin{bmatrix} 1 & 1 & 0 \\ \sqrt{2} & 0 & 1 \end{bmatrix}$. DCI Format for UE configured in PUSCH transmission mode for uplink single-user MIMO shall be used.

Table 6.2D.1-1: UE Power Class for UL MIMO in closed loop spatial multiplexing scheme

NR band	Class 1 (dBm)	Tolerance (dB)	Class 2 (dBm)	Tolerance (dB)	Class 3 (dBm)	Tolerance (dB)	Class 5 (dBm)	Tolerance (dB)
n41			26	+2/-3 ¹	23	+2/-3 ¹		
n77			26	+2/-3	23	+2/-3		
n78			26	+2/-3	23	+2/-3		
n79			26	+2/-3	23	+2/-3		
NOTE 1: The transmission bandwidths confined within FUL_low and FUL_low + 4 MHz or FUL_high – 4 MHz and FUL_high, the maximum output power requirement is relaxed by reducing the lower tolerance limit by 1.5 dB								
NOTE 2: Power class 3 is the default power class unless otherwise stated								

Table 6.2D.1-2: UL MIMO configuration in closed-loop spatial multiplexing scheme

Transmission scheme	DCI format	Codebook Index
Codebook based uplink	DCI format 0_1	Codebook index 0

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.2.1 apply for the power class as indicated by the *ue-PowerClass* field in capability signalling.

6.2D.2 UE maximum output power reduction for UL MIMO

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the allowed Maximum Power Reduction (MPR) for the maximum output power in Table 6.2D.1-1 is specified in Table 6.2.2-1. The requirements shall be met with UL MIMO configurations defined in Table 6.2D.1-2. For UE supporting UL MIMO, the maximum output power is defined as the sum of the maximum output power from both UE antenna connector for the power class as indicated by the *ue-PowerClass* field in capability signalling.

For the UE maximum output power modified by MPR, the power limits specified in clause 6.2D.4 apply.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.2.2 apply.

6.2D.3 UE additional maximum output power reduction for UL MIMO

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the A-MPR values specified in clause 6.2.3 shall apply to the maximum output power specified in Table 6.2D.1-1. The requirements shall be met with the UL MIMO configurations specified in Table 6.2D.1-2. For UE supporting UL MIMO, the maximum output power is defined as the sum of the maximum output power from both UE antenna connector. Unless stated otherwise, an A-MPR of 0 dB shall be used.

For the UE maximum output power modified by A-MPR, the power limits specified in clause 6.2D.4 apply.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.2.3 apply for the power class as indicated by the *ue-PowerClass* field in capability signalling.

6.2D.4 Configured transmitted power for UL MIMO

For UE supporting UL MIMO, the transmitted power is configured per each UE.

The definitions of configured maximum output power $P_{\text{CMAX},c}$, the lower bound $P_{\text{CMAX}_L,c}$, and the higher bound $P_{\text{CMAX}_H,c}$ specified in clause 6.2.4 shall apply to UE supporting UL MIMO, where

- $P_{\text{PowerClass}}$, $\Delta P_{\text{PowerClass}}$ and $\Delta T_{C,c}$ are specified in clause 6.2D.1;
- MPR_c is specified in clause 6.2D.2;

- A-MPR_c is specified in clause 6.2D.3.

The measured configured maximum output power $P_{\text{UMAX},c}$ for serving cell c shall be within the following bounds:

$$P_{\text{CMAX}_L,c} - \text{MAX}\{T_L, T_{\text{LOW}}(P_{\text{CMAX}_L,c})\} \leq P_{\text{UMAX},c} \leq P_{\text{CMAX}_H,c} + T_{\text{HIGH}}(P_{\text{CMAX}_H,c})$$

where $T_{\text{LOW}}(P_{\text{CMAX}_L,c})$ and $T_{\text{HIGH}}(P_{\text{CMAX}_H,c})$ are defined as the tolerance and applies to $P_{\text{CMAX}_L,c}$ and $P_{\text{CMAX}_H,c}$ separately, while T_L is the absolute value of the lower tolerance in Table 6.2D.1-1 for the applicable operating band.

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the tolerance is specified in Table 6.2D.4-1. The requirements shall be met with UL MIMO configurations specified in Table 6.2D.1-2.

Table 6.2D.4-1: $P_{\text{CMAX},c}$ tolerance in closed-loop spatial multiplexing scheme

$P_{\text{CMAX},c}$ (dBm)	Tolerance $T_{\text{LOW}}(P_{\text{CMAX}_L,c})$ (dB)	Tolerance $T_{\text{HIGH}}(P_{\text{CMAX}_H,c})$ (dB)
$P_{\text{CMAX},c} = 26$	3.0	2.0
$23 \leq P_{\text{CMAX},c} < 26$	3.0	2.0
$22 \leq P_{\text{CMAX},c} < 23$	5.0	2.0
$21 \leq P_{\text{CMAX},c} < 22$	5.0	3.0
$20 \leq P_{\text{CMAX},c} < 21$	5.0	4.0
$16 \leq P_{\text{CMAX},c} < 20$	5.0	
$11 \leq P_{\text{CMAX},c} < 16$	6.0	
$-40 \leq P_{\text{CMAX},c} < 11$	7.0	

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.2.4 apply for the power class as indicated by the *ue-PowerClass* field in capability signalling.

6.3 Output power dynamics

6.3.1 Minimum output power

The minimum controlled output power of the UE is defined as the power in the channel bandwidth for all transmit bandwidth configurations (resource blocks), when the power is set to a minimum value.

The minimum output power is defined as the mean power in at least one sub-frame 1 ms. The minimum output power shall not exceed the values specified in Table 6.3.1-1.

Table 6.3.1-1: Minimum output power

Channel bandwidth (MHz)	Minimum output power (dBm)	Measurement bandwidth (MHz)
5	-40	4.515
10	-40	9.375
15	-40	14.235
20	-40	19.095
25	-39	23.955
30	-38.2	28.815
40	-37	38.895
50	-36	48.615
60	-35.2	58.35
80	-34	78.15
90	-33.5	88.23
100	-33	98.31

6.3.2 Transmit OFF power

Transmit OFF power is defined as the mean power in the channel bandwidth when the transmitter is OFF. The transmitter is considered OFF when the UE is not allowed to transmit on any of its ports..

The transmit OFF power is defined as the mean power in a duration of at least one sub-frame (1 ms) excluding any transient periods. The transmit OFF power shall not exceed the values specified in Table 6.3.2-1.

Table 6.3.2-1: Transmit OFF power

Channel bandwidth (MHz)	Transmit OFF power (dBm)	Measurement bandwidth (MHz)
----------------------------	-----------------------------	--------------------------------

5	-50	4.515
10	-50	9.375
15	-50	14.235
20	-50	19.095
25	-50	23.955
30	-50	28.815
40	-50	38.895
50	-50	48.615
60	-50	58.35
80	-50	78.15
90	-50	88.23
100	-50	98.31

6.3.3 Transmit ON/OFF time mask

6.3.3.1 General

The transmit power time mask defines the transient period(s) allowed

- between transmit OFF power as defined in clause 6.3.2 and transmit ON power symbols (transmit ON/OFF)
- between continuous ON-power transmissions with powerchange or RB hopping is applied.

In case of RB hopping, transition period is shared symmetrically.

Unless otherwise stated the requirements in clause 6.5 apply also in transient periods.

In the following clauses, following definitions apply:

- A slot or long subslot transmission is a transmission with more than 2 symbols.
- A short subslot transmission is a transmission with 1 or 2 symbols.

6.3.3.2 General ON/OFF time mask

The general ON/OFF time mask defines the observation period between transmit OFF and ON power and between transmit ON and OFF power for each SCS. ON/OFF scenarios include: contiguous, and non-contiguous transmission, etc

The OFF power measurement period is defined in a duration of at least one slot excluding any transient periods. The ON power is defined as the mean power over one slot excluding any transient period.



Figure 6.3.3.2-1: General ON/OFF time mask for NR UL transmission in FR1

6.3.3.3 Transmit power time mask for slot and short or long subslot boundaries

The transmit power time mask for slot and a long subslot transmission boundaries defines the transient periods allowed between slot and long subslot PUSCH transmissions. For PUSCH-PUCCH and PUSCH-SRS transitions and multiplexing the time masks in clause 6.3.3.7 apply.

The transmit power time mask for slot or long subslot and short subslot transmission boundaries defines the transient periods allowed between slot or long subslot and short subslot transmissions. The time masks in clause 6.3.3.8 apply.

The transmit power time mask for short subslot transmission boundaries defines the transient periods allowed between short subslot transmissions. The time masks in clause 6.3.3.9 apply.

6.3.3.4 PRACH time mask

The PRACH ON power is specified as the mean power over the PRACH measurement period excluding any transient periods as shown in Figure 6.3.3.4-1. The measurement period for different PRACH preamble format is specified in Table 6.3.3.4-1.

Table 6.3.3.4-1: PRACH ON power measurement period

PRACH preamble format	SCS (kHz)	Measurement period (ms)
0	1.25	0.903125

1	1.25	2.284375
2	1.25	3.352604
3	5	0.903125
A1	15	0.142708
	30	0.071354
A2	15	0.285417
	30	0.142708
A3	15	0.428125
	30	0.2140625
B1	15	0.140365
	30	0.070182
B4	15	0.83046875
	30	0.415234375
A1/B1	15	0.142708 ms for first six occasion 0.140365 ms for the last occasion
	30	0.071354 ms for first six occasion 0.070182 ms for the last occasion
A2/B2	15	0.285417 ms for first two occasion 0.278385 ms for the third occasion
	30	0.142708 ms for first two occasion 0.1391925 ms for the third occasion
A3/B3	15	0.428125 ms for the first occasion 0.41640625 ms for the second occasion
	30	0.2140625 ms for the first occasion 0.208203125 ms for the second occasion
C0	15	0.10703125
	30	0.053515625
C2	15	0.333333
	30	0.166667
NOTE: For PRACH on PRACH occasion start from the beginning of 0.5 ms or span the boundary of 0.5 ms of the subframe, the measurement period will plus 0.032552 μ s		

Figure 6.3.3.4-1: PRACH ON/OFF time mask

6.3.3.5 Void

6.3.3.6 SRS time mask

For SRS transmission mapped to one OFDM symbol, the ON power is defined as the mean power over the symbol duration excluding any transient period; See Figure 6.3.3.6-1

Figure 6.3.3.6-1: Single SRS time mask for NR UL transmission

For SRS transmission mapped to two or more OFDM symbols the ON power is defined as the mean power for each symbol duration excluding any transient period. For consecutive SRS transmissions without power change, Figure 6.3.3.6-2 applies.

Figure 6.3.3.6-2: Consecutive SRS time mask for the case when no power change is required with SRS usage other than antenna switching.

When power change between consecutive SRS transmissions is required, then Figure 6.3.3.6-3 and Figure 6.3.3.6-4 apply.

Figure 6.3.3.6-3: Consecutive SRS time mask for the case when power change is required and when 15 kHz and 30 kHz SCS is used in FR1 with SRS usage other than antenna switching.

Figure 6.3.3.6-4: Consecutive SRS time mask for the case when power change is required and when 60 kHz SCS is used in FR1

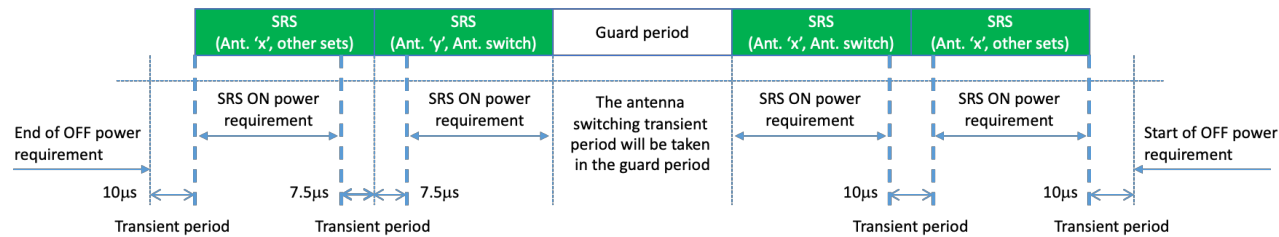


Figure 6.3.3.6-5: FR1 Time mask for 15 kHz and 30 kHz SCS for the case when consecutive SRS switching usage is between antenna switching & other sets

where "other sets" belongs to a "usage set" other than the set for antenna switching. The usage sets for SRS switching are defined in clause 6.2.1 of TS 38.214 [10].

NOTE: Guard period of one symbol is defined between two SRS resources of an SRS resource set for antenna switching for 15kHz, 30kHz and 60kHz SCS in Table 6.2.1.2-1 of TS 38.214 [10].

The above transient period applies to all the transmit CCs in CA with the CC sounding SRS. UE RF requirements do not apply during this transient period.

6.3.3.7 PUSCH-PUCCH and PUSCH-SRS time masks

The PUCCH/PUSCH/SRS time mask defines the observation period between sounding reference symbol (SRS) and an adjacent PUSCH/PUCCH symbol and subsequent UL transmissions. The time masks apply for all types of frame structures and their allowed PUCCH/PUSCH/SRS transmissions unless otherwise stated.

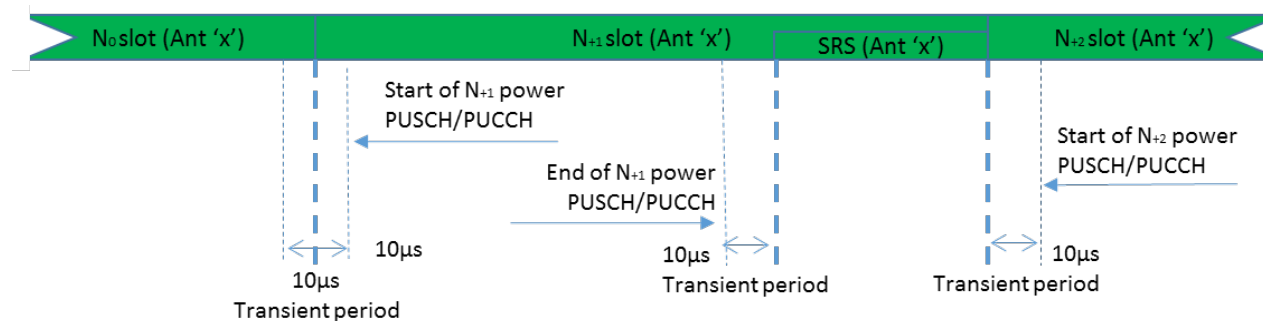


Figure 6.3.3.7-1: PUCCH/PUSCH/SRS time mask when there is a transmission before or after or both before and after SRS, when sounded on the same antenna (Ant 'x')

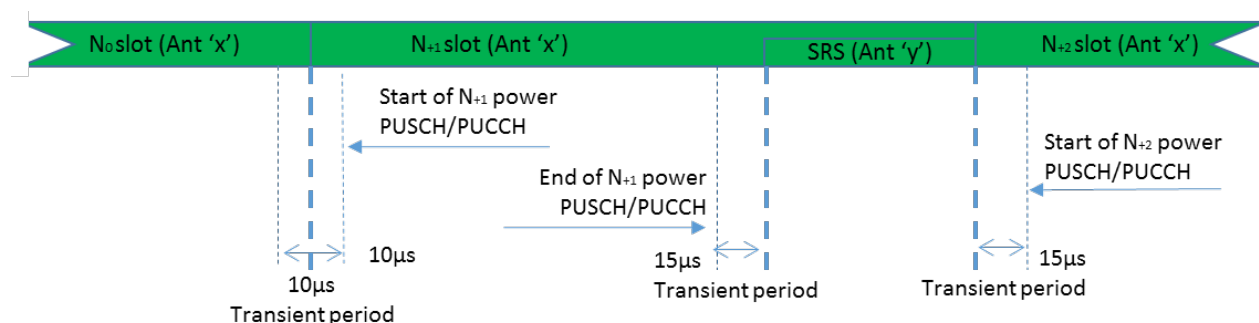


Figure 6.3.3.7-2: PUCCH/PUSCH/SRS time mask when there is a transmission before or after or both before and after SRS, when sounded on a different antenna (Ant 'x' and Ant 'y' are different antenna ports)

This transient period of 15 usec applies before and after SRS transmission to all the transmit CCs in CA with the CC sounding SRS. UE RF requirements do not apply during this transient period.

When there is no transmission preceding SRS transmission or succeeding SRS transmission, then the same time mask applies as shown in Figure 6.3.3.7-1.

6.3.3.8 Transmit power time mask for consecutive slot or long subslot transmission and short subslot transmission boundaries

The transmit power time mask for consecutive slot or long subslot transmission and short slot transmission boundaries defines the transient periods allowed between such transmissions.

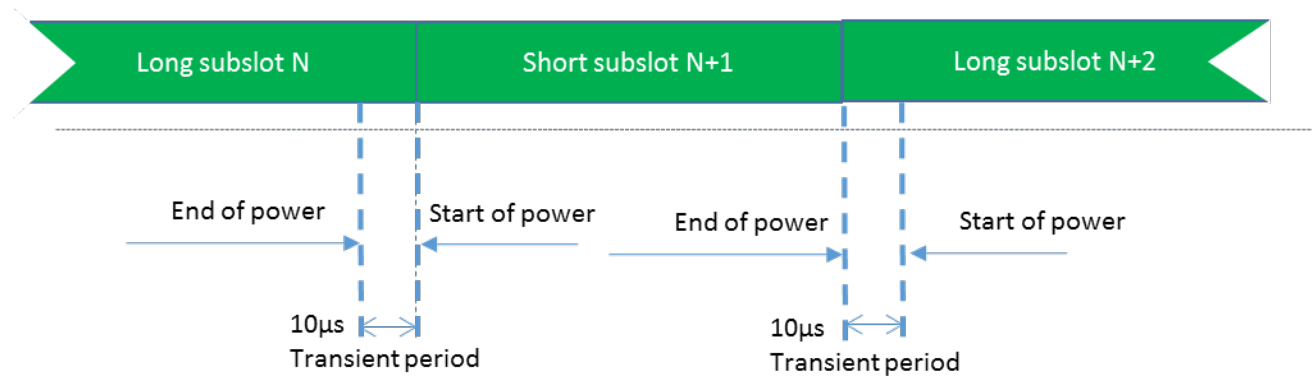


Figure 6.3.3.8-1: Consecutive slot or long subslot transmission and short subslot transmission time mask

6.3.3.9 Transmit power time mask for consecutive short subslot transmissions boundaries

The transmit power time mask for consecutive short subslot transmission boundaries defines the transient periods allowed between short subslot transmissions.

The transient period shall be equally shared as shown on Figure 6.3.3.9-2.

Figure 6.3.3.9-1: Void

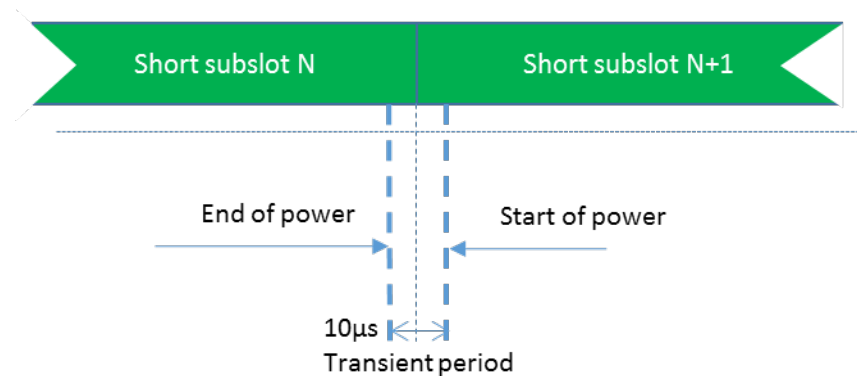


Figure 6.3.3.9-2: Consecutive short subslot transmissions time mask

Figure 6.3.3.9-3: Consecutive short subslot (1 symbol gap) time mask for the case when transient period is required on both sides of the symbol and when 60 kHz SCS is used in FR1

6.3.4 Power control

6.3.4.1 General

The requirements on power control accuracy apply under normal conditions.

6.3.4.2 Absolute power tolerance

The absolute power tolerance is the ability of the UE transmitter to set its initial output power to a specific value for the first sub-frame (1 ms) at the start of a contiguous transmission or non-contiguous transmission with a transmission gap larger than 20 ms. The tolerance includes the channel estimation error.

The minimum requirement specified in Table 6.3.4.2-1 apply in the power range bounded by the minimum output power as specified in clause 6.3.1 and the maximum output power as specified in clause 6.2.1.

Table 6.3.4.2-1: Absolute power tolerance

Conditions	Tolerance
Normal	± 9.0 dB

6.3.4.3 Relative power tolerance

The relative power tolerance is the ability of the UE transmitter to set its output power in a target sub-frame (1 ms) relatively to the power of the most recently transmitted reference sub-frame (1 ms) if the transmission gap between these sub-frames is less than or equal to 20 ms.

The minimum requirements specified in Table 6.3.4.3-1 apply when the power of the target and reference sub-frames are within the power range bounded by the minimum output power as defined in clause 6.3.1 and the measured P_{UMAX} as defined in clause 6.2.4.

To account for RF Power amplifier mode changes, 2 exceptions are allowed for each of two test patterns. The test patterns are a monotonically increasing power sweep and a monotonically decreasing power sweep over a range bounded by the requirements of minimum power and maximum power specified in clauses 6.3.1 and 6.2.1, respectively. For those exceptions, the power tolerance limit is a maximum of ± 6.0 dB in Table 6.3.4.3-1.

Table 6.3.4.3-1: Relative power tolerance

Power step ΔP (Up or down) (dB)	All combinations of PUSCH and PUCCH transitions (dB)	All combinations of PUSCH/PUCCH and SRS transitions between sub-frames (dB)	PRACH (dB)
$\Delta P < 2$	± 2.0 (NOTE)	± 2.5	± 2.0
$2 \leq \Delta P < 3$	± 2.5	± 3.5	± 2.5
$3 \leq \Delta P < 4$	± 3.0	± 4.5	± 3.0
$4 \leq \Delta P < 10$	± 3.5	± 5.5	± 3.5
$10 \leq \Delta P < 15$	± 4.0	± 7.0	± 4.0
$15 \leq \Delta P$	± 5.0	± 8.0	± 5.0
NOTE: For PUSCH to PUSCH transitions with the allocated resource blocks fixed in frequency and no transmission gaps other than those generated by downlink subframes, DwPTS fields or Guard Periods: for a power step $\Delta P \leq 1$ dB, the relative power tolerance for transmission is ± 0.7 dB.			

6.3.4.4 Aggregate power tolerance

The aggregate power control tolerance is the ability of the UE transmitter to maintain its power in a sub-frame (1 ms) during non-contiguous transmissions within 21 ms in response to 0 dB commands with respect to the first UE transmission and all other power control parameters as specified in TS 38.213 [8] kept constant.

The minimum requirement specified in Table 6.3.4.4-1 apply in the power range bounded by the minimum output power as specified in clause 6.3.1 and the maximum output power as specified in clause 6.2.1.

Table 6.3.4.4-1: Aggregate power tolerance

TPC command	UL channel	Aggregate power tolerance within 21 ms
0 dB	PUCCH	± 2.5 dB
0 dB	PUSCH	± 3.5 dB

6.3A Output power dynamics for CA

6.3A.1 Minimum output power for CA

6.3A.1.1 Void

6.3A.1.2 Void

6.3A.1.3 Minimum output power for inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the minimum output power requirements in clause 6.3.1 apply.

For inter-band carrier aggregation with uplink assigned to two NR bands, the minimum output power is defined per carrier and the requirement is specified in clause 6.3.1.

6.3A.2 Transmit OFF power for CA

6.3A.2.1 Void

6.3A.2.2 Void

6.3A.2.3 Transmit OFF power for inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the transmit OFF power requirements in subclause 6.3.2 apply.

For inter-band carrier aggregation with uplink assigned to two NR bands, the transmit OFF power specified in clause 6.3.2 is applicable for each component carrier when the transmitter is OFF on all component carriers. The transmitter is considered to be OFF when the UE is not allowed to transmit on any of its ports.

6.3A.3 Transmit ON/OFF time mask for CA

6.3A.3.1 Void

6.3A.3.2 Void

6.3A.3.3 Transmit ON/OFF time mask for inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the transmit ON/OFF time mask requirements in subclause 6.3.3 apply.

For inter-band carrier aggregation with uplink assigned to two NR bands, the general output power ON/OFF time mask specified in clause 6.3.3.1 is applicable for each component carrier during the ON power period and the transient periods. The OFF period as specified in clause 6.3.3.1 shall only be applicable for each component carrier when all the component carriers are OFF.

6.3A.4 Power control for CA

6.3A.4.1 Void

6.3A.4.2 Void

6.3A.4.3 Power control for inter-band CA

No requirements unique to CA operation are defined.

6.3D Output power dynamics for UL MIMO

6.3D.1 Minimum output power for UL MIMO

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the minimum output power is defined as the sum of the mean power at each transmit connector in one sub-frame (1 ms). The minimum output power shall not exceed the values specified in Table 6.3.1-1.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.3.1 apply.

6.3D.2 Transmit OFF power for UL MIMO

The transmit OFF power is defined as the mean power at each transmit antenna connector in a duration of at least one sub-frame (1 ms) excluding any transient periods.

The transmit OFF power at each transmit antenna connector shall not exceed the values specified in Table 6.3.2-1.

6.3D.3 Transmit ON/OFF time mask for UL MIMO

For UE supporting UL MIMO, the ON/OFF time mask requirements in clause 6.3.3 apply at each transmit antenna connector. For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the general ON/OFF time mask requirements specified in clause 6.3.3.1 apply to each transmit antenna connector. The requirements shall be met with the UL MIMO configurations described in clause 6.2D.1.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.3.3 apply.

6.3D.4 Power control for UL MIMO

For UE supporting UL MIMO, the power control tolerance applies to the sum of output power at each transmit antenna connector.

The power control requirements specified in clause 6.3.4 apply to UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme. The requirements shall be met with UL MIMO configurations described in clause 6.2D.1.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.3.4 apply.

6.4 Transmit signal quality

6.4.1 Frequency error

The UE basic measurement interval of modulated carrier frequency is 1 UL slot. The mean value of basic measurements of UE modulated carrier frequency shall be accurate to within ± 0.1 PPM observed over a period of 1 ms of cumulated measurement intervals compared to the carrier frequency received from the NR Node B.

6.4.2 Transmit modulation quality

6.4.2.0 General

Transmit modulation quality defines the modulation quality for expected in-channel RF transmissions from the UE. The transmit modulation quality is specified in terms of:

- Error Vector Magnitude (EVM) for the allocated resource blocks (RBs)
- EVM equalizer spectrum flatness derived from the equalizer coefficients generated by the EVM measurement process

- Carrier leakage
- In-band emissions for the non-allocated RB

All the parameters defined in clause 6.4.2 are defined using the measurement methodology specified in Annex F. In case the parameter 3300 or 3301 is reported from UE via the parameter *txDirectCurrentLocation* in *UplinkTxDirectCurrentList* IE (as defined in TS 38.331 [7]), carrier leakage measurement requirement in clause 6.4.2.2 and 6.4.2.3 shall be waived, and the RF correction with regard to the carrier leakage and IQ image shall be omitted during the calculation of transmit modulation quality.

6.4.2.1 Error Vector Magnitude

The Error Vector Magnitude is a measure of the difference between the reference waveform and the measured waveform. This difference is called the error vector. Before calculating the EVM the measured waveform is corrected by the sample timing offset and RF frequency offset. Then the carrier leakage shall be removed from the measured waveform before calculating the EVM. The measured waveform is further equalised using the channel estimates subjected to the EVM equaliser spectrum flatness requirement specified in clause 6.4.2.4. For DFT-s-OFDM waveforms, the EVM result is defined after the front-end FFT and IDFT as the square root of the ratio of the mean error vector power to the mean reference power expressed as a %. For CP-OFDM waveforms, the EVM result is defined after the front-end FFT as the square root of the ratio of the mean error vector power to the mean reference power expressed as a %.

The basic EVM measurement interval in the time domain is one preamble sequence for the PRACH and one slot for PUCCH and PUSCH in the time domain. The EVM measurement interval is reduced by any symbol that contains an allowable power transient in the measurement interval, as defined in clause 6.3.3.

The RMS average of the basic EVM measurements over 10 subframes for the average EVM case, and over 60 subframes for the reference signal EVM case, for the different modulation schemes shall not exceed the values specified in Table 6.4.2.1-1 for the parameters defined in Table 6.4.2.1-2. For EVM evaluation purposes, all 13 PRACH preamble formats and all 5 PUCCH formats are considered to have the same EVM requirement as QPSK modulated.

Table 6.4.2.1-1: Requirements for Error Vector Magnitude

Parameter	Unit	Average EVM Level
Pi/2-BPSK	%	30
QPSK	%	17.5

16 QAM	%	12.5
64 QAM	%	8
256 QAM	%	3.5

Table 6.4.2.1-2: Parameters for Error Vector Magnitude

Parameter	Unit	Level
UE Output Power	dBm	Table 6.3.1-1
UE Output Power for 256 QAM	dBm	Table 6.3.1-1 + 10 dB
Operating conditions		Normal conditions

6.4.2.2 Carrier leakage

Carrier leakage is an additive sinusoid waveform whose frequency is the same as the modulated waveform carrier frequency. The measurement interval is one slot in the time domain.

In the case that uplink sharing, the carrier leakage may have 7.5 kHz shift with the carrier frequency.

The relative carrier leakage power is a power ratio of the additive sinusoid waveform and the modulated waveform. The relative carrier leakage power shall not exceed the values specified in Table 6.4.2.2-1.

Table 6.4.2.2-1: Requirements for Carrier Leakage

Parameter	Relative Limit (dBc)
Output power > 10 dBm	-28
0 dBm ≤ Output power ≤ 10 dBm	-25
-30 dBm ≤ Output power < 0 dBm	-20
-40 dBm ≤ Output power < -30 dBm	-10

6.4.2.3 In-band emissions

The in-band emission is defined as the average emission across 12 sub-carriers and as a function of the RB offset from the edge of the allocated UL transmission bandwidth. The in-band emission is measured as the ratio of the UE output power in a non-allocated RB to the UE output power in an allocated RB.

The basic in-band emissions measurement interval is defined over one slot in the time domain; however, the minimum requirement applies when the in-band emission measurement is averaged over 10 sub-frames. When the PUSCH or PUCCH transmission slot is shortened due to multiplexing with SRS, the in-band emissions measurement interval is reduced by one or more symbols, accordingly.

The average of the basic in-band emission measurement over 10 sub-frames shall not exceed the values specified in Table 6.4.2.3-1.

Table 6.4.2.3-1: Requirements for in-band emissions

Parameter description	Unit	Limit (NOTE 1)		Applicable Frequencies
General	dB			Any non-allocated (NOTE 2)
IQ Image	dB	-28	Image frequencies when output power > 10 dBm	Image frequencies (NOTES 2, 3)
		-25	Image frequencies when output power ≤ 10 dBm	
Carrier leakage	dBc	-28	Output power > 10 dBm	Carrier leakage frequency (NOTES 4, 5)
		-25	0 dBm ≤ Output power ≤ 10 dBm	
		-20	-30 dBm ≤ Output power < 0 dBm	
		-10	-40 dBm ≤ Output power < -30 dBm	

NOTE 1: An in-band emissions combined limit is evaluated in each non-allocated RB. For each such RB, the minimum requirement is calculated as the higher of - 30 dB and the power sum of all limit values (General, IQ Image or Carrier leakage) that apply. is defined in NOTE 10.

NOTE 2: The measurement bandwidth is 1 RB and the limit is expressed as a ratio of measured power in one non-allocated RB to the measured average power per allocated RB, where the averaging is done across all allocated RBs. For pi/2 BPSK with Spectrum Shaping, the limit is expressed as a ratio of measured power in one non-allocated RB to the measured power in the allocated RB with highest PSD.

NOTE 3: The applicable frequencies for this limit are those that are enclosed in the reflection of the allocated bandwidth, based on symmetry with respect to the carrier leakage frequency, but excluding any allocated RBs.

NOTE 4: The measurement bandwidth is 1 RB and the limit is expressed as a ratio of measured power in one non-allocated RB to the measured total power in all allocated RBs.

NOTE 5: The applicable frequencies for this limit depend on the parameter *txDirectCurrentLocation* in *UplinkTxDirectCurrent* IE, and are those that are enclosed either in the RB containing the carrier leakage frequency, or in the two RBs immediately adjacent to the carrier leakage frequency but excluding any

	allocated RB.
NOTE 6:	L_{CRB} is the Transmission Bandwidth (see clause 5.3).
NOTE 7:	N_{RB} is the Transmission Bandwidth Configuration (see clause 5.3).
NOTE 8:	EVM is the limit specified in Table 6.4.2.1-1 for the modulation format used in the allocated RBs.
NOTE 9:	is the starting frequency offset between the allocated RB and the measured non-allocated RB (e.g. $\Delta_{RB} = 1$ or $\Delta_{RB} = -1$ for the first adjacent RB outside of the allocated bandwidth).
NOTE 10:	is an average of the transmitted power over 10 sub-frames normalized by the number of allocated RBs, measured in dBm.
NOTE 11:	For almost contiguous allocations defined in sub-clause 6.2.2, $L_{CRB} = N_{RB_alloc} + N_{RB_gap}$ with no in-gap emission requirement.

6.4.2.4 EVM equalizer spectrum flatness

The zero-forcing equalizer correction applied in the EVM measurement process (as described in Annex F) must meet a spectral flatness requirement for the EVM measurement to be valid. The EVM equalizer spectrum flatness is defined in terms of the maximum peak-to-peak ripple of the equalizer coefficients (dB) across the allocated uplink block. The basic measurement interval is the same as for EVM.

The peak-to-peak variation of the EVM equalizer coefficients contained within the frequency range of the uplink allocation shall not exceed the maximum ripple specified in Table 6.4.2.4-1 for normal conditions. For uplink allocations contained within both Range 1 and Range 2, the coefficients evaluated within each of these frequency ranges shall meet the corresponding ripple requirement and the following additional requirement: the relative difference between the maximum coefficient in Range 1 and the minimum coefficient in Range 2 must not be larger than 5 dB, and the relative difference between the maximum coefficient in Range 2 and the minimum coefficient in Range 1 must not be larger than 7 dB (see Figure 6.4.2.4-1).

The EVM equalizer spectral flatness shall not exceed the values specified in Table 6.4.2.4-2 for extreme conditions. For uplink allocations contained within both Range 1 and Range 2, the coefficients evaluated within each of these frequency ranges shall meet the corresponding ripple requirement and the following additional requirement: the relative difference between the maximum coefficient in Range 1 and the minimum coefficient in Range 2 must not be larger than 6 dB, and the relative difference between the maximum coefficient in Range 2 and the minimum coefficient in Range 1 must not be larger than 10 dB (see Figure 6.4.2.4-1).

Table 6.4.2.4-1: Requirements for EVM equalizer spectrum flatness (normal conditions)

Frequency range	Maximum ripple (dB)
-----------------	---------------------

< 8(12) dB _{p-p}	$F_{UL_Low} \geq 3 \text{ MHz}$ and $F_{UL_High} - F_{UL_Meas} \geq 3 \text{ MHz}$ (Range 1)	4 (p-p)
	$F_{UL_Low} < 3 \text{ MHz}$ or $F_{UL_High} - F_{UL_Meas} < 3 \text{ MHz}$ (Range 2)	8 (p-p)
	F_{UL_Meas} refers to the sub-carrier frequency for which the equalizer coefficient is evaluated NOTE 2: F_{UL_Low} and F_{UL_High} refer to each NR frequency band specified in Table 5.2-1	

Table 6.4.2.4-2: Minimum requirements for EVM equalizer spectrum flatness (extreme conditions)

Frequency range	Maximum Ripple (dB)
$F_{UL_Meas} - F_{UL_Low} \geq 5 \text{ MHz}$ and $F_{UL_High} - F_{UL_Meas} \geq 5 \text{ MHz}$ (Range 1)	4 (p-p)
$F_{UL_Meas} - F_{UL_Low} < 5 \text{ MHz}$ or $F_{UL_High} - F_{UL_Meas} < 5 \text{ MHz}$ (Range 2)	12 (p-p)
NOTE 1: F_{UL_Meas} refers to the sub-carrier frequency for which the equalizer coefficient is evaluated	
NOTE 2: F_{UL_Low} and F_{UL_High} refer to each NR frequency band specified in Table 5.2-1	

Figure 6.4.2.4-1: The limits for EVM equalizer spectral flatness with the maximum allowed variation of the coefficients indicated (the ETC minimum requirement are within brackets).

6.4.2.4.1 Requirements for Pi/2 BPSK modulation

These requirements apply if the IE *powerBoostPi2BPSK* is set to 1 for power class 3 UE operating in TDD bands n40, n41, n77, n78 and n79 with Pi/2 BPSK modulation and UE indicates support for UE capability *powerBoosting-pi2BPSK* and 40 % or less slots in radio frame are used for UL transmission. Otherwise the requirements for EVM equalizer spectrum flatness defined in clause 6.4.2.4 apply.

The EVM equalizer coefficients across the allocated uplink block shall be modified to fit inside the mask specified in Table 6.4.2.4.1-1 for normal conditions, prior to the calculation of EVM. The limiting mask shall be placed to minimize the change in equalizer coefficients in a sum of squares sense.

Table 6.4.2.4.1-1: Mask for EVM equalizer coefficients for Pi/2 BPSK, normal conditions

Frequency range	Parameter	Maximum ripple (dB)
$ F_{UL_Meas} - F_{center} \leq X$ MHz (Range 1)	X1	6 (p-p)
$ F_{UL_Meas} - F_{center} > X$ MHz (Range 2)	X2	14 (p-p)
NOTE 1: F_{UL_Meas} refers to the sub-carrier frequency for which the equalizer coefficient is evaluated NOTE 2: F_{center} refers to the center frequency of an allocated block of PRBs NOTE 3: X, in MHz, is equal to 25% of the bandwidth of the PRB allocation NOTE 4: See Figure 6.4.2.4.1-1 for description of X1, X2		

Figure 6.4.2.4.1-1: The limits for EVM equalizer spectral flatness with the maximum allowed variation.

For Pi/2 BPSK modulation the UE shall be allowed to employ spectral shaping and the shaping filter shall be restricted so that the impulse response of the shaping filter itself shall meet

$$|\tilde{a}_t(t,0)| \geq |\tilde{a}_t(t,\tau)| \quad \forall \tau \neq 0$$

$$20\log_{10} |\tilde{a}_t(t,\tau)| < -15 \text{ dB} \quad 1 < \tau < M-1,$$

where $|\tilde{a}_t(t,\tau)| = \text{IDFT}\{|\tilde{a}_t(t,f)| e^{j\phi(t,f)}\}$, f is the frequency of the M allocated subcarriers, $\tilde{a}(t,f)$ and $\phi(t,f)$ are the amplitude and phase response.

0 dB reference is defined as $20\log_{10} |\tilde{a}_t(t,0)|$.

6.4A Transmit signal quality for CA

6.4A.1 Frequency error for CA

6.4A.1.1 Void

6.4A.1.2 Void

6.4A.1.3 Frequency error for inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the frequency error requirements in subclause 6.4.1 apply.

For inter-band carrier aggregation with uplink assigned to two NR bands, the frequency error requirements defined in clause 6.4.1 shall apply on each component carrier with all component carriers active.

6.4A.2 Transmit modulation quality for CA

6.4A.2.1 Void

6.4A.2.2 Void

6.4A.2.3 Transmit modulation quality for inter-band CA

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the transmit modulation quality requirements in subclause 6.4.2 apply.

For inter-band carrier aggregation with uplink assigned to two NR bands, the transmit modulation quality requirements shall apply on each component carrier as defined in clause 6.4.2 with all component carriers active: PCC with PRB allocation and SCC without PRB allocation and without CSI reporting and SRS configured.

6.4D Transmit signal quality for UL MIMO

6.4D.0 General

For a UE supporting UL MIMO, the requirements in this section are defined per layer or as the sum of emissions from both antennas to account for the UL MIMO scheme.

Alternatively, when applicable, requirements may be verified per antenna connector using an UL MIMO transmission with codebook of $\begin{bmatrix} 1 & 1 & 0 \\ \sqrt{2} & 0 & 1 \end{bmatrix}$ and a configuration defined in Table 6.4D.0-1.

Table 6.4D.0-1: UL MIMO configuration for per connector measurements

Transmission scheme	DCI format	Codebook Index
Codebook based uplink	DCI format 0_1	Codebook index 0

6.4D.1 Frequency error for UL MIMO

For UE(s) supporting UL MIMO, the basic measurement interval of modulated carrier frequency is 1 UL slot. The mean value of basic measurements of UE modulated carrier frequency per layer shall be accurate to within ± 0.1 PPM observed over a period of 1 ms of cumulated measurement intervals compared to the carrier frequency received from the NR Node B.

6.4D.2 Transmit modulation quality for UL MIMO

6.4D.2.0 General

For UE supporting UL MIMO, the transmit modulation quality requirements are specified based on measurements made at each transmit antenna connector.

If UE is configured for transmission on single-antenna port, the requirements specified for single carrier apply.

The transmit modulation quality is specified in terms of:

- Error Vector Magnitude (EVM) for the allocated resource blocks (RBs)
- EVM equalizer spectrum flatness derived from the equalizer coefficients generated by the EVM measurement process
- Carrier leakage (caused by IQ offset)
- In-band emissions for the non-allocated RB

In case the parameter 3300 or 3301 is reported from UE via the parameter *txDirectCurrentLocation* in *UplinkTxDirectCurrentList* IE (as defined in TS 38.331 [7]), carrier leakage measurement requirement in clause 6.4D.2.2 and 6.4D.2.3 shall be waived, and the RF correction with regard to the carrier leakage and IQ image shall be omitted during the calculation of transmit modulation quality.

6.4D.2.1 Error Vector Magnitude

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the Error Vector Magnitude requirements specified in clause 6.4.2.1 apply per layer. The requirements shall be met with the UL MIMO configurations specified in Table 6.2D.1-2

6.4D.2.2 Carrier leakage

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the Relative Carrier Leakage Power requirements specified in Table 6.4.2.2-1 which is defined in clause 6.4.2.2 apply per layer. The requirements shall be met with the UL MIMO configurations specified in Table 6.2D.1-2.

6.4D.2.3 In-band emissions

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the In-band Emission requirements specified in Table 6.4.2.3-1 which is defined in clause 6.4.2.3 apply at each transmit antenna connector. The requirements shall be met with the uplink MIMO configurations specified in Table 6.2D.1-2

6.4D.2.4 EVM equalizer spectrum flatness for UL MIMO

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the EVM Equalizer Spectrum Flatness requirements specified in clause 6.4.2.4 apply per layer. The requirements shall be met with the UL MIMO configurations specified in Table 6.2D.1-2

6.4D.3 Time alignment error for UL MIMO

For UE(s) with multiple transmit antenna connectors supporting UL MIMO, this requirement applies to frame timing differences between transmissions on multiple transmit antenna connectors in the closed-loop spatial multiplexing scheme.

The time alignment error (TAE) is defined as the average frame timing difference between any two transmissions on different transmit antenna connectors.

For UE(s) with multiple transmit antenna connectors, the Time Alignment Error (TAE) shall not exceed 130 ns.

6.4D.4 Requirements for coherent UL MIMO

For coherent UL MIMO, Table 6.4D.4-1 lists the maximum allowable difference between the measured relative power and phase errors between different antenna connectors in any slot within the specified time window from the last transmitted SRS on the same antenna connectors, for the purpose of uplink transmission (codebook or non-codebook usage) and those measured at that last SRS. The requirements in Table 6.4D.4-1 apply when the UL transmission power at each antenna connector is larger than 0 dBm for SRS transmission and for the duration of time window.

Table 6.4D.4-1: Maximum allowable difference of relative phase and power errors in a given slot compared to those measured at last SRS transmitted

Difference of relative phase error	Difference of relative power error	Time window
40 degrees	4 dB	20 msec

The above requirements when all the following conditions are met within the specified time window:

- UE is not signaled with a change in number of SRS ports in SRS-config, or a change in PUSCH-config
- UE remains in DRX active time (UE does not enter DRX OFF time)
- No measurement gap occurs
- No instance of SRS transmission with the usage antenna switching occurs
- Active BWP remains the same
- EN-DC and CA configuration is not changed for the UE (UE is not configured or de-configured with PCell or SCell(s))

6.5 Output RF spectrum emissions

6.5.1 Occupied bandwidth

Occupied bandwidth is defined as the bandwidth containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel. The occupied bandwidth for all transmission bandwidth configurations (Resources Blocks) shall be less than the channel bandwidth specified in Table 6.5.1-1.

Table 6.5.1-1: Occupied channel bandwidth

	NR channel bandwidth											
	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
Occupied channel bandwidth (MHz)	5	10	15	20	25	30	40	50	60	80	90	100

6.5.2 Out of band emission

6.5.2.1 General

The Out of band emissions are unwanted emissions immediately outside the assigned channel bandwidth resulting from the modulation process and non-linearity in the transmitter but excluding spurious emissions. This out of band emission limit is specified in terms of a spectrum emission mask and an adjacent channel leakage power ratio.

To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

6.5.2.2 Spectrum emission mask

The spectrum emission mask of the UE applies to frequencies (Δf_{OoB}) starting from the edge of the assigned NR channel bandwidth. For frequencies offset greater than Δf_{OoB} , the spurious requirements in clause 6.5.3 are applicable.

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

The power of any UE emission shall not exceed the levels specified in Table 6.5.2.2-1 for the specified channel bandwidth.

Table 6.5.2.2-1: General NR spectrum emission mask

Spectrum emission limit (dBm) / Channel bandwidth													
Δf_{OOB} (MHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz	Measurement bandwidth
$\pm 0\text{-}1$	-13	-13	-13	-13	-13	-13	-13						1 % channel bandwidth
$\pm 0\text{-}1$								-24	-24	-24	-24	-24	30 kHz
$\pm 1\text{-}5$	-10	-10	-10	-10	-10	-10	-10	-10	-10	-10	-10	-10	1 MHz
$\pm 5\text{-}6$	-13	-13	-13	-13	-13	-13	-13	-13	-13	-13	-13		
$\pm 6\text{-}10$	-25												
$\pm 10\text{-}15$		-25											
$\pm 15\text{-}20$			-25										

± 20-25				-25									
± 25-30					-25								
± 30-35						-25							
± 35-40													
± 40-45							-25						
± 45-50													
± 50-55								-25					
± 55-60													
± 60-65									-25				
± 65-80													
± 80-85										-25			
± 85-90													
± 90-95											-25		
± 95-100													
± 100-105												-25	

6.5.2.3 Additional spectrum emission mask

6.5.2.3.1 Requirements for network signalling value "NS_35"

Additional spectrum emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

When "NS_35" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.2.3.1-1.

Table 6.5.2.3.1-1: Additional requirements for "NS_35"

Δf_{OoB} (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)				Measurement bandwidth
	5	10	15	20	
0-0.1	-15	-18	-20	-21	30 kHz
0.1-6	-13	-13	-13	-13	100 kHz
6-10	-25 ¹	-13	-13	-13	100 kHz
10-15		-25 ¹	-13	-13	100 kHz
15-20			-25 ¹	-13	100 kHz
20-25				-25	1 MHz

NOTE 1: The measurement bandwidth shall be 1 MHz

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

6.5.2.3.2 Requirements for network signalling value "NS_04"

Additional spectrum emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message. The additional spectrum emission requirements in NS_04 are based on FCC rule 47 CFR 27.53(m)(4).

The n41 SEM transition point from -13 dBm/MHz to -25 dBm/MHz is based on the emission bandwidth. The emission bandwidth is defined as the width of the signal between two points, one below the carrier center frequency and one above the carrier center frequency, outside of which all emissions are attenuated at least 26 dB below the transmitter power based on FCC rule 47 CFR 27.53(m)(6). Since the 26-dB emission bandwidth is implementation dependent, the maximum transmission bandwidth of the channel bandwidth in MHz (for CP-OFDM, this bandwidth equals $N_{RB} * SCS * 12 / 1,000$, and for DFT-S-OFDM, this bandwidth equals the maximum applicable $L_{CRB} * SCS * 12 / 1,000$) contained within the 26 dB emission bandwidth is used for determining the applicable SEM indicated by NS_04 as specified in Table 6.5.2.3.2-1 and Table 6.5.2.3.2-2.

Table 6.5.2.3.2-1: transmission bandwidth determining the NS_04 SEM for CP-OFDM

SCS (kHz)	Channel bandwidth (MHz) / Maximum transmission bandwidth (MHz)								
	10	15	20	40	50	60	80	90	100
15	9.36	14.22	19.08	38.88	48.6	N/A	N/A	N/A	N/A
30	8.64	13.68	18.36	38.16	47.88	58.32	78.12	88.02	98.28
60	7.92	12.96	17.28	36.72	46.8	56.88	77.04	87.12	97.20

Table 6.5.2.3.2-2: transmission bandwidth determining the NS_04 SEM for DFT-S-OFDM

SCS (kHz)	Channel bandwidth (MHz) / Maximum transmission bandwidth (MHz)								
	10	15	20	40	50	60	80	90	100
15	9.00	13.50	18.00	38.88	48.60	N/A	N/A	N/A	N/A

30	8.64	12.96	18.00	36.00	46.08	58.32	77.76	87.48	97.20
60	7.20	12.96	17.28	36.00	46.08	54.00	72.00	86.40	97.20

When "NS_04" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.2.3.2-3.

Table 6.5.2.3.2-3: n41 SEM with "NS_04"

Δf_{OOB} MHz	Channel bandwidth (MHz) / Spectrum emission limit (dBm)									Measurement bandwidth
	10	15	20	40	50	60	80	90	100	
$\pm 0 - 1$	-10	-10	-10	-10						2 % channel bandwidth
						-10				1 MHz
$\pm 1 - 5$		-10								1 MHz
$\pm 5 - X$		-13								
$\pm X - (BW_{\text{Channel}} + 5 \text{ MHz})$		-25								
NOTE: X is defined in Table 6.5.2.3.2-1 for CP-OFDM and 6.5.2.3.2-2 for DFT-S-OFDM										

6.5.2.3.3 Requirements for network signalling values "NS_03" and "NS_03U"

Additional spectrum emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

When "NS_03" or "NS_03U" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.2.3.3-1.

Table 6.5.2.3.3-1: Additional requirements for "NS_03" and "NS_03U"

Δf_{OoB} (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)					Measurement bandwidth
	5	10	15	20	40	
0-1	-13	-13	-13	-13	-13	1 % of channel BW
1-6	-13	-13	-13	-13	-13	1 MHz
6-10	-25	-13	-13	-13	-13	1 MHz
10-15		-25	-13	-13	-13	1 MHz
15-20			-25	-13	-13	1 MHz

20-25				-25	-13	1 MHz
25-40					-13	1 MHz
40-45					-25	1 MHz

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

Table 6.5.2.3.3-2: Void

6.5.2.3.4 Requirements for network signalling value "NS_06"

Additional spectrum emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

When "NS_06" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.2.3.4-1.

Table 6.5.2.3.4-1: Additional requirements for "NS_06"

Δf_{OOB} (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)			Measurement bandwidth
	5	10	15	
$\pm 0 - 0.1$	-15	-18	-20	30 kHz
$\pm 0.1 - 1$	-13	-13	-13	100 kHz
$\pm 1 - 6$	-13	-13	-13	1 MHz
$\pm 6 - 10$	-25			
$\pm 10 - 15$				
$\pm 15 - 20$		-25	-25	

NOTE: As a general rule, the resolution bandwidth of the measuring equipment should be equal to the measurement bandwidth. However, to improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

6.5.2.3.5 Void

6.5.2.3.6 Void

6.5.2.3.7 Void

6.5.2.4 Adjacent channel leakage ratio

Adjacent Channel Leakage power Ratio (ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency.

To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

6.5.2.4.1 NR ACLR

NR Adjacent Channel Leakage power Ratio (NR_{ACLR}) is the ratio of the filtered mean power centred on the assigned NR channel frequency to the filtered mean power centred on an adjacent NR channel frequency at nominal channel spacing.

The assigned NR channel power and adjacent NR channel power are measured with rectangular filters with measurement bandwidths specified in Table 6.5.2.4.1-1.

If the measured adjacent channel power is greater than -50 dBm then the NR_{ACLR} shall be higher than the value specified in Table 6.5.2.4.1-2.

Table 6.5.2.4.1-1: NR ACLR measurement bandwidth

NR channel bandwidth / NR ACLR measurement bandwidth												
	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
NR ACLR measurement bandwidth (MHz)	4.515	9.375	14.235	19.095	23.955	28.815	38.895	48.615	58.35	78.15	88.23	98.31

Table 6.5.2.4.1-2: NR ACLR requirement

	Power class 2	Power class 3
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NR ACLR	31 dB	30 dB
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6.5.2.4.2 UTRA ACLR

UTRA adjacent channel leakage power ratio (UTRA_{ACLR}) is the ratio of the filtered mean power centred on the assigned NR channel frequency to the filtered mean power centred on an adjacent(s) UTRA channel frequency.

UTRA_{ACLR} is specified for the first adjacent UTRA channel (UTRA_{ACLR1}) which center frequency is ± 2.5 MHz from NR channel edge and for the 2nd adjacent UTRA channel (UTRA_{ACLR2}) which center frequency is ± 7.5 MHz from NR channel edge.

The UTRA channel power is measured with a RRC filter with roll-off factor $\frac{F_{3dB}}{B_{3dB}} = 0.22$ and bandwidth of 3.84 MHz. The assigned NR channel power is measured with a rectangular filter with measurement bandwidth specified in Table 6.5.2.4.1-1.

If the measured adjacent channel power is greater than - 50 dBm then the UTRA_{ACLR1} and UTRA_{ACLR2} shall be higher than the value specified in Table 6.5.2.4.2-1.

Table 6.5.2.4.2-1: UTRA ACLR requirement

	Power class 3
UTRA _{ACLR1}	33 dB
UTRA _{ACLR2}	36 dB

UTRA ACLR requirement is applicable when the network signalling value NS_03U, NS_05U, NS_43U or NS_100 is signalled by the network in the field *additionalSpectrumEmission*.

6.5.3 Spurious emissions

6.5.3.0 General

Spurious emissions are emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emissions, intermodulation products and frequency conversion products, but exclude out of band emissions unless otherwise stated. The spurious emission limits are specified in terms of general requirements in line with SM.329 [9] and NR operating band requirement to address UE co-existence.

To improve measurement accuracy, sensitivity and efficiency, the resolution bandwidth may be smaller than the measurement bandwidth. When the resolution bandwidth is smaller than the measurement bandwidth, the result should be integrated over the measurement bandwidth in order to obtain the equivalent noise bandwidth of the measurement bandwidth.

NOTE: For measurement conditions at the edge of each frequency range, the lowest frequency of the measurement position in each frequency range should be set at the lowest boundary of the frequency range plus MBW/2. The highest frequency of the measurement position in each frequency range should be set at the highest boundary of the frequency range minus MBW/2. MBW denotes the measurement bandwidth defined for the protected band.

6.5.3.1 General spurious emissions

Unless otherwise stated, the spurious emission limits apply for the frequency ranges that are more than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth. The spurious emission limits in Table 6.5.3.1-2 apply for all transmitter band configurations (N_{RB}) and channel bandwidths.

Table 6.5.3.1-1: Boundary between NR out of band and general spurious emission domain

Channel bandwidth	OOB boundary F_{OOB} (MHz)
BW_{Channel}	$BW_{\text{Channel}} + 5$

Table 6.5.3.1-2: Requirement for general spurious emissions limits

Frequency Range	Maximum Level	Measurement bandwidth	NOTE
$9 \text{ kHz} \leq f < 150 \text{ kHz}$	-36 dBm	1 kHz	
$150 \text{ kHz} \leq f < 30 \text{ MHz}$	-36 dBm	10 kHz	
$30 \text{ MHz} \leq f < 1000 \text{ MHz}$	-36 dBm	100 kHz	
$1 \text{ GHz} \leq f < 12.75 \text{ GHz}$	-30 dBm	1 MHz	4
	-25 dBm	1 MHz	3
$12.75 \text{ GHz} \leq f < 5^{\text{th}}$ harmonic of the upper frequency edge of the UL operating band in GHz	-30 dBm	1 MHz	1
$12.75 \text{ GHz} < f < 26 \text{ GHz}$	-30 dBm	1 MHz	2
NOTE 1: Applies for Band for which the upper frequency edge of the UL Band is greater than 2.55 GHz and less than or equal to 5.2 GHz			
NOTE 2: Applies for Band that the upper frequency edge of the UL Band more than 5.2 GHz			
NOTE 3: Applies for Band n41, CA configurations including Band n41, and EN-DC			

NOTE 4:	configurations that include n41 specified in clause 5.2B of TS 38.101-3 [3] when NS_04 is signalled.
	Does not apply for Band n41, CA configurations including Band n41, and EN-DC configurations that include n41 specified in subclause 5.2B of TS 38.101-3 [3] when NS_04 is signalled.

6.5.3.2 Spurious emissions for UE co-existence

This clause specifies the requirements for NR bands for coexistence with protected bands.

Table 6.5.3.2-1: Requirements for spurious emissions for UE co-existence

NR Band	Spurious emission for UE co-existence						
	Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	NOTE
n1, n84	E-UTRA Band 1, 5, 7, 8, 11, 18, 19, 20, 21, 22, 26, 27, 28, 31, 32, 38, 40, 41, 42, 43, 44, 45, 50, 51, 52, 65, 67, 68, 69, 72, 73, 74, 75, 76, NR Band n78, n79	F _{DL_low}	-	F _{DL_high}	-50	1	
	NR Band n77	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA Band 3, 34	F _{DL_low}	-	F _{DL_high}	-50	1	15
	Frequency range	1880	-	1895	-40	1	15, 27
	Frequency range	1895	-	1915	-15.5	5	15, 26, 27
	Frequency range	1915	-	1920	+1.6	5	15, 26, 27
n2	E-UTRA Band 4, 5, 12, 13, 14, 17, 24, 26, 27, 28, 29, 30, 41, 42, 50, 51, 53, 66, 70, 71, 74, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 2, 25	F _{DL_low}	-	F _{DL_high}	-50	1	15
	E-UTRA Band 43, 48	F _{DL_low}	-	F _{DL_high}	-50	1	2
n3, n80	E-UTRA Band 1, 5, 7, 8, 20, 26, 27, 28, 31, 32, 33, 34, 38, 39, 40, 41, 43, 44, 45, 50, 51, 65, 67, 68, 69, 72, 73, 74, 75, 76, NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 3	F _{DL_low}	-	F _{DL_high}	-50	1	15
	E-UTRA Band 11, 18, 19, 21	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 22, 42, 52, NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	2

	Frequency range	1884.5	-	1915.7	-41	0.3	8
n5	E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 12, 13, 14, 17, 18, 19, 24, 25, 26, 28, 29, 30, 31, 34, 38, 40, 42, 43, 45, 48, 50, 51, 65, 66, 70, 71, 73, 74, 85 NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 41, 52, 53 NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA Band 11, 21	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	8
n7	E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 12, 13, 14, 17, 20, 22, 26, 27, 28, 29, 30, 31, 32, 33, 34, 40, 42, 43, 50, 51, 52, 65, 66, 67, 68, 72, 74, 75, 76, 85, NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	
	NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	2
	Frequency range	2570	-	2575	+1.6	5	15, 21, 26
	Frequency range	2575	-	2595	-15.5	5	15, 21, 26
	Frequency range	2595	-	2620	-40	1	15, 21
n8, n81	E-UTRA Band 1, 20, 28, 31, 32, 33, 34, 38, 39, 40, 45, 50, 51, 65, 67, 68, 69, 72, 73, 74, 75, 76	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA band 3, 7, 22, 41, 42, 43, 52, NR Band n77, n78, n79	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA 8	F _{DL_low}	-	F _{DL_high}	-50	1	15
	E-UTRA Band 11, 21	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	8
n12	E-UTRA Band 2, 5, 13, 14, 17, 24, 25, 26, 27, 30, 41, 53, 70, 71, 74	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 4, 48, 50, 51, 66	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA Band 12, 85	F _{DL_low}	-	F _{DL_high}	-50	1	15
n20, n82	E-UTRA Band 1, 3, 7, 8, 22, 31, 32, 33, 34, 40, 43, 50, 51, 65, 67, 68, 72, 74, 75, 76	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 20	F _{DL_low}	-	F _{DL_high}	-50	1	15
	E-UTRA Band 38, 42, 52, 69, NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	2
	Frequency range	758	-	788	-50	1	

n25	E-UTRA Band 4, 5, 12, 13, 14, 17, 24, 26, 27, 28, 29, 30, 41, 42, 53, 66, 70, 71, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 2	F _{DL_low}	-	F _{DL_high}	-50	1	15
	E-UTRA Band 25	F _{DL_low}	-	F _{DL_high}	-50	1	15
	E-UTRA Band 43, 48	F _{DL_low}	-	F _{DL_high}	-50	1	2
n28, n83	E-UTRA Band 1, 4, 22, 32, 42, 43, 50, 51, 65, 66, 74, 75, 76, NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA Band 1	F _{DL_low}	-	F _{DL_high}	-50	1	19, 25
	E-UTRA Band 2, 3, 5, 7, 8, 18, 19, 20, 25, 26, 27, 31, 34, 38, 40, 41, 52, 72, 73 NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 11, 21	F _{DL_low}	-	F _{DL_high}	-50	1	19, 24
	Frequency range	470	-	694	-42	8	15, 35
	Frequency range	470	-	710	-26.2	6	34
	Frequency range	662	-	694	-26.2	6	15
	Frequency range	758	-	773	-32	1	15
	Frequency range	773	-	803	-50	1	
n34	Frequency range	1884.5	-	1915.7	-41	0.3	8, 19
	E-UTRA Band 1, 3, 7, 8, 11, 18, 19, 20, 21, 22, 26, 28, 31, 32, 33, 38, 39, 40, 41, 42, 43, 44, 45, 50, 51, 52, 65, 67, 69, 72, 74, 75, 76, NR Band n78, n79	F _{DL_low}	-	F _{DL_high}	-50	1	5
	NR Band n77	F _{DL_low}	-	F _{DL_high}	-50	1	2
	Frequency range	1884.5	-	1915.7	-41	0.3	8
n38	E-UTRA Band 1, 2, 3, 4, 5, 8, 12, 13, 14, 17, 20, 22, 27, 28, 29, 30, 31, 32, 33, 34, 40, 42, 43, 50, 51, 52, 65, 66, 67, 68, 72, 74, 75, 76, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	2620	-	2645	-15.5	5	15, 22, 26
	Frequency range	2645	-	2690	-40	1	15, 22
n39	E-UTRA Band 1, 8, 22, 26, 34, 40, 41, 42, 44, 45, 50, 51, 52, 74, NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	
	NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	2
	Frequency range	1805	-	1855	-40	1	33

	Frequency range	1855	-	1880	-15.5	5	15, 26, 33
n40	E-UTRA Band 1, 3, 5, 7, 8, 11, 18, 19, 20, 21, 22, 26, 27, 28, 31, 32, 33, 34, 38, 39, 42, 43, 44, 45, 50, 51, 52, 65, 67, 68, 69, 72, 74, 75, 76, NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	
	NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	2
	Frequency range	1884.5		1915.7	-41	0.3	8
n41	E-UTRA Band 1, 2, 3, 4, 5, 8, 12, 13, 14, 17, 24, 25, 26, 27, 28, 29, 30, 34, 39, 42, 44, 45, 48, 50, 51, 52, 65, 66, 70, 71, 73, 74, 85, NR Band n77, n78	F _{DL_low}	-	F _{DL_high}	-50	1	
	NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA Band 11, 18, 19, 21	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	1884.5		1915.7	-41	0.3	8
n50	E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 12, 13, 17, 20, 26, 28, 29, 31, 34, 38, 39, 40, 41, 42, 43, 48, 52, 65, 66, 67, 68, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
n51	E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 12, 13, 17, 20, 26, 28, 29, 31, 34, 38, 39, 40, 41, 42, 43, 48, 65, 66, 67, 68	F _{DL_low}	-	F _{DL_high}	-50	1	
n66, n86	E-UTRA Band 2, 4, 5, 7, 12, 13, 14, 17, 25, 26, 27, 28, 29, 30, 38, 41, 43, 50, 51, 53, 66, 70, 71, 74, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 42, 48	F _{DL_low}	-	F _{DL_high}	-50	1	2
n70	E-UTRA Band 2, 4, 5, 12, 13, 14, 17, 24, 25, 26, 29, 30, 41, 48, 66, 70, 71, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
n71	E-UTRA Band 4, 5, 12, 13, 14, 17, 24, 26, 30, 48, 53, 66, 85	F _{DL_low}	-	F _{DL_high}	-50	1	
	E-UTRA Band 2, 25, 41, 70	F _{DL_low}	-	F _{DL_high}	-50	1	2
	E-UTRA Band 29	F _{DL_low}	-	F _{DL_high}	-38	1	15
	E-UTRA Band 71	F _{DL_low}	-	F _{DL_high}	-50	1	15
n74	E-UTRA Band 1, 2, 3, 4, 5, 7, 8, 12, 13, 17, 18, 19, 20, 26, 28, 29, 31, 34, 38, 39, 40, 41, 42, 43, 48, 52, 65, 66, 67, 68, 85	F _{DL_low}	-	F _{DL_high}	-50	1	

	NR Band n77, n78						
	NR Band n79	F _{DL_low}	-	F _{DL_high}	-50	1	2
	Frequency range	1884.5	-	1915.7	-41	0.3	8
	Frequency range	1400	-	1427	-32	27	15, 41
	Frequency range	1475	-	1488	-50	1	42
	Frequency range	1475	-	1488	-28	1	15, 42
	Frequency range	1475	-	1488	-50	1	15, 45
	Frequency range	1475.9	-	1510.9	-35	1	15, 46
n77	E-UTRA Band 1, 3, 5, 7, 8, 11, 18, 19, 20, 21, 26, 28, 34, 39, 40, 41, 65, 74	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	8
n78	E-UTRA Band 1, 3, 5, 7, 8, 11, 18, 19, 20, 21, 26, 28, 32, 34, 39, 40, 41, 65, 74, 75, 76	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	8
n79	E-UTRA Band 1, 3, 5, 7, 8, 11, 18, 19, 21, 28, 34, 39, 40, 41, 42, 65, 74	F _{DL_low}	-	F _{DL_high}	-50	1	
	Frequency range	1884.5	-	1915.7	-41	0.3	8

NOTE 1: F_{DL_low} and F_{DL_high} refer to each frequency band specified in Table 5.2-1 in TS 38.101-1 or Table 5.5-1 in TS 36.101

NOTE 2: As exceptions, measurements with a level up to the applicable requirements defined in Table 6.5.3.1-2 are permitted for each assigned NR carrier used in the measurement due to 2nd, 3rd, 4th or 5th harmonic spurious emissions. Due to spreading of the harmonic emission the exception is also allowed for the first 1 MHz frequency range immediately outside the harmonic emission on both sides of the harmonic emission. This results in an overall exception interval centred at the harmonic emission of (2 MHz + N x L_{CRB} x RB_{size} kHz), where N is 2, 3, 4, 5 for the 2nd, 3rd, 4th or 5th harmonic respectively. The exception is allowed if the measurement bandwidth (MBW) totally or partially overlaps the overall exception interval.

NOTE 3: 15 kHz SCS is assumed when RB is mentioned in the note when channel bandwidth is less than or equal to 50MHz, lowest SCS is assumed when channel bandwidth is larger than 50 MHz. The transmission bandwidth in terms of RB position and range is not limited to 15 kHz SCS and shall scale with SCS accordingly.

NOTE 4: Void

NOTE 5: For non-synchronised TDD operation to meet these requirements some restriction will be needed for either the operating band or protected band

NOTE 6: N/A

NOTE 7: Void

NOTE 8: Applicable when co-existence with PHS system operating in 1884.5 - 1915.7 MHz.

NOTE 9: Void

NOTE 10: Void

NOTE 11: Void

NOTE 12: Void

NOTE 13: Void

NOTE 14: Void

NOTE 15: These requirements also apply for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

NOTE 16: Void

NOTE 17: Void

NOTE 18: Void

NOTE 19: Applicable when the assigned NR carrier is confined within 718 MHz and 748 MHz and when the channel bandwidth used is 5 or 10 MHz.

NOTE 20: Void

NOTE 21: This requirement is applicable for any channel bandwidths within the range 2500 - 2570 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2560.5 - 2562.5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2552 - 2560 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.

NOTE 22: This requirement is applicable for power class 3 UE for any channel bandwidths within the range 2570 - 2615 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 2605.5 - 2607.5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 2597 - 2605 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB. For carriers overlapping the frequency range 2615 - 2620 MHz the requirement applies with the maximum output power configured to +19 dBm in the IE P-Max.

NOTE 23: Void

NOTE 24: As exceptions, measurements with a level up to the applicable requirement of -38 dBm/MHz is permitted for each assigned NR carrier used in the measurement due to 2nd harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.3.1-1) for which the 2nd harmonic totally or partially overlaps the measurement bandwidth (MBW).

NOTE 25: As exceptions, measurements with a level up to the applicable requirement of -36 dBm/MHz is permitted for each assigned NR carrier used in the measurement due to 3rd harmonic spurious emissions. An exception is allowed if there is at least one individual RB within the transmission bandwidth (see Figure 5.3.1-1) for which the 3rd harmonic totally or partially overlaps the measurement bandwidth (MBW).

NOTE 26: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band.

NOTE 27: This requirement is applicable for any channel bandwidths within the range 1920 - 1980 MHz with the following restriction: for carriers of 15 MHz bandwidth when carrier centre frequency is within the range 1927.5 - 1929.5 MHz and for carriers of 20 MHz bandwidth when carrier centre frequency is within the range 1930 - 1938 MHz the requirement is applicable only for an uplink transmission bandwidth less than or equal to 54 RB.

NOTE 28: Void

NOTE 29: Void

NOTE 30: Void

NOTE 31: Void

NOTE 32: Void

NOTE 33:	This requirement is only applicable for carriers with bandwidth up to 20MHz and confined within 1885-1920 MHz (requirement for carriers with at least 1RB confined within 1880 - 1885 MHz is not specified). This requirement applies for an uplink transmission bandwidth less than or equal to 54 RB for carriers of 15 MHz bandwidth when carrier center frequency is within the range 1892.5 - 1894.5 MHz and for carriers of 20 MHz bandwidth when carrier center frequency is within the range 1895 - 1903 MHz.
NOTE 34:	This requirement is applicable for 5 and 10 MHz NR channel bandwidth allocated within 718-728 MHz. For carriers of 10 MHz bandwidth, this requirement applies for an uplink transmission bandwidth less than or equal to 30 RB with $RB_{start} > 1$ and $RB_{start} < 48$.
NOTE 35:	This requirement is applicable in the case of a 10 MHz NR carrier confined within 703 MHz and 733 MHz, otherwise the requirement of -25 dBm with a measurement bandwidth of 8 MHz applies.
NOTE 36:	Void
NOTE 37:	Void
NOTE 38:	Void
NOTE 39:	Void
NOTE 40:	Void
NOTE 41:	Applicable for cases and when the lower edge of the assigned NR UL channel bandwidth frequency is greater than or equal to 1427 MHz + the channel BW assigned for 5 and 10 MHz bandwidth, and when the lower edge of the assigned NR UL channel bandwidth frequency is greater than or equal to 1440 MHz for 15 and 20 MHz bandwidth. This requirement shall be verified with UE transmission power configured as high as possible but no higher than 15 dBm.
NOTE 42:	Applicable when upper edge of the assigned NR UL channel bandwidth frequency is more than 1460 MHz and less than or equal to 1470 MHz for 5 MHz bandwidth, and when the upper edge of the assigned NR UL channel bandwidth frequency is more than 1460 MHz and less than or equal to 1465 MHz for 10 MHz bandwidth.
NOTE 43:	Void
NOTE 44:	Void
NOTE 45:	Applicable when upper edge of the assigned NR UL channel bandwidth frequency is equal to or less than 1460 MHz.
NOTE 46:	Applicable for 5 MHz bandwidth and when the NR carrier is within 1447.9 – 1462.9 MHz.

NOTE: To simplify Table 6.5.3.2-1, E-UTRA band numbers are listed for bands which are specified only for E-UTRA operation or both E-UTRA and NR operation. NR band numbers are listed for bands which are specified only for NR operation.

6.5.3.3 Additional spurious emissions

These requirements are specified in terms of an additional spectrum emission requirement. Additional spurious emission requirements are signalled by the network to indicate that the UE shall meet an additional requirement for a specific deployment scenario as part of the cell handover/broadcast message.

6.5.3.3.1 Requirement for network signalling value "NS_04"

When "NS_04" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.1-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.1-1: Additional requirements for "NS_04"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	10, 15, 20, 40, 50, 60, 80, 90, 100	
$2495 \leq f < 2496$	-13	1 % of Channel BW
$2490.5 \leq f < 2495$	-13	1 MHz
$0.009 < f < 2490.5$	-25	1 MHz

6.5.3.3.2 Requirement for network signalling value "NS_17"

When "NS_17" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.2-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.2-1: Additional requirements for "NS_17"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth	NOTE
	5, 10		
$470 \leq f \leq 710$	-26.2	6 MHz	1
NOTE 1: Applicable when the assigned NR carrier is confined within 718 MHz and 748 MHz and when the channel bandwidth used is 5 or 10 MHz.			

6.5.3.3.3 Requirement for network singalling value "NS_18"

When "NS_18" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.3-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.3-1: Additional requirements for "NS_18"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth	
	5, 10, 15, 20		
692-698	-26.2	6 MHz	

6.5.3.3.4 Requirement for network signalling values "NS_05" and "NS_05U"

When "NS_05" or "NS_05U" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.4-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.4-1: Additional requirements for "NS_05" and "NS_05U"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth	
	5, 10, 15, 20		
1884.5 ≤ $f_{26.4}$ ≤ 1915.7	-41	300 kHz	

6.5.3.3.5 Requirement for network signalling values "NS_43" and "NS_43U"

When "NS_43" or "NS_43U" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.5-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.5-1: Additional requirement for "NS_43" and "NS_43U"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15	
$860 \leq f \leq 890$	-40	1 MHz
NOTE 1: Applicable for 5 MHz and 15 MHz channel BW confined between 900 MHz and 915 MHz and for 10 MHz channel BW confined between 905 MHz and 915 MHz		

6.5.3.3.6 Requirement for network signalling value "NS_37"

When "NS_37" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.6-1. This requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.6-1: Additional requirement for "NS_37"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15	
$1475.9 \leq f \leq 1510.9$	-35	1 MHz

6.5.3.3.7 Requirement for network signalling value "NS_38"

When "NS_38" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.7-1. This requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.7-1: Additional requirements for NR channels assigned within 1430-1452MHz for "NS_38"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20	
$1400 \leq f \leq 1427$	-32	27 MHz
NOTE 1: This requirement shall be verified with UE transmission power configured as high as possible but no higher than 15 dBm.		

6.5.3.3.8 Requirement for network signalling value "NS_39"

When "NS_39" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.8-1. This requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.8-1: Additional requirements for "NS_39"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20	
$1475 \leq f \leq 1488$	-28	1 MHz

6.5.3.3.9 Requirement for network signalling value "NS_40"

When "NS_40" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.9-1. This requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.9-1: Additional requirements for NR channels assigned within 1427-1432MHz for "NS_40"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5	
$1400 \leq f \leq 1427$	-32	27 MHz
NOTE 1: This requirement shall be verified with UE transmission power configured as high as possible but no higher than 15 dBm.		

6.5.3.3.10 Requirement for network signalling value "NS_41"

When "NS_41" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.10-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.10-1: Additional requirements for NR channels assigned within 1432-1517 MHz for "NS_41"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20, 40, 50, 60	
$1400 \leq f \leq 1427$	-32	27 MHz
NOTE 1: This requirement shall be verified with UE transmission power configured as high as possible but no higher than 15 dBm.		

6.5.3.3.11 Requirement for network signalling value "NS_42"

When "NS_42" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.11-1. This requirement also applies for the frequency ranges that are less than F_{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.11-1: Additional requirements for NR channels assigned within 1432-1517 MHz for "NS_42"

Frequency range (MHz)	Channel bandwidth (MHz) / Spectrum emission limit (dBm)	Measurement bandwidth
	5, 10, 15, 20, 40, 50, 60	
$1518 \leq f \leq 1520$	-0.8	1 MHz
$1520 < f \leq 1559$	-30	1 MHz

6.5.3.3.12 Void

6.5.3.3.13 Void

6.5.3.3.14 Void

6.5.3.3.15 Void

6.5.3.3.16 Requirement for network signalling value "NS_50"

When "NS_50" is indicated in the cell, the power of any UE emission shall not exceed the levels specified in Table 6.5.3.3.16-1. This requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.

Table 6.5.3.3.16-1: Additional requirements for "NS_50"

Protected band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	NOTE
Frequency range	1805	-	1855	-40	1	1
Frequency range	1855	-	1880	-15.5	5	1, 2, 3
NOTE 1: This requirement is applicable for carriers with aggregated channel bandwidths confined in 1885-1920 MHz for 25MHz and 30MHz channel BWs and confined in 1880-1920 MHz for 40MHz channel BW.						
NOTE 2: The requirement also applies for the frequency ranges that are less than F_{OoB} (MHz) in Table 6.6.3.1-1 and Table 6.6.3.1A-1 from the edge of the channel bandwidth.						
NOTE 3: For these adjacent bands, the emission limit could imply risk of harmful interference to UE(s) operating in the protected operating band.						

6.5.4 Transmit intermodulation

The transmit intermodulation performance is a measure of the capability of the transmitter to inhibit the generation of signals in its non linear elements caused by presence of the wanted signal and an interfering signal reaching the transmitter via the antenna.

UE transmit intermodulation is defined by the ratio of the mean power of the wanted signal to the mean power of the intermodulation product when an interfering CW signal is added at a level below the wanted signal at each transmitter antenna port with the other antenna port(s) if any terminated. Both the wanted signal power and the intermodulation product power are measured through NR rectangular filter with measurement bandwidth shown in Table 6.5.4-1.

The requirement of transmit intermodulation is specified in Table 6.5.4-1.

Table 6.5.4-1: Transmit Intermodulation

Wanted signal channel bandwidth	BW_{Channel}	
Interference signal frequency offset from channel center	BW_{Channel}	$2*BW_{\text{Channel}}$

Interference CW signal level	-40 dBc	
Intermodulation product	< -29 dBc	< -35 dBc
Measurement bandwidth	The maximum transmission bandwidth configuration among the different SCS's for the channel BW as defined in Table 6.5.2.4.1-1	
Measurement offset from channel center	BW_{Channel} and $2*BW_{\text{Channel}}$	$2*BW_{\text{Channel}}$ and $4*BW_{\text{Channel}}$

6.5A Output RF spectrum emissions for CA

6.5A.0 General

For inter-band carrier aggregation with one uplink carrier assigned to one NR band, the output RF spectrum emissions requirements in clause 6.5 apply.

6.5A.1 Occupied bandwidth for CA

6.5A.1.1 Void

6.5A.1.2 Void

6.5A.1.3 Occupied bandwidth for Inter-band CA

For inter-band carrier aggregation with uplink assigned to two NR bands, the occupied bandwidth is defined per component carrier. Occupied bandwidth is the bandwidth containing 99 % of the total integrated mean power of the transmitted spectrum on assigned channel bandwidth on the component carrier. The occupied bandwidth shall be less than the channel bandwidth specified in Table 6.5.1-1.

6.5A.2 Out of band emission for CA

6.5A.2.1 General

This clause contains requirements for out of band emissions for UE configured of carrier aggregation.

6.5A.2.2 Spectrum emission mask

- 6.5A.2.2.1 Void
- 6.5A.2.2.2 Void
- 6.5A.2.2.3 Spectrum emission mask for Inter-band CA

For inter-band carrier aggregation with uplink assigned to two NR bands, the spectrum emission mask of the UE is defined per component carrier while both component carriers are active and the requirements are specified in clauses 6.5.2.1 and 6.5.2.2. If for some frequency spectrum emission masks of component carriers overlap then spectrum emission mask allowing higher power spectral density applies for that frequency. If for some frequency a component carrier spectrum emission mask overlaps with the channel bandwidth of another component carrier, then the emission mask does not apply for that frequency.

- 6.5A.2.3 Additional spectrum emission mask
 - 6.5A.2.3.1 Void
 - 6.5A.2.3.2 Void
 - 6.5A.2.3.3 Additional spectrum emission mask for Inter-band CA
- 6.5A.2.4 Adjacent channel leakage ratio
 - 6.5A.2.4.1 NR ACLR
 - 6.5A.2.4.1.1 Void
 - 6.5A.2.4.1.2 Void
 - 6.5A.2.4.1.3 NR ACLR for Inter-band CA

For inter-band carrier aggregation with uplink assigned to two NR bands, the NR Adjacent Channel Leakage power Ratio (NRACLR) is defined per component carrier while both component carriers are active and the requirement is specified in clause 6.5.2.4.1.

- 6.5A.2.4.2 UTRA ACLR
 - 6.5A.2.4.2.1 Void
 - 6.5A.2.4.2.2 Void

6.5A.2.4.2.3 UTRA ACLR for Inter-band CA

For inter-band carrier aggregation with uplink assigned to two NR bands, the UTRA Adjacent Channel Leakage power Ratio (UTRAACLR) is defined per component carrier while both component carrier are active and the requirement is specified in clause 6.5.2.4.2.

6.5A.3 Spurious emission for CA

6.5A.3.1 General spurious emissions

For inter-band carrier aggregation with uplink assigned to two NR bands, the spurious emission requirement Table 6.5.3.1-2 apply for the frequency ranges that are more than F_{OoB} as defined in Table 6.5.3.1-1 away from edges of the assigned channel bandwidth on a component carrier. If for some frequency a spurious emission requirement of individual component carrier overlaps with the spectrum emission mask or channel bandwidth of another component carrier then it does not apply.

NOTE: For inter-band carrier aggregation with uplink assigned to two NR bands the requirements in Table 6.5.3.1-2 could be verified by measuring spurious emissions at the specific frequencies where second and third order intermodulation products generated by the two transmitted carriers can occur; in that case, the requirements for remaining applicable frequencies in Table 6.5.3.1-2 would be considered to be verified by the measurements verifying the one uplink inter-band CA spurious emission requirement.

6.5A.3.2 Spurious emissions for UE co-existence

6.5A.3.2.1 Void

6.5A.3.2.2 Void

6.5A.3.2.3 Spurious emissions for UE co-existence for Inter-band CA

This clause specifies the additional requirements for inter-band uplink carrier aggregation configurations with the single CC uplink assigned to two NR bands for coexistence with protected bands for the specified uplink carrier aggregation configurations in Table 6.5A.3.2.3-1. The intersection of the requirements for the individual bands specified in clause 6.5.3.2 shall also apply for the specified uplink carrier aggregation configurations. Intersection of a requirement means that both UL constituent bands have the same protected band requirement specified and if one or both protected bands have note(s) associated those note(s) also apply.

For inter-band carrier aggregation with the uplink assigned to two NR bands, the requirements in Table 6.5A.3.2.3-1 apply on each component carrier with all component carriers are active.

NOTE: For inter-band carrier aggregation with uplink assigned to two NR bands the requirements in Table 6.5A.3.2.3-1 could be verified by measuring spurious emissions at the specific frequencies where second and third order intermodulation products generated by the two transmitted carriers can occur; in that case, the requirements for remaining applicable frequencies in Table 6.5A.3.2.3-1 would be considered to be verified by the measurements verifying the one uplink inter-band CA UE to UE co-existence requirements.

Table 6.5A.3.2.3-1: Requirements for uplink inter-band carrier aggregation (two bands)

NR CA combination	Spurious emission						
	Protected Band	Frequency range (MHz)			Maximum Level (dBm)	MBW (MHz)	NOTE
CA_n3-n78	Frequency range	1884.5	-	1915.7	-41	0.3	3
CA_n8-n78	Frequency range	1884.5	-	1915.7	-41	0.3	3
NOTE 1: Void							
NOTE 2: -Void							
NOTE 3: Applicable when co-existence with PHS system operating in 1884.5 -1915.7 MHz							
NOTE 4: These requirements also apply for the frequency ranges that are less than F _{OOB} (MHz) in Table 6.5.3.1-1 from the edge of the channel bandwidth.							
NOTE 5: Void							

NOTE: To simplify Table 6.5A.3.2.3-1, E-UTRA band numbers are listed for bands which are specified only for E-UTRA operation or both E-UTRA and NR operation. NR band numbers are listed for bands which are specified only for NR operation.

6.5A.4 Transmit intermodulation for CA

6.5A.4.1 Void

6.5A.4.2 Void

6.5A.4.3 Transmit intermodulation for Inter-band CA

For inter-band carrier aggregation with uplink assigned to two NR bands, the transmit intermodulation requirement is specified in Table 6.5.4-1 which shall apply on each component carrier with both component carriers active.

6.5D Output RF spectrum emissions for UL MIMO

6.5D.1 Occupied bandwidth for UL MIMO

For UE supporting UL MIMO, the requirements for occupied bandwidth apply to the sum of the powers from both UE transmit antenna connectors. The occupied bandwidth is defined as the bandwidth containing 99 % of the total integrated mean power of the transmitted spectrum on the assigned channel.

For UE with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the occupied bandwidth at each transmitter antenna shall be less than the channel bandwidth specified in table 6.5.1-1. The requirements shall be met with UL MIMO configurations described in clause 6.2D.1.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.5.1 apply.

6.5D.2 Out of band emission for UL MIMO

For UE supporting UL MIMO, the requirements for Out of band emissions resulting from the modulation process and non-linearity in the transmitters is defined as the sum of the emissions from both UE transmit antenna connectors.

For UEs with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the requirements in subclause 6.5.2 apply. The requirements shall be met with UL MIMO configurations described in clause 6.2D.1.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.5.2 apply.

6.5D.3 Spurious emission for UL MIMO

For UE supporting UL MIMO, the requirements for Spurious emissions which are caused by unwanted transmitter effects such as harmonics emission, parasitic emissions, intermodulation products and frequency conversion products are defined as the sum of the emissions from both UE transmit antenna connectors.

For UEs with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the requirements specified in subclause 6.5.3 apply. The requirements shall be met with the UL MIMO configurations described in clause 6.2D.1.

If UE is configured for transmission on single-antenna port, the requirements in clause 6.5.3 apply.

6.5D.4 Transmit intermodulation for UL MIMO

For UE supporting UL MIMO, the transmit intermodulation requirements are specified at each transmit antenna connector and the wanted signal is defined as the sum of output powers from both transmit antenna connector.

For UEs with two transmit antenna connectors in closed-loop spatial multiplexing scheme, the requirements specified in clause 6.5.4 apply to each transmit antenna connector. The requirements shall be met with the UL MIMO configurations described in clause 6.2D.1.

If UE is scheduled for single antenna-port PUSCH transmission by DCI format 0_0 or by DCI format 0_1 for single antenna port codebook based transmission, the requirements in clause 6.5.4 apply.

7 Receiver characteristics

7.1 General

Unless otherwise stated the receiver characteristics are specified at the antenna connector(s) of the UE. For UE(s) with an integral antenna only, a reference antenna(s) with a gain of 0 dBi is assumed for each antenna port(s). UE with an integral antenna(s) may be taken into account by converting these power levels into field strength requirements, assuming a 0 dBi gain antenna. For UEs with more than one receiver antenna connector, identical interfering signals shall be applied to each receiver antenna port if more than one of these is used (diversity).

The levels of the test signal applied to each of the antenna connectors shall be as defined in the respective clauses below.

With the exception of clause 7.3, the requirements shall be verified with the network signalling value NS_01 configured (Table 6.2.3-1).

All the parameters in clause 7 are defined using the UL reference measurement channels specified in Annexes A.2.2, the DL reference measurement channels specified in Annex A.3.2 and using the set-up specified in Annex C.3.1.

7.2 Diversity characteristics

The UE is required to be equipped with a minimum of two Rx antenna ports in all operating bands except for the bands n7, n38, n41, n77, n78, n79 where the UE is required to be equipped with a minimum of four Rx antenna ports. This requirement applies when the band is used as a standalone band or as part of a band combination.

For the single carrier REFSENS requirements in Clause 7, the UE shall be verified with two Rx antenna ports in all supported frequency bands, additional requirements for four Rx ports shall be verified in operating bands where the UE is equipped with four Rx antenna ports.

For Rx requirements other than single carrier REFSENS in Clause 7, the UE shall be verified with four Rx antenna ports and skip two Rx antenna ports requirements in operating bands where the UE is equipped with four Rx antenna ports, otherwise, the UE shall be verified with two Rx antenna ports.

The above rules apply for all clauses with the exception of clause 7.9.

7.3 Reference sensitivity

7.3.1 General

The reference sensitivity power level REFSENS is the minimum mean power applied to each one of the UE antenna ports for all UE categories, at which the throughput shall meet or exceed the requirements for the specified reference measurement channel.

In later clauses of Clause 7 where the value of REFSENS is used as a reference to set the corresponding requirement:

- when the UE is verified with 2 Rx antenna ports, it shall be verified against those requirements by applying the REFSENS value in Table 7.3.2-1 with 2 Rx antenna ports tested;
- when the UE is verified with 4 Rx antenna ports, it shall be verified against those requirements by applying the resulting REFSENS value derived from the requirement in Table 7.3.2-2 with 4 Rx antenna ports tested.

7.3.2 Reference sensitivity power level

The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.3.2-1 and Table 7.3.2-2.

Table 7.3.2-1: Two antenna port reference sensitivity QPSK PREFSENS

Operating band / SCS / Channel bandwidth / Duplex-mode														
Operating Band	SCS kHz	5 MHz (dBm)	10 MHz (dBm)	15 MHz (dBm)	20 MHz (dBm)	25 MHz (dBm)	30 MHz (dBm)	40 MHz (dBm)	50 MHz (dBm)	60 MHz (dBm)	80 MHz (dBm)	90 MHz (dBm)	100 MHz (dBm)	Duplex Mode
n1	15	-100.0	-96.8	-95.0	-93.8									FDD
	30		-97.1	-95.1	-94.0									
	60		-97.5	-95.4	-94.2									
n2	15	-98.0	-94.8	-93.0	-91.8									FDD
	30		-95.1	-93.1	-92.0									
	60		-95.5	-93.4	-92.2									
n3	15	-97.0	-93.8	-92.0	-90.8	-89.7	-88.9							FDD
	30		-94.1	-92.1	-91.0	-89.8	-89.0							
	60		-94.5	-92.4	-91.2	-90.0	-89.1							
n5	15	-98.0	-94.8	-93.0	-86.8									FDD
	30		-95.1	-93.1	-88.6									
	60													
n7 ¹	15	-98.0	-94.8	-93.0	-91.8									FDD
	30		-95.1	-93.1	-92.0									
	60		-95.5	-93.4	-92.2									
n8	15	-97.0	-93.8	-91.4	-85.8									FDD
	30		-94.1	-91.7	-87.2									
	60													
n12	15	-97.0	-93.8	-84.0										FDD
	30		-94.1	-84.1										
	60													
n20	15	-97.0	-93.8	-91.0	-89.8									FDD
	30		-94.1	-91.1	-90.0									
	60													
n25	15	-96.5	-93.3	-91.5	-90.3									FDD
	30		-93.6	-91.6	-90.5									
	60		-94.0	-91.9	-90.7									

n28	15	-98.5	-95.5	-93.5	-90.8									FDD
	30		-95.6	-93.6	-91.0									
	60													
n34	15	-100.0	-96.8	-95.0										TDD
	30		-97.1	-95.1										
	60		-97.5	-95.4										
n38 ¹	15	-100.0	-96.8	-95.0	-93.8									TDD
	30		-97.1	-95.1	-94.0									
	60		-97.5	-95.4	-94.2									
n39	15	-100.0	-96.8	-95.0	-93.8	-92.7	-91.9	-90.6						TDD
	30		-97.1	-95.1	-94.0	-92.8	-92.0	-90.7						
	60		-97.5	-95.4	-94.2	-93.0	-92.1	-90.9						
n40	15	-100.0	-96.8	-95.0	-93.8	-92.7	-91.9	-90.6	-89.6					TDD
	30		-97.1	-95.1	-94.0	-92.8	-92.0	-90.7	-89.7	-88.9	-87.6			
	60		-97.5	-95.4	-94.2	-93.0	-92.1	-90.9	-89.8	-89.1	-87.6			
n41 ¹	15		-94.8	-93.0	-91.8			-88.6	-87.6					TDD
	30		-95.1	-93.1	-92.0			-88.7	-87.7	-86.9	-85.6	-85.1	-84.7	
	60		-95.5	-93.4	-92.2			-88.9	-87.8	-87.1	-85.6	-85.1	-84.7	
n50	15	-100.0	-96.8	-95.0	-93.8			-90.6	-89.6					TDD
	30		-97.1	-95.1	-94.0			-90.7	-89.7	-88.9	-87.6			
	60		-97.5	-95.4	-94.2			-90.9	-89.8	-89.1	-87.6			
n51	15	-100.0												TDD
	30													
	60													
n66	15	-99.5	-96.3	-94.5	-93.3			-90.1						FDD
	30		-96.6	-94.6	-93.5			-90.2						
	60		-97.0	-94.9	-93.7			-90.4						
n70	15	-100.0	-96.8	-95.0	-93.8	-92.7								FDD
	30		-97.1	-95.1	-94.0	-92.8								
	60		-97.5	-95.4	-94.2	-93.0								
n71	15	-97.2	-94.0	-91.6	-86.0									FDD
	30		-94.3	-91.9	-87.4									

	60													
n74	15	-99.5 ³	-96.3 ³	-94.5 ³	-89.3 ³									FDD
	30		-96.6 ³	-94.6 ³	-89.5 ³									
	60		-97.0 ³	-94.9 ³	-89.6 ³									
n75 ⁷	15	-100	-96.8	-95.0	-93.8									SDL
	30		-97.1	-95.1	-94.0									
	60		-97.5	-95.4	-94.2									
n76 ⁷	15	-100												SDL
	30													
	60													
n77 ^{1,4}	15		-95.3	-93.5	-92.2			-89.1	-88.1					TDD
	30		-95.6	-93.6	-92.4			-89.2	-88.2	-87.4	-86.1	-85.6	-85.1	
	60		-96.0	-93.9	-92.6			-89.4	-88.3	-87.5	-86.2	-85.7	-85.2	
n78 ¹	15		-95.8	-94.0	-92.7			-89.6	-88.6					TDD
	30		-96.1	-94.1	-92.9			-89.7	-88.7	-87.9	-86.6	-86.1	-85.6	
	60		-96.5	-94.4	-93.1			-89.9	-88.8	-88.0	-86.7	-86.2	-85.7	
n79 ¹	15							-89.6	-88.6					TDD
	30							-89.7	-88.7	-87.9	-86.6		-85.6	
	60							-89.9	-88.8	-88.0	-86.7		-85.7	

NOTE 1: Four Rx antenna ports shall be the baseline for this operating band except for two Rx vehicular UE.

NOTE 2: The transmitter shall be set to P_{UMAX} as defined in clause 6.2.4

NOTE 3: The requirement is modified by -0.5 dB when the assigned NR channel bandwidth is confined within 1475.9 - 1510.9 MHz.

NOTE 4: The requirement is modified by -0.5 dB when the assigned UE channel bandwidth is confined within 3300 - 3800 MHz.

NOTE 6: Void.

NOTE 7: For SDL bands, the reference sensitivity requirements shall be verified by inter-band CA combinations with SDL band, which are supported by UE.

n8	15	25	25 ¹	20 ¹	20 ¹									FDD
	30		12 ¹	10 ¹	10 ¹									
	60													
n12	15	20 ¹	20 ¹	20 ¹										FDD
	30		10 ¹	10 ¹										
	60													
n20	15	25	20 ¹	20 ²	20 ²									FDD
	30		10 ¹	10 ²	10 ²									
	60													
n25	15	25	50 ¹	50 ¹	50 ¹									FDD
	30		24	24 ¹	24 ¹									
	60		10 ¹	10 ¹	10 ¹									
n28	15	25	25 ¹	25 ¹	25 ¹									FDD
	30		10 ¹	10 ¹	10 ¹									
	60													
n34	15	25	50	75										TDD
	30		24	36										
	60		10	18										
n38	15	25	50	75	100									TDD
	30		24	36	50									
	60		10	18	24									
n39	15	25	50	75	100	128	160	216						TDD
	30		24	36	50	64	75	100						
	60		10	18	24	30	36	50						
n40	15	25	50	75	100	128	160	216	270					TDD
	30		24	36	50	64	75	100	128	162	216			
	60		10	18	24	30	36	50	64	75	100			
n41	15		50	75	100			216	270					TDD
	30		24	36	50			100	128	162	216	243	270	
	60		10	18	24			50	64	75	100	120	135	
n50	15	25	50	75	100			216	270					TDD
	30		24	36	50			100	128	162	NOTE			

											3			
	60		10	18	24			50	64	75	NOTE 3			
n51	15	25												TDD
	30													
	60													
n66	15	25	50 ¹	75 ¹	100 ¹			216						FDD
	30		24	36 ¹	50 ¹			100 ¹						
	60		10 ¹	18	24			50 ¹						
n70	15	25	50 ¹	75 ¹	NOTE 3	NOTE 3								FDD
	30		24	36 ¹	NOTE 3	NOTE 3								
	60		10 ¹	18	NOTE 3	NOTE 3								
n71	15	25	25 ¹	20 ¹	20 ¹									FDD
	30		12 ¹	10 ¹	10 ¹									
	60													
n74	15	25	25 ¹	25 ¹	25 ¹									FDD
	30		10 ¹	10 ¹	10 ¹									
	60		5 ¹	5 ¹	5 ¹									
n77	15		50	75	100			216	270					TDD
	30		24	36	50			100	128	162	216	243	270	
	60		10	18	24			50	64	75	100	120	135	
n78	15		50	75	100			216	270					TDD
	30		24	36	50			100	128	162	216	243	270	
	60		10	18	24			50	64	75	100	120	135	
n79	15							216	270					TDD
	30							100	128	162	216		270	
	60							50	64	75	100		135	

NOTE 1: UL resource blocks shall be located as close as possible to the downlink operating band but confined within the transmission bandwidth configuration for the channel bandwidth (Table 5.3.2-1).

NOTE 2: For band n20; for 15 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 11 and in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 16; for 30 kHz SCS, in the case of 15 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 6 and

NOTE 3: in the case of 20 MHz channel bandwidth, the UL resource blocks shall be located at RB_{start} 8;
For DL channel bandwidths that do not have symmetric UL channel bandwidth, highest valid UL configuration with lowest duplex distance shall be used.

Unless given by Table 7.3.2-4, the minimum requirements specified in Tables 7.3.2-1 and 7.3.2-2 shall be verified with the network signalling value NS_01 (Table 6.2.3-1) configured.

Table 7.3.2-4: Network signaling value for reference sensitivity

Operating band	Network Signalling value
n2	NS_03
n12	NS_06
n25	NS_03
n66	NS_03
n70	NS_03
n71	NS_35

7.3.3 $\Delta R_{IB,c}$

For a UE supporting CA, SUL or DC band combination, the minimum requirement for reference sensitivity in Table 7.3.2-1 shall be increased by the amount given by $\Delta R_{IB,c}$ defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3] for the applicable operating bands.

In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then

- When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the average value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3], truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta R_{IB,c}$ among the different supported band combinations involving such band shall be applied
- When the operating band frequency range is > 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the maximum value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3] for the applicable operating bands.

7.3A Reference sensitivity for CA

7.3A.1 General

The reference sensitivity power level REFSENS is the minimum mean power applied to each one of the UE antenna ports for all UE categories, at which the throughput shall meet or exceed the requirements for the specified reference measurement channel.

For operations with 4 Rx antenna ports, the MSD in the applicable bands shall be increased by the absolute value of $\Delta R_{IB,4R}$ in Table 7.3.2-2 when MSD > 0.

For reference sensitivity exception test points where the specified carrier frequency does not correspond to a valid NR-ARFCN, the closest NR-ARFCN as specified in clause 5.4.2 applies.

7.3A.2 Reference sensitivity power level for CA

7.3A.2.1 Reference sensitivity power level for Intra-band contiguous CA

For intra-band contiguous carrier aggregation, the throughput of each component carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2.2, A.2.3.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.3.2-1, Table 7.3.2-2, and Table 7.3.2-3.

7.3A.2.2 Void

7.3A.2.3 Reference sensitivity power level for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band the throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1 with parameters specified in Table 7.3.2-1, Table 7.3.2-2 and Table 7.3.2-3 modified in accordance with clause 7.3A.3.2. The reference sensitivity is defined to be met with all downlink component carriers active and one of the uplink carriers active. Exceptions to reference sensitivity are allowed in accordance with clause 7.3A.4.

7.3A.2.4 Void

7.3A.3 $\Delta R_{IB,c}$ for CA

7.3A.3.1 General

For a UE supporting a CA configuration, the $\Delta R_{IB,c}$ applies for both SC and CA operation.

7.3A.3.2 $\Delta R_{IB,c}$ for Inter-band CA

For the UE which supports inter-band carrier aggregation, the minimum requirement for reference sensitivity in clause 7.3A.2 shall be increased by the amount given by $\Delta R_{IB,c}$ defined in clause 7.3A.3.2 for the applicable operating bands. Unless otherwise stated, $\Delta R_{IB,c}$ is set to zero.

In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then

- When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the average value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3], truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta R_{IB,c}$ among the different supported band combinations involving such band shall be applied
- When the operating band frequency range is > 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the maximum value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3] for the applicable operating bands.

7.3A.3.2.1 $\Delta R_{IB,c}$ for two bands

Table 7.3A.3.2.1-1: $\Delta R_{IB,c}$ due to CA (two bands)

Inter-band CA combination	NR Band	$\Delta R_{IB,c}$ (dB)
CA_n3-n77	n3	0.2
	n77	0.5
CA_n3-n78	n3	0.2
	n78	0.5
CA_n3-n79	n79	0.5
CA_n8-n78	n8	0.2
	n78	0.5
CA_n8-n79	n79	0.5

CA n28-n75	n28	0.2
CA_n28-n78	n28	0.2
	n78	0.5
CA_n41-n78 ¹	n78	0.5
CA_n75-n78	n78	0.5
CA_n76-n78	n78	0.5
NOTE 1: The requirements only apply when the sub-frame and Tx-Rx timings are synchronized between the component carriers. In the absence of synchronization, the requirements are not within scope of these specifications.		

7.3A.3.2.2 Void

7.3A.4 Reference sensitivity exceptions due to UL harmonic interference for CA

Sensitivity degradation is allowed for a band in frequency range 1 if it is impacted by UL harmonic interference from another band in frequency range 1 of the same CA configuration. Reference sensitivity exceptions are specified in Table 7.3A.4-1 with uplink configuration specified in Table 7.3A.4-2.

Table 7.3A.4-1: Reference sensitivity exceptions due to UL harmonic for NR CA FR1

MSD due to harmonic exception for the DL band													
UL band	DL band	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
		dB	dB	dB	dB	dB	dB	dB	dB	dB	dB	dB	dB
n3	n77 ^{1,2}		23.9	22.1	20.9			17.9	16.9	16.1	14.8	14.3	13.8
	n77 ³		1.1	0.8	0.3								
n3	n78 ^{1,2}		23.9	22.1	20.9			17.9	16.9	16.1	14.8	14.3	13.8
	n78 ³		1.1	0.8	0.3								
n8	n78 ^{4,5}		10.8	9.1	8.0			5.1	4.2	3.5	2.3	2.1	1.4
n8	n79 ^{6,7}							[6.8]	6.2	[5.6]	4.9		4.4
n28	n75 ^{1,2}	28.1	25.3	24.0	22.8								
	n78 ^{6,7}		[10.4]	[8.9]	[7.8]			[4.7]	[3.7]	[3]	[1.7]	[1.2]	[0.7]

Table 7.3.2-3 applies.
NOTE 3: Unless stated otherwise, UL resource blocks shall be centred within the transmission bandwidth configuration for the channel bandwidth.

Table 7.3A.4-3: Void

Table 7.3A.4-3a: Void

Sensitivity degradation is allowed for a band if it is impacted by receiver harmonic mixing due to another band part of the same CA configuration. Reference sensitivity exceptions are specified in Table 7.3A.4-4 with uplink configuration specified in Table 7.3A.4-4a. Sensitivity degradation is not required for receiver even order harmonic mixing with aggressor 3rd order and above harmonic interference.

Table 7.3A.4-4: Reference sensitivity exceptions due to harmonic mixing for CA in NR FR1

NR Band / Channel bandwidth of the affected DL band												
UL band	DL band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	40 MHz (dB)	50 MHz (dB)	60 MHz (dB)	80 MHz (dB)	90 MHz (dB)	100 MHz (dB)
n78	n28 ³	31	28	26.2	25							
n78	n41 ²		10.4	10.4	10.4		7.2	6.2	5.5	4.5		4.5
n79	n8 ³	25	21.8	19.4	13.8							
NOTE 1: Void												
NOTE 2: The requirements should be verified for DL NR-ARFCN of the Victim (lower) band (superscript LB) such that and with the UL carrier frequency and the channel bandwidth configured in the higher band, both in MHz.												
NOTE 3: The requirements should be verified for DL EARFCN of the victim (lower) band (superscript LB) such that and with the UL carrier frequency and the channel bandwidth configured in the higher band, both in MHz.												

Table 7.3A.4-4a: Uplink configuration for reference sensitivity exceptions due to receiver harmonic mixing for CA in NR FR1

NR Band / SCS / Channel bandwidth of the affected DL band

verified when this is not the case (the requirements specified in clause 7.3 apply).
NOTE 4: This band is subject to IMD5 also which MSD is not specified.
NOTE 5: Void.

7.3A.6 Reference sensitivity exceptions due to cross band isolation for CA

Sensitivity degradation is allowed for a band if it is impacted by UL of another band part of the same NR CA configuration due to cross band isolation issues. Reference sensitivity exceptions for the victim band are specified in Table 7.3A.6-1 with uplink configuration of the aggressor band specified in Table 7.3A.6-2.

Table 7.3A.6-1: Reference sensitivity exceptions (MSD) due to cross band isolation for NR CA FR1

NR Band / Channel bandwidth of the affected DL band													
UL band	DL band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	40 MHz (dB)	50 MHz (dB)	60 MHz (dB)	80 MHz (dB)	90 MHz (dB)	100 MHz (dB)
n41	n78		8.3	8.3	8.3			6.3	5.3	4.5	4.0	3.9	3.8
n78	n41 ¹		4.5	4.5	4.5			4.5	4.5	4.5	4.5	4.5	4.5
n78	n79							2	2	2	2		2
n79	n78		2.6	2.6	2.6			2.6	2.6	2.6	2.6	2.6	2.6

NOTE 1: Applicable only when harmonic mixing MSD for this combination is not applied.
NOTE 2: The requirements only apply for UEs supporting inter-band carrier aggregation with simultaneous Rx/Tx capability. Simultaneous Rx/Tx capability does not apply for UEs supporting band n78 with a n77 implementation.

Table 7.3A.6.2: Uplink configuration for reference sensitivity exceptions due to cross band isolation for NR CA FR1

NR Band / SCS / Channel bandwidth of the affected DL band														
UL band	DL band	SCS of UL band (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
n41	n78	15		100	100	100			100	100	100	100	100	100

n78	n41	30		270	270	270			270	270	270	270	270	270
n78	n79	30							270 ²	270 ²	270 ²	270 ²		270 ²
n79	n78	30		270 ²	270 ²	270 ²			270 ²	270 ²	270 ²	270 ²	270 ²	270 ²
NOTE 1: The UL configuration applies regardless of the channel bandwidth of the UL band unless the UL resource blocks exceed that specified in Table 7.3.2-3 for the uplink bandwidth in which case the allocation according to Table 7.3.2-3 applies. NOTE 2: The requirements only apply for UEs supporting inter-band carrier aggregation with simultaneous Rx/Tx capability. Simultaneous Rx/Tx capability does not apply for UEs supporting band n78 with a n77 implementation.														

7.3B Void

7.3C Reference sensitivity for SUL

7.3C.1 General

The reference sensitivity power level REFSENS is the minimum mean power applied to each one of the UE antenna ports for all UE categories, at which the throughput shall meet or exceed the requirements for the specified reference measurement channel.

7.3C.2 Reference sensitivity power level for SUL

For SUL operation, the reference receive sensitivity (REFSENS) requirement for downlink bands specified in Table 7.3.2-1 and Table 7.3.2-2 shall be met for an uplink transmission bandwidth less than or equal to that specified in Table 7.3.2-3 or supplementary uplink transmission bandwidth less than or equal to that specified in Table 7.3C.2-1 with reference measurement channels as specified in Annexes A.2.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1), unless sensitivity degradation is allowed in this clause of this specification. These exceptions also apply to any higher order CA or DC combination containing one of the exception combinations in this clause as subset.

Table 7.3C.2-1: Supplementary uplink configuration for reference sensitivity

		NR Band / SCS of SUL band / Channel bandwidth of the DL band / N _{RB}													
DL band	SUL band	SCS of SUL band (kHz)	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz	

n78	n80	15		160	160	160			160	160	160	160	160	160
n78	n81	15		100	100	100			100	100	100	100	100	100
n78	n82	15		100	100	100			100	100	100	100	100	100
n78	n83	15		100	100	100			100	100	100	100	100	100
n78	n84	15		100	100	100			100	100	100	100	100	100
n78	n86	15		100	100	100			100	100	100	100	100	100
n79	n80	15							160	160	160	160		160
n79	n81	15							100	100	100	100		100

For the UE that supports any of the SUL operation given in Table 7.3C.2-2, exceptions to the requirements specified in Table 7.3.2-1 are allowed when the uplink is active in a lower frequency band and is within a specified frequency range such that transmitter harmonics fall within the downlink transmission bandwidth assigned in a higher band as noted in Table 7.3C.2-2. For these exceptions, the UE shall meet the requirements specified in Table 7.3C.2-2 and Table 7.3C.2-3.

Table 7.3C.2-2: Reference sensitivity for SUL operation (exceptions due to harmonic issue)

NR Band / Channel bandwidth of the high band													
UL band	DL band	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
		dB	dB	dB	dB	dB	dB	dB	dB	dB	dB	dB	dB
n80	n78 ^{1,2}		23.9	22.1	20.9			17.9	16.8	16.0	14.8	14.3	13.8
	n78 ³		1.1	0.8	0.3								
n82	n78 ^{4,5}		10.8	9.1	8			6	4.0	3.2	2.0	1.5	1.0
n81	n78 ^{4,5}		10.8	9.1	8			5.1	4.2	3.5	2.3	1.5	1.4
n83	n78 ^{6,7}		10.4	8.9	7.8			4.7	3.7	3	1.7	1.2	0.7
n86	n78 ^{1,2}		23.9	22.1	20.9			17.9	16.8	16.0	14.8	14.3	13.8
	n78 ³		1.1	0.8	0.3								
n81	n79 ^{6,7}							[6.8]	6.2	[5.6]	4.9		4.4

NOTE 1: These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) band for which the 2nd transmitter harmonic is within the downlink transmission bandwidth of a victim (higher) band and a range ΔF_{HD} above and below the edge of this downlink transmission bandwidth. The value ΔF_{HD} depends on the band combination: $\Delta F_{HD} = 10$ MHz for SUL_n78-n80, SUL_n78-n86.

NOTE 2: The requirements should be verified for UL EARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with f_{DL}^{NB} carrier frequency in the victim (higher) band in MHz and $BW_{Channel}^{LB}$ the channel

	bandwidth configured in the lower band.
NOTE 3:	The requirements are only applicable to channel bandwidths no larger than 20 MHz and with a carrier frequency at MHz offset from in the victim (higher band) with , where $BW_{Channel}^{LB}$ are the channel bandwidths configured in the aggressor (lower) and victim (higher) bands in MHz, respectively.
NOTE 4:	These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) band for which the 4 th transmitter harmonic is within the downlink transmission bandwidth of a victim (higher) band.
NOTE 5:	The requirements should be verified for UL EARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with f_{DL}^{nB} carrier frequency in the victim (higher) band in MHz and $BW_{Channel}^{LB}$ the channel bandwidth configured in the lower band.
NOTE 6:	These requirements apply when there is at least one individual RE within the uplink transmission bandwidth of the aggressor (lower) band for which the 5th transmitter harmonic is within the downlink transmission bandwidth of a victim (higher) band.
NOTE 7:	The requirements should be verified for UL NR-ARFCN of the aggressor (lower) band (superscript LB) such that in MHz and with f_{DL}^{nB} carrier frequency in the victim (higher) band in MHz and $BW_{Channel}^{LB}$ the channel bandwidth configured in the lower band.

Table 7.3C.2-3: Supplementary uplink configuration (exceptions due to harmonic issue)

NR Band / Channel bandwidth of the high band													
UL band	DL band	5 MHz (N _{RB})	10 MHz (N _{RB})	15 MHz (N _{RB})	20 MHz (N _{RB})	25 MHz (N _{RB})	30 MHz (N _{RB})	40 MHz (N _{RB})	50 MHz (N _{RB})	60 MHz (N _{RB})	80 MHz (N _{RB})	90 MHz (N _{RB})	100 MHz (N _{RB})
n80	n78		25	36	50			50	50	50	50	50	50
n81	n78		16	25	25			25	25	25	25	25	25
n81	n79							25	25	25	25		25
n82	n78		16	20	20			20	20	20	20	20	20
n83	n78		10	15	20			25	25	25	25	25	25
n86	n78		25	36	50			100	100	100	100	100	100
NOTE 1: 15 kHz SCS is assumed for UL band.													
NOTE 2: The UL configuration applies regardless of the channel bandwidth of the low band.													
NOTE 3: Unless stated otherwise, UL resource blocks shall be centered within the transmission bandwidth configuration for the channel bandwidth.													

7.3C.3 $\Delta R_{IB,c}$ for SUL

7.3C.3.1 General

For a UE supporting a SUL configuration, the $\Delta R_{IB,c}$ applies for both SC and SUL operation.

7.3C.3.2 SUL band combination

For the UE which supports SUL band combination, the minimum requirement for reference sensitivity in clause 7.3C.2 shall be increased by the amount given in $\Delta R_{IB,c}$ defined in clause 7.3C.3.2 for the applicable operating bands. Unless otherwise stated, $\Delta R_{IB,c}$ is set to zero.

In case the UE supports more than one of band combinations for CA, SUL or DC, and an operating band belongs to more than one band combinations then

- When the operating band frequency range is ≤ 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the average value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3], truncated to one decimal place that apply for that operating band among the supported band combinations. In case there is a harmonic relation between low band UL and high band DL, then the maximum $\Delta R_{IB,c}$ among the different supported band combinations involving such band shall be applied
- When the operating band frequency range is > 1 GHz, the applicable additional $\Delta R_{IB,c}$ shall be the maximum value for all band combinations defined in clause 7.3A, 7.3B, 7.3C in this specification and 7.3A, 7.3B in TS 38.101-3 [3] for the applicable operating bands.

7.3C.3.2.1 $\Delta R_{IB,c}$ for two bands

Table 7.3C.3.2.1-1: $\Delta R_{IB,c}$ due to SUL (two bands)

Band combination for SUL	NR Band	$\Delta R_{IB,c}$ (dB)
SUL_n78-n80	n78	0.5
SUL_n78-n81	n78	0.5
SUL_n78-n82	n78	0.5
SUL_n78-n83	n78	0.5
SUL_n78-n84	n78	0.5
SUL_n78-n86	n78	0.5

7.3D Reference sensitivity for UL MIMO

For UE with two transmitter antenna connectors in closed-loop spatial multiplexing scheme, the minimum requirements specified in clause 7.3 shall be met with the UL MIMO configurations described in clause 6.2D.1 and the reference measurement channels as specified in Annexes A.2.2 for CP-OFDM waveforms shall apply. For UL MIMO, the parameter P_{UMAX} is the total transmitter power over the two transmits power over the two transmit antenna connectors.

7.4 Maximum input level

Maximum input level is defined as the maximum mean power received at the UE antenna port, at which the specified relative throughput shall meet or exceed the minimum requirements for the specified reference measurement channel. The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.4-1.

Table 7.4-1: Maximum input level

Rx Parameter	Units	Channel bandwidth											
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
Power in Transmission Bandwidth Configuration	dBm	-25 ²				-24 ²	-23 ²	-22 ²	-21 ²	-20 ²			
		-27 ³				-26 ³	-25 ³	-24 ³	-23 ³	-22 ³			
NOTE 1: The transmitter shall be set to 4 dB below P _{CMAX_L,f,c} at the minimum uplink configuration specified in Table 7.3.2-3 with P _{CMAX_L,f,c} as defined in clause 6.2.4.													
NOTE 2: Reference measurement channel is A.3.2.3 or A.3.3.3 for 64 QAM.													
NOTE 3: Reference measurement channel is A.3.2.4 or A.3.3.4 for 256 QAM.													

7.4A Maximum input level for CA

7.4A.1 Maximum input level for Intra-band contiguous CA

For intra-band contiguous carrier aggregation maximum input level is defined as the maximum mean power received at the UE antenna port, over the Transmission bandwidth configuration of each CC.

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in Annexes A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.4A.1-1 for each component carrier.

Table 7.4A.1-1: Maximum input level for Intra-band contiguous CA

Rx Parameter	Units	NR CA Bandwidth Class		
		C		
Power in largest transmission bandwidth configuration CC, $P_{\text{largest BW}}$	dBm	-23 ²		
		-25 ³		
Power in each other CC	dBm	$P_{\text{largest BW}} + 10 \cdot \log\{(N_{\text{RB},c} \cdot \text{SCS}_c) / (N_{\text{RB},\text{largest BW}} \cdot \text{SCS}_{\text{largest BW}})\}$		
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum uplink configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ as defined in clause 6.2.4.				
NOTE 2: Reference measurement channel is A.3.2.3 or A.3.3.3 for 64 QAM.				
NOTE 3: Reference measurement channel is A.3.2.4 or A.3.3.4 for 256 QAM.				

7.4A.2 Void

7.4A.3 Maximum input level for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the maximum input level is defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.4 for each component carrier while all downlink carriers are active.

The throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in Annexes A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1) for each component carrier.

7.4D Maximum input level for UL MIMO

For UE with two transmitter antenna connectors in closed-loop spatial multiplexing, the minimum requirements specified in clause 7.4 shall be met with the UL MIMO configurations described in clause 6.2D.1. For UL MIMO, the parameter P_{CMAX_L} is defined as the total transmitter power over the two transmit antenna connectors.

7.5 Adjacent channel selectivity

Adjacent channel selectivity (ACS) is a measure of a receiver's ability to receive an NR signal at its assigned channel frequency in the presence of an adjacent channel signal at a given frequency offset from the centre frequency of the assigned channel. ACS is the ratio of the receive filter attenuation on the assigned channel frequency to the receive filter attenuation on the adjacent channel(s).

The UE shall fulfil the minimum requirements specified in Table 7.5-1 for NR bands with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz and the minimum requirements specified in Table 7.5-2 for NR bands with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz. These requirements apply for all values of an adjacent channel interferer up to -25 dBm and for any SCS specified for the channel bandwidth of the wanted signal. However, it is not possible to directly measure the ACS; instead the lower and upper range of test parameters are chosen as in Table 7.5-3 and Table 7.5-4 for verification of the requirements specified in Table 7.5-1, and as in Table 7.5-5 and Table 7.5-6 for verification of the requirements specified in Table 7.5-2. For these test parameters, the throughput shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1). For operating bands with an unpaired DL part (as noted in Table 5.2-1), the requirements only apply for carriers assigned in the paired part.

Table 7.5-1: ACS for NR bands with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz

RX parameter	Units	Channel bandwidth				
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz
ACS	dB	33	33	30	27	26
RX parameter	Units	Channel bandwidth				
		30 MHz	40 MHz	50 MHz	60 MHz	80 MHz
ACS	dB	25.5	24	23	22.5	21
RX parameter	Units	Channel bandwidth				
		90 MHz	100 MHz			
ACS	dB	20.5	20			

Table 7.5-2: ACS for NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

RX parameter	Units	Channel bandwidth				
		10 MHz	15 MHz	20 MHz	40 MHz	50 MHz
ACS	dB	33	33	33	33	33
RX parameter	Units	Channel bandwidth				
		60 MHz	80 MHz	90 MHz	100 MHz	
ACS	dB	33	33	33	33	

Table 7.5-3: Test parameters for NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz, case 1

RX parameter	Units	Channel bandwidth				
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + 14 dB				
$P_{interferer}$	dBm	REFSENS + 45.5 dB	REFSENS + 45.5 dB	REFSENS + 42.5 dB	REFSENS + 39.5 dB	REFSENS + 38.5 dB
$BW_{interferer}$	MHz	5	5	5	5	5
$F_{interferer}$ (offset)	MHz	5 / -5	7.5 / -7.5	10 / -10	12.5 / -12.5	15 / -15
RX parameter	Units	Channel bandwidth				
		30 MHz	40 MHz	50 MHz	60 MHz	80 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + 14 dB				
$P_{interferer}$	dBm	REFSENS + 38 dB	REFSENS + 36.5 dB	REFSENS + 35.5 dB	REFSENS + 35 dB	REFSENS + 33.5 dB
$BW_{interferer}$	MHz	5	5	5	5	5
$F_{interferer}$ (offset)	MHz	17.5 / -17.5	22.5 / -22.5	27.5 / -27.5	32.5 / -32.5	42.5 / -42.5
RX parameter	Units	Channel bandwidth				
		90 MHz	100 MHz			
Power in transmission	dBm	REFSENS + 14 dB				

bandwidth configuration					
$P_{\text{interferer}}$	dBm	REFSENS + 33 dB	REFSENS + 32.5 dB		
$BW_{\text{interferer}}$	MHz	5	5		
$F_{\text{interferer}}$ (offset)	MHz	47.5 / -47.5	52.5 / -52.5		
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{C}_{\text{MAX_L,f,c}}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{C}_{\text{MAX_L,f,c}}}$ defined in clause 6.2.4. NOTE 2: The absolute value of the interferer offset $F_{\text{interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with 15 kHz SCS. NOTE 3: The interferer consists of the NR interferer RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A. 5.2.1.					

Table 7.5-4: Test parameters for NR bands with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz, case 2

RX parameter	Units	Channel bandwidth				
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz
Power in transmission bandwidth configuration	dBm	-56.5	-56.5	-53.5	-50.5	-49.5
$P_{\text{interferer}}$	dBm	-25				
$BW_{\text{interferer}}$	MHz	5	5	5	5	5
$F_{\text{interferer}}$ (offset)	MHz	5 / -5	7.5 / -7.5	10 / -10	12.5 / -12.5	15 / -15
RX parameter	Units	Channel bandwidth				
		30 MHz	40 MHz	50 MHz	60 MHz	80 MHz
Power in transmission bandwidth configuration	dBm	-49	-47	-46.5	-46	-44.5
$P_{\text{interferer}}$	dBm	-25				
$BW_{\text{interferer}}$	MHz	5	5	5	5	5
$F_{\text{interferer}}$ (offset)	MHz	17.5 / /	22.5 / /	27.5 / /	32.5 / /	42.5 / /

		-17.5	-22.5	-27.5	-32.5	-42.5
RX parameter	Units	Channel bandwidth				
		90 MHz	100 MHz			
Power in transmission bandwidth configuration	dBm	-44	-43.5			
$P_{\text{interferer}}$	dBm	-25				
$BW_{\text{interferer}}$	MHz	5	5			
$F_{\text{interferer}}$ (offset)	MHz	47.5 / -47.5	52.5 / -52.5			
NOTE 1: The transmitter shall be set to 24 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.						
NOTE 2: The absolute value of the interferer offset $F_{\text{interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with 15 kHz SCS.						
NOTE 3: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1						

Table 7.5-5: Test parameters for NR bands with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz, case 1

RX parameter	Units	Channel bandwidth				
		10 MHz	15 MHz	20 MHz	40 MHz	50 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + 14 dB				
$P_{\text{interferer}}$	dBm	REFSENS + 45.5 dB				
$BW_{\text{interferer}}$	MHz	10	15	20	40	50
$F_{\text{interferer}}$ (offset)	MHz	10 / -10	15 / -15	20 / -20	40 / -40	50 / -50
RX parameter	Units	Channel bandwidth				
		60 MHz	80 MHz	90 MHz	100 MHz	
Power in transmission bandwidth configuration	dBm	REFSENS + 14 dB				
$P_{\text{interferer}}$	dBm	REFSENS + 45.5 dB	REFSENS + 45.5 dB	REFSENS + 45.5 dB	REFSENS + 45.5 dB	

NOTE 2:	The absolute value of the interferer offset $F_{\text{interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with an SCS equal to that of the wanted signal.
NOTE 3:	The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1.

7.5A Adjacent channel selectivity for CA

7.5A.1 Adjacent channel selectivity for Intra-band contiguous CA

For intra-band contiguous carrier aggregation the downlink SCC(s) shall be configured at nominal channel spacing to the PCC. The UE shall fulfil the minimum requirement specified in Table 7.5A.1-1 and 7.5A.1-1a for an adjacent channel interferer on either side of the aggregated downlink signal at a specified frequency offset and for an interferer power up to -25 dBm. The throughput of each carrier shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Tables 7.5A.1-2, 7.5A.1-2a, 7.5A.1-3 and 7.5A.1-3a.

Table 7.5A.1-1: ACS for intra-band contiguous CA with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz

		NR CA bandwidth class		
Rx Parameter	Units	C		
ACS	dB	33.0		

Table 7.5A.1-1a: ACS for intra-band contiguous CA with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz

		NR CA bandwidth class
Rx Parameter	Units	C
ACS	dB	17.0

Table 7.5A.1-2: Test parameters for intra-band contiguous CA with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz, case 1

Rx Parameter	Units	NR CA bandwidth class
		C
Pw in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + 14 dB
$P_{\text{Interferer}}$	dBm	Aggregated power + 31.5 dB
$BW_{\text{Interferer}}$	MHz	$BW_{\text{channel CA}}$
$F_{\text{Interferer}}$ (offset)	MHz	$\frac{BW_{\text{channel CA}}/2 + F_{\text{offset}}}{-BW_{\text{channel CA}}/2 - F_{\text{offset}}}$
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4 . NOTE 2: The absolute value of the interferer offset $F_{\text{Interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the carrier closest to the interferer in MHz. The interferer is an NR signal with an SCS equal to that of the closest carrier. NOTE 3: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1.		

Table 7.5A.1-2a: Test parameters for intra-band contiguous CA with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz, case 1

Rx Parameter	Units	NR CA bandwidth class
		C
Pw in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + 14 dB
$P_{\text{Interferer}}$	dBm	Aggregated power + 15.5 dB
$BW_{\text{Interferer}}$	MHz	5
$F_{\text{Interferer}}$ (offset)	MHz	$\frac{2.5 + F_{\text{offset}}}{-2.5 - F_{\text{offset}}}$
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4. NOTE 2: The absolute value of the interferer offset $F_{\text{Interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the carrier closest to the interferer in MHz. The interferer is an NR signal with 15 kHz SCS. NOTE 3: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1.		

Table 7.5A.1-3: Test parameters for intra-band contiguous CA with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz, case 2

Rx Parameter	Units	NR CA bandwidth class		
		C		
Pw in Transmission Bandwidth Configuration, per CC	dBm	-56.5		
$P_{\text{Interferer}}$	dBm	-25		
$BW_{\text{Interferer}}$	MHz	$BW_{\text{channel CA}}$		
$F_{\text{Interferer}}$ (offset)	MHz	$BW_{\text{channel CA}}/2 + F_{\text{offset}}$ / $-BW_{\text{channel CA}}/2 - F_{\text{offset}}$		
NOTE 1: The transmitter shall be set to 24 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4. NOTE 2: The absolute value of the interferer offset $F_{\text{Interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the carrier closest to the interferer in MHz. The interferer is an NR signal with an SCS equal to that of the closest carrier. NOTE 3: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1.				

Table 7.5A.1-3a: Test parameters for intra-band contiguous CA with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz, case 2

Rx Parameter	Units	NR CA Bandwidth Class
		C
Pw in Transmission Bandwidth Configuration, per CC	dBm	$-40.5 + 10\log(N_{\text{RB,c}}/N_{\text{RB_agg}})$
$P_{\text{Interferer}}$	dBm	-25
$BW_{\text{Interferer}}$	MHz	5
$F_{\text{Interferer}}$ (offset)	MHz	$2.5 + F_{\text{offset}}$ / $-2.5 - F_{\text{offset}}$
NOTE 1: The transmitter shall be set to 24 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4. NOTE 2: The absolute value of the interferer offset $F_{\text{Interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the carrier closest to the interferer in MHz. The interferer is an NR signal with 15 kHz SCS. NOTE 3: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1.		

7.5A.2 Void

7.5A.3 Adjacent channel selectivity Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the adjacent channel requirements are defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.5 for each component carrier while all downlink carriers are active.

The throughput of each carrier shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

7.5D Adjacent channel selectivity for UL MIMO

For UE(s) with two transmitter antenna connectors in closed-loop spatial multiplexing scheme, the minimum requirements specified in clause 7.5 shall be met with the UL MIMO configurations described in clause 6.2D.1. For UL MIMO, the parameter P_{CMAX_L} is defined as the total transmitter power over the two transmit antenna connectors.

7.6 Blocking characteristics

7.6.1 General

The blocking characteristic is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency in the presence of an unwanted interferer on frequencies other than those of the spurious response or the adjacent channels, without this unwanted input signal causing a degradation of the performance of the receiver beyond a specified limit. The blocking performance shall apply at all frequencies except those at which a spurious response occurs.

7.6.2 In-band blocking

For NR bands with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz in-band blocking (IBB) is defined for an unwanted interfering signal falling into the UE receive band or into the first 15 MHz below or above the UE receive band. The throughput of the wanted

NOTE 2: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1 and 15 kHz SCS.

Table 7.6.2-2: In-band blocking for NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz

NR band	Parameter	Unit	Case 1	Case 2	Case 3
	$P_{interferer}$	dBm	-56	-44	-15
n1, n2, n3, n5, n7, n8, n12, n20, n25, n28, n34, n38, n39, n40, n41, n50, n51, n66, n70, n74, n75, n76	$F_{interferer}$ (offset)	MHz	$-BW_{Channel}/2 - F_{offset, case 1}$ and $BW_{Channel}/2 + F_{offset, case 1}$	$\leq -BW_{Channel}/2 - F_{offset, case 2}$ and $\geq BW_{Channel}/2 + F_{offset, case 2}$	
	$F_{interferer}$	MHz	NOTE 2	$F_{DL_low} - 15$ to $F_{DL_high} + 15$	
n71	$F_{interferer}$	MHz	NOTE 2	$F_{DL_low} - 12$ to $F_{DL_high} + 15$	$F_{DL_low} - 12$
NOTE 1: The absolute value of the interferer offset $F_{interferer}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the wanted signal in MHz. The interferer is an NR signal with 15 kHz SCS.					
NOTE 2: For each carrier frequency, the requirement applies for two interferer carrier frequencies: a: $-BW_{Channel}/2 - F_{offset, case 1}$; b: $BW_{Channel}/2 + F_{offset, case 1}$					

For NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz in-band blocking (IBB) is defined for an unwanted interfering signal falling into the UE receive band or into an immediately adjacent frequency range up to $3 \cdot BW_{Channel}$ below or above the UE receive band where $BW_{Channel}$ is the bandwidth of the wanted signal. The throughput of the wanted signal shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1)] with parameters specified in Table 7.6.2-3 and Table 7.6.2-4. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal.

Table 7.6.2-3: In-band blocking parameters for NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

RX parameter	Units	Channel bandwidth
--------------	-------	-------------------

		10 MHz	15 MHz	20 MHz	40 MHz	50 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	6				
$BW_{\text{interferer}}$	MHz	10	15	20	40	50
$F_{\text{offset, case 1}}$	MHz	15	22.5	30	60	75
$F_{\text{offset, case 2}}$	MHz	25	37.5	50	100	125
RX parameter	Units	Channel bandwidth				
		60 MHz	80 MHz	90 MHz	100 MHz	
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	6				
$BW_{\text{interferer}}$	MHz	60	80	90	100	
$F_{\text{offset, case 1}}$	MHz	90	120	135	150	
$F_{\text{offset, case 2}}$	MHz	150	200	225	250	
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.						
NOTE 2: The interferer consists of the RMC specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A. 5.2.1						

Table 7.6.2-4: In-band blocking for NR bands with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz

NR band	Parameter	Unit	Case 1	Case 2
	$P_{\text{interferer}}$	dBm	-56	-44
n77, n78, n79	$F_{\text{interferer}}$ (offset)	MHz	$-BW_{\text{Channel}}/2 - F_{\text{offset, case 1}}$ and $BW_{\text{Channel}}/2 + F_{\text{offset, case 1}}$	$\leq -BW_{\text{Channel}}/2 - F_{\text{offset, case 2}}$ and $\geq BW_{\text{Channel}}/2 + F_{\text{offset, case 2}}$
	$F_{\text{interferer}}$		NOTE 2	$F_{\text{DL_low}} - 3 \cdot BW_{\text{Channel}}$ to $F_{\text{DL_high}} + 3 \cdot BW_{\text{Channel}}$
NOTE 1: The absolute value of the interferer offset $F_{\text{interferer}}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the wanted				

	signal in MHz. The interferer is an NR signal with an SCS equal to that of the wanted signal.
NOTE 2:	For each carrier frequency, the requirement applies for two interferer carrier frequencies: a: $-BW_{\text{Channel}}/2 - F_{\text{offset, case 1}}$; b: $BW_{\text{Channel}}/2 + F_{\text{offset, case 1}}$
NOTE 3:	BW_{Channel} denotes the channel bandwidth of the wanted signal.

7.6.3 Out-of-band blocking

For NR bands with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz out-of-band band blocking is defined for an unwanted CW interfering signal falling outside a frequency range 15 MHz below or above the UE receive band. The throughput of the wanted signal shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.6.3-1 and Table 7.6.3-2. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal. For operating bands with an unpaired DL part (as noted in Table 5.2-1), the requirements only apply for carriers assigned in the paired part.

Table 7.6.3-1: Out-of-band blocking parameters for NR bands with $F_{\text{DL_high}} < 2700$ MHz and $F_{\text{UL_high}} < 2700$ MHz

Channel bandwidth	Power in transmission bandwidth configuration [dBm]
5 MHz	REFSENS + 6.0 dB
10 MHz	REFSENS + 6.0 dB
15 MHz	REFSENS + 7.0 dB
20 MHz	REFSENS + 9.0 dB
25 MHz	REFSENS + 10.0 dB
30 MHz	REFSENS + 11.0 dB
40 MHz	REFSENS + 12.0 dB
50 MHz	REFSENS + 13.0 dB
60 MHz	REFSENS + 14.0 dB
80 MHz	REFSENS + 15.0 dB
90 MHz	REFSENS + 15.5 dB
100 MHz	REFSENS + 16.0 dB
NOTE: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$	

defined in clause 6.2.4.

Table 7.6.3-2: Out of-band blocking for NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz

NR band	Parameter	Unit	Range 1	Range 2	Range 3
n1, n2, n3, n5, n7, n8, n12, n20, n25, n28, n34, n38, n39, n40, n41, n50, n51, n66, n70, n71, n74, n75, n76	$P_{interferer}$	dBm	-44	-30	-15
	$F_{interferer}$ (CW)	MHz	$-60 < f - F_{DL_low} < -15$ or $15 < f - F_{DL_high} < 60$	$-85 < f - F_{DL_low} \leq -60$ or $60 \leq f - F_{DL_high} < 85$	$1 \leq f \leq F_{DL_low} - 85$ or $F_{DL_high} + 85 \leq f \leq 12750$
NOTE 1: The power level of the interferer ($P_{interferer}$) for Range 3 shall be modified to -20 dBm for $F_{interferer} > 6000$ MHz. NOTE 2: For band 51 the F_{DL_high} of band 50 is applied as F_{DL_high} for band 51. For band 50, the F_{DL_low} of band 51 is applied as F_{DL_low} for band 50. NOTE 3: For band 76 the F_{DL_high} of band 75 is applied as F_{DL_high} for band 76. For band 75, the F_{DL_low} of band 76 is applied as F_{DL_low} for band 75. NOTE 4: For UEs supporting both bands 38 and 41, the F_{DL_high} and F_{DL_low} of band 41 is applied as F_{DL_high} and F_{DL_low} for band 38. NOTE 5: Void. NOTE 6: Void. NOTE 7: For UE supporting both bands 25 and 70, the F_{DL_high} of band 70 is applied as F_{DL_high} for band 25, and the F_{DL_low} of band 25 is applied as F_{DL_low} for band 70.					

For interferer frequencies across ranges 1, 2 and 3 in Table 7.6.3-2, a maximum of

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of MHz with the number of resource blocks in the downlink transmission bandwidth configuration, $BW_{channel}$ is the bandwidth of the frequency channel in MHz and $n = 1, 2, 3$ for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in subclause 7.7 apply.

For NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz out-of-band band blocking is defined for an unwanted CW interfering signal falling outside a frequency range up to $3 \cdot BW_{Channel}$ below or from $3 \cdot BW_{Channel}$ above the UE receive band, where $BW_{Channel}$ is the channel bandwidth. The throughput of the wanted signal shall be ≥ 95 % of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.6.3-3 and Table 7.6.3-4. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal.

Table 7.6.3-3: Out-of-band blocking parameters for NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

RX parameter	Units	Channel bandwidth				
		10 MHz	15 MHz	20 MHz	40 MHz	50 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	6	7	9	9	9
RX parameter	Units	Channel bandwidth				
		60 MHz	80 MHz	90 MHz	100 MHz	
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	9	9	9	9	
NOTE: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4.						

Table 7.6.3-4: Out of-band blocking for NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

NR band	Parameter	Unit	Range1	Range 2	Range 3
n77, n78 (NOTE 3)	$P_{interferer}$	dBm	-44	-30	-15
	$F_{interferer}$ (CW)	MHz	$-60 < f - F_{DL_low} \leq -3 \cdot BW_{Channel}$ or $3 \cdot BW_{Channel} \leq f - F_{DL_high} < 60$	$-200 < f - F_{DL_low} \leq -\text{MAX}(60, 3 \cdot BW_{Channel})$ or $\text{MAX}(60, 3 \cdot BW_{Channel}) \leq f - F_{DL_high} < 200$	$1 \leq f \leq F_{DL_low} - \text{MAX}(200, 3 \cdot BW_{Channel})$ or $F_{DL_high} + \text{MAX}(200, 3 \cdot BW_{Channel}) \leq f \leq 12750$
n79 (NOTE 4)	$F_{interferer}$ (CW)	MHz	N/A	$-150 < f - F_{DL_low} \leq -\text{MAX}(60, 3 \cdot BW_{Channel})$ or 	$1 \leq f \leq F_{DL_low} - \text{MAX}(150, 3 \cdot BW_{Channel})$ or

				$\text{MAX}(60, 3 \cdot \text{BW}_{\text{Channel}}) \leq f - F_{\text{DL_high}} < 150$	$F_{\text{DL_high}} + \text{MAX}(150, 3 \cdot \text{BW}_{\text{Channel}}) \leq f \leq 12750$
NOTE 1: The power level of the interferer ($P_{\text{Interferer}}$) for Range 3 shall be modified to -20 dBm for $F_{\text{Interferer}} > 6000$ MHz. NOTE 2: $\text{BW}_{\text{Channel}}$ denotes the channel bandwidth of the wanted signal NOTE 3: The power level of the interferer ($P_{\text{Interferer}}$) for Range 3 shall be modified to -20 dBm, for $F_{\text{Interferer}} > 2700$ MHz and $F_{\text{Interferer}} < 4800$ MHz. For $\text{BW}_{\text{Channel}} > 15$ MHz, the requirement for Range 1 is not applicable and Range 2 applies from the frequency offset of $3 \cdot \text{BW}_{\text{Channel}}$ from the band edge. For $\text{BW}_{\text{Channel}} > 60$ MHz, the requirement for Range 2 is not applicable and Range 3 applies from the frequency offset of $3 \cdot \text{BW}_{\text{Channel}}$ from the band edge. NOTE 4: The power level of the interferer ($P_{\text{Interferer}}$) for Range 3 shall be modified to -20 dBm, for $F_{\text{Interferer}} > 3650$ MHz and $F_{\text{Interferer}} < 5750$ MHz. For $\text{BW}_{\text{Channel}} > 40$ MHz, the requirement for Range 2 is not applicable and Range 3 applies from the frequency offset of $3 \cdot \text{BW}_{\text{Channel}}$ from the band edge.					

For interferer frequencies across ranges 1, 2 and 3 in Table 7.6.3-4, a maximum of

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of MHz with the number of resource blocks in the downlink transmission bandwidth configuration, $\text{BW}_{\text{Channel}}$ the bandwidth of the frequency channel in MHz and $n = 1, 2, 3$ for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in subclause 7.7 apply.

7.6.4 Narrow band blocking

This requirement is measure of a receiver's ability to receive a NR signal at its assigned channel frequency in the presence of an unwanted narrow band CW interferer at a frequency, which is less than the nominal channel spacing.

The relative throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.6.4-1. For operating bands with an unpaired DL part (as noted in Table 5.2-1), the requirements only apply for carriers assigned in the paired part.

Table 7.6.4-1: Narrow Band Blocking

NR band	Parameter	Unit	Channel Bandwidth											
			5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
n1, n2, n3, n5, n7, n8, n12, n20, n25, n28, n34, n38, n39, n40, n41, n50, n51, n66, n70, n71, n74, n75, n76	P_w	dBm		P_{REFSENS} + channel-bandwidth specific value below										
			16	13	14	16	16	16	16	16	16	16	16	16
	$P_{\text{uw}}(\text{CW})$	dBm	-55	-55	-55	-55	-55	-55	-55	-55	-55	-55	-55	-55
	F_{uw} (offset SCS= 15 kHz)	MHz	2.7075	5.2125	7.7025	10.2075	13.0275	15.6075	20.5575	25.7025	NA	NA	NA	NA
	F_{uw} (offset SCS= 30 kHz)	MHz	NA	NA	NA	NA	NA	NA	NA	NA	30.855	40.935	45.915	50.865

NOTE 1: The transmitter shall be set a 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4

NOTE 2: Reference measurement channel is specified in Annexes A.3.2 and A.3.3 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1.

NOTE 3: The P_{REFSENS} power level is specified in Table 7.3.2-1 and Table 7.3.2-2 for two and four antenna ports, respectively.

7.6A Blocking characteristics for CA

7.6A.1 General

7.6A.2 In-band blocking for CA

7.6A.2.1 In-band blocking for Intra-band contiguous CA

For intra-band contiguous carrier aggregation the downlink SCC(s) shall be configured at nominal channel spacing to the PCC. The UE shall fulfil the minimum requirement specified in Table 7.6A.2.1-1 and 7.6A.2.1-1a for an adjacent channel interferer on either side of the aggregated downlink signal at a specified frequency offset and for an interferer power up to -25 dBm. The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

Table 7.6A.2.1-1: In-band blocking parameters for intra-band contiguous CA with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

Rx Parameter	Units	NR CA bandwidth class		
		C		
Pw in Transmission Bandwidth Configuration, per CC	dB	REFSENS + CA bandwidth class specific value below		
		6		
$BW_{interferer}$	MHz	$BW_{channel\ CA}$		
$F_{offset, case\ 1}$	MHz	$BW_{channel\ CA} + BW_{channel\ CA}/2$		
$F_{offset, case\ 2}$	MHz	$BW_{interferer} + F_{offset, case\ 1}$		
NOTE 1: The transmitter shall be set to 4dB below $P_{C_{MAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{C_{MAX_L,f,c}}$ defined in clause 6.2.4.				
NOTE 2: The interferer consists of the Reference measurement channel specified in Annexes A.3.2 and A.3.3 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1				

Table 7.6A.2.1-1a: In-band blocking parameters for intra-band contiguous CA with $F_{DL_low} < 2700$ MHz and $F_{UL_low} < 2700$ MHz

Rx Parameter	Units	NR CA bandwidth class		
		C		

Pw in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + NR CA bandwidth class specific value below
		19.0
$BW_{\text{Interferer}}$	MHz	5
$F_{\text{offset, case 1}}$	MHz	7.5
$F_{\text{offset, case 2}}$	MHz	12.5
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.		
NOTE 2: The interferer consists of the Reference measurement channel specified in Annexes A.3.2 and A.3.3 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1 and set-up according to Annex C.3.1		

Table 7.6A.2.1-2: In-band blocking for intra-band contiguous CA with $F_{\text{DL_low}} \geq 3300$ MHz and $F_{\text{UL_low}} \geq 3300$ MHz

NR band	Parameter	Unit	Case 1	Case 2
	$P_{\text{interferer}}$	dBm	-56	-44
n77, n78, n79	$F_{\text{interferer}} \text{ (offset)}$	MHz	$-F_{\text{offset}} - F_{\text{offset, case 1}}$ and $F_{\text{offset}} + F_{\text{offset, case 1}}$	$\leq -F_{\text{offset}} - F_{\text{offset, case 2}}$ and $\geq F_{\text{offset}} + F_{\text{offset, case 2}}$
	$F_{\text{interferer}}$	MHz	NOTE 2	$F_{\text{DL_low}} - 3BW_{\text{channel CA}}$ to $F_{\text{DL_high}} + 3BW_{\text{channel CA}}$
NOTE 1: The absolute value of the interferer offset $F_{\text{interferer}} \text{ (offset)}$ shall be further adjusted to MHz with SCS the sub-carrier spacing of the carrier closest to the interferer in MHz. The interferer is an NR signal with an SCS equal to that of the closest carrier.				
NOTE 2: For each carrier frequency, the requirement applies for two interferer carrier frequencies: a: $-F_{\text{offset}} - F_{\text{offset, case 1}}$; b: $F_{\text{offset}} + F_{\text{offset, case 1}}$				
NOTE 3: $BW_{\text{channel CA}}$ denotes the aggregated channel bandwidth of the wanted signal				

Table 7.6A.2.1-2a: In-band blocking for intra-band contiguous CA with $F_{\text{DL_low}} < 2700$ MHz and $F_{\text{UL_low}} < 2700$ MHz

NR band	Parameter	Unit	Case 1	Case 2
	$P_{\text{interferer}}$	dBm	-56	-44
n41	$F_{\text{interferer}} \text{ (offset)}$	MHz	$-F_{\text{offset}} - F_{\text{offset, case 1}}$ and $F_{\text{offset}} + F_{\text{offset, case 1}}$	$\leq -F_{\text{offset}} - F_{\text{offset, case 2}}$ and $\geq F_{\text{offset}} + F_{\text{offset, case 2}}$
	$F_{\text{interferer}}$	MHz	NOTE 2	$F_{\text{DL_low}} - 15$ to

				$F_{DL_high} + 15$
NOTE 1:	The absolute value of the interferer offset $F_{interferer}$ (offset) shall be further adjusted to MHz with SCS the sub-carrier spacing of the carrier closest to the interferer in MHz. The interferer is an NR signal with 15 kHz SCS.			
NOTE 2:	For each carrier frequency, the requirement applies for two interferer carrier frequencies: a: $-F_{offset} - F_{offset, case 1}$; b: $F_{offset} + F_{offset, case 1}$			
NOTE 3:	$BW_{channel CA}$ denotes the aggregated channel bandwidth of the wanted signal			

7.6A.2.2 Void

7.6A.2.3 In-band blocking for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the in-band blocking requirements are defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.6.2 for each component carrier while all downlink carriers are active.

For the UE which supports inter-band CA configuration in Table 7.3A.3.2, $P_{interferer}$ power defined in Table 7.6.2-2 and 7.6.2-4 is increased by the amount given by $\Delta R_{IB,c}$ in Table 7.3A.3.2.

The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

7.6A.3 Out-of-band blocking for CA

7.6A.3.1 Out-of-band blocking for Intra-band contiguous CA

For intra-band contiguous carrier aggregation the downlink SCC(s) shall be configured at nominal channel spacing to the PCC.

For FDD, the PCC shall be configured closest to the uplink band. All downlink carriers shall be active throughout the test.

The UE shall fulfil the minimum requirement in presence of an interfering signal specified in Tables 7.6A.3-1 and Tables 7.6A.3-2 being on either side of the aggregated signal. The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

Table 7.6A.3-1: Out-of-band blocking parameters for intra-band contiguous CA

RX parameter	Units	CA bandwidth class		
		C		
Power in transmission bandwidth configuration	dBm	REFSENS + CA bandwidth class specific value below		
	dB	9		
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.				

Table 7.6A.3-2: Out of-band blocking for intra-band contiguous CA

NR band	Parameter	Unit	Range1	Range 2	Range 3
	$P_{\text{interferer}}$	dBm	-45	-30	-15
n41	$F_{\text{interferer}}$ (CW)	MHz	$-60 < f - F_{\text{DL_low}} < -15$ or $15 < f - F_{\text{DL_high}} < 60$	$-85 < f - F_{\text{DL_low}} \leq -60$ or $60 \leq f - F_{\text{DL_high}} < 85$	$1 \leq f \leq F_{\text{DL_low}} - 85$ or $F_{\text{DL_high}} + 85 \leq f \leq 12750$
n77, n78 (NOTE 3)	$F_{\text{interferer}}$ (CW)	MHz	N/A	N/A	$1 \leq f \leq F_{\text{DL_low}} - \text{MAX}(200, 3 \cdot \text{BW}_{\text{Channel_CA}})$ or $F_{\text{DL_high}} + \text{MAX}(200, 3 \cdot \text{BW}_{\text{Channel_CA}}) \leq f \leq 12750$
n79 (NOTE 4)	$F_{\text{interferer}}$ (CW)	MHz	N/A	N/A	$1 \leq f \leq F_{\text{DL_low}} - \text{MAX}(150, 3 \cdot \text{BW}_{\text{Channel_CA}})$ or $F_{\text{DL_high}} + \text{MAX}(150, 3 \cdot \text{BW}_{\text{Channel_CA}}) \leq f \leq 12750$
NOTE 1: The power level of the interferer ($P_{\text{interferer}}$) for Range 3 shall be modified to -20 dBm for $F_{\text{interferer}} > 6000$ MHz.					
NOTE 2: $\text{BW}_{\text{Channel_CA}}$ denotes the aggregated channel bandwidth of the wanted signal					
NOTE 3: The power level of the interferer ($P_{\text{interferer}}$) for Range 3 shall be modified to -20 dBm, for $F_{\text{interferer}} > 2700$ MHz and $F_{\text{interferer}} < 4800$ MHz. For $\text{BW}_{\text{Channel_CA}} > 15$ MHz, the requirement for Range 1 is not applicable and Range 2 applies from the frequency offset of $3 \cdot \text{BW}_{\text{Channel_CA}}$ from the band edge. For $\text{BW}_{\text{Channel_CA}} > 60$ MHz, the requirement for Range 2 is not applicable and Range 3 applies from the frequency offset of $3 \cdot \text{BW}_{\text{Channel_CA}}$ from the band edge.					
NOTE 4: The power level of the interferer ($P_{\text{interferer}}$) for Range 3 shall be modified to -20 dBm, for $F_{\text{interferer}} > 3650$ MHz and $F_{\text{interferer}} < 5750$ MHz. For $\text{BW}_{\text{Channel_CA}} > 40$ MHz, the requirement for Range 2 is not applicable and Range 3 applies from the					

frequency offset of $3 \cdot BW_{\text{Channel_CA}}$ from the band edge
--

For interferer frequencies across ranges 1, 2 and 3 in Table 7.6A.3-2, a maximum of

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of MHz with the number of resource blocks in the downlink transmission bandwidth configuration, BW_{Channel} is the bandwidth of the frequency channel in MHz and $n = 1, 2, 3$ for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in subclause 7.7A.1 apply.

7.6A.3.2 Void

7.6A.3.3 Out-of-band blocking for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the out-of-band blocking requirements are defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.6.3 for each component carrier while all downlink carriers are active.

For inter-band carrier aggregation with component carriers in operating bands $< 2.7\text{GHz}$ including n48, and for $F_{\text{DL_Low}(j)} - 15\text{ MHz} \leq f \leq F_{\text{DL_High}(j)} + 15\text{ MHz}$, the appropriate adjacent channel selectivity and in-band blocking requirements in the respective subclauses 7.5 and 7.6.2 shall be applied for carrier j . For inter-band carrier aggregation with component carriers in operating bands $> 2.7\text{GHz}$ excluding n48, and for $F_{\text{DL_Low}(j)} - 3 \cdot BW_{\text{channel}} \leq f \leq F_{\text{DL_High}(j)} + 3 \cdot BW_{\text{channel}}$, the appropriate adjacent channel selectivity and in-band blocking requirements in the respective subclauses 7.5 and 7.6.2 shall be applied for carrier j . $F_{\text{DL_Low}(j)}$ and $F_{\text{DL_High}(j)}$ denote the respective lower and upper frequency limits of the operating band containing carrier j , $j = 1, \dots, X$, with carriers numbered in increasing order of carrier frequency and X the number of component carriers in the band combination. BW_{channel} denotes the channel bandwidth of the wanted signal component carrier j . If CW interferer falls in a gap between $F_{\text{DL_High}(j)}$ and $F_{\text{DL_Low}(j+1)}$ where the corresponding OOB ranges 1 and 2 overlap, then the lower level interferer limit of the overlapping OOB ranges applies.

For inter-band carrier aggregation with uplink assigned to two NR bands, the out-of-band blocking requirements specified in clause 7.6.3 shall be met with the transmitter power for the uplink set to 7 dB below $P_{\text{CMAX_L,f,c}}$ for each serving cell c.

For the UE which supports inter-band CA configuration in Table 7.3A.3.2.1-1, $P_{\text{interferer}}$ power defined in Table 7.6.3-2 and 7.6.3-4 is increased by the amount given by $\Delta R_{\text{IB,c}}$ in Table 7.3A.3.2.1-1.

For inter-band CA combination listed in Table 7.6A.3.3-1, exceptions to the requirement specified in Table 7.6A.3.3-2 are allowed when the second order intermodulation product of the lower frequency band UL carrier and the CW interfering signal fully or partially overlaps with the higher frequency band DL carrier.

Table 7.6A.3.3-1: CA band combination with exceptions allowed

CA band combination
CA_n8-n78
CA_n8-n79
CA_n28-n78

Table 7.6A.3.3-2: Requirement for out-of-band blocking exceptions

Parameter	Unit	Level
$P_{\text{interferer}}$ (CW)	dBm	-44 ¹
NOTE 1: The requirement applies when , where and are the carrier frequencies for lower frequency band UL and higher frequency band DL, respectively. and are the channel bandwidths configured for lower frequency band UL carrier and higher frequency band DL carrier in MHz, respectively.		

For all interferer frequency ranges specified in clause 7.6.3 a maximum of

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of MHz with N_{RB} the number of resource blocks in the downlink transmission bandwidth configuration, BW_{Channel} the bandwidth of the frequency channel in MHz and n = 1, 2, 3 for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in subclause 7.7 apply.

The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

7.6A.4 Narrow band blocking for CA

7.6A.4.1 Narrow band blocking for Intra-band contiguous CA

For intra-band contiguous carrier aggregation, the downlink SCC(s) shall be configured at nominal channel spacing to the PCC. For FDD, the PCC shall be configured closest to the uplink band. All downlink carriers shall be active throughout the test. The uplink output power shall be set as specified in Table 7.6A.4.1-1 with the uplink configuration. For UE(s) supporting one uplink, the uplink configuration of the PCC shall be in accordance with Table 7.3.2-3. The UE shall fulfil the minimum requirement in presence of an interfering signal specified in Table 7.6A.4.1-1 being on either side of the aggregated signal. The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.6A.4.1-1.

Table 7.6A.4.1-1: Narrow-band blocking for intra-band contiguous CA

NR band	Parameter	Unit	NR CA bandwidth class
			C
n41	P _w in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + NA CA Bandwidth Class specific value below
			16
	P _{uw} (CW)	dBm	-55
	F _{uw} (offset 100 kHz 15 kHz, 30 kHz)	MHz	$- F_{\text{offset}} - 0.2$ $/$ $+ F_{\text{offset}} + 0.2$

NOTE 1: The transmitter shall be set a 4 dB below P_{C_{MAX}_L,f,c} at the minimum UL configuration specified in Table 7.3.2-3 with P_{C_{MAX}_L,f,c} defined in clause 6.2.4.

NOTE 2: Reference measurement channel is specified in Annexes A.3.2 and A3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD as described in Annex A.5.1.1/A.5.2.1.

NOTE 3: The PREFSENS power level is specified in Table 7.3.2-1 and Table 7.3.2-2 for two and four antenna ports, respectively.

NOTE 4: The F_{uw} (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the interferer and shall be further adjusted to MHz to be offset from the sub-carrier raster.

7.6A.4.2 Void

7.6A.4.3 Narrow band blocking for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the narrow band blocking requirements are defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.6.4 for each component carrier while all downlink carriers are active.

For the UE which supports inter-band CA configuration in Table 7.3A.3.2.1-1, P_{UW} power defined in Table 7.6.4-1 is increased by the amount given by $\Delta R_{\text{IB},c}$ in Table 7.3A.3.2.1-1.

The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

7.6C Blocking characteristics for SUL

7.6C.1 General

7.6C.2 In-band blocking for SUL

For SUL operation, the in-band blocking requirement for downlink bands specified in clause 7.6.2 shall be met.

7.6C.3 Out-of-band blocking for SUL

For SUL operation, the out-of-band blocking requirement for downlink bands specified in clause 7.6.3 shall be met. For operation band combination listed in Table 7.6C.3-1, exceptions to the requirement specified in Table 7.6C.3-2 are allowed when the second order intermodulation product of the SUL carrier and the CW interfering signal fully or partially overlaps with the DL carrier.

Table 7.6C.3-1: SUL operating band combination with exceptions allowed

NR Band combination for SUL
SUL_n78-n81
SUL_n78-n82

SUL_n78-n83
SUL_n79-n81

Table 7.6C.3-2: Requirement for out-of-band blocking exceptions

Parameter	Unit	Level
$P_{\text{Interferer}} \text{ (CW)}$	dBm	-44 ¹
NOTE 1: The requirement applies when, where and are the channel bandwidths configured for SUL and DL (victim) bands in MHz, respectively.		

For all interferer frequency ranges specified in clause 7.6.3 a maximum of

exceptions are allowed for spurious response frequencies in each assigned frequency channel when measured using a step size of MHz with N_{RB} the number of resource blocks in the downlink transmission bandwidth configuration, BW_{Channel} the bandwidth of the frequency channel in MHz and $n = 1, 2, 3$ for SCS = 15, 30, 60 kHz, respectively. For these exceptions, the requirements in subclause 7.7 apply.

7.6C.4 Narrow band blocking for SUL

Narrow band blocking is not specified for SUL band combination.

7.6D Blocking characteristics for UL MIMO

For UE with two transmitter antenna connectors in closed-loop spatial multiplexing scheme, the minimum requirements specified in clause 7.6 shall be met with the UL MIMO configurations described in clause 6.2D.1. For UL MIMO, the parameter P_{CMAX_L} is defined as the total transmitter power over the two transmit antenna connectors.

7.7 Spurious response

Spurious response is a measure of the ability of the receiver to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency for which a response is obtained, i.e. for which the out-of-band blocking limit as specified in clause 7.6.3 is not met.

The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters for the wanted signal as specified in Table 7.7-1 for NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz and in Table 7.7-1a for NR bands with $F_{DL_high} \geq 3300$ MHz and $F_{UL_high} \geq 3300$ MHz and for the interferer as specified in Table 7.7-2. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal. For operating bands with an unpaired DL part (as noted in Table 5.2-1), the requirements only apply for carriers assigned in the paired part.

Table 7.7-1: Spurious response parameters for NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz

RX parameter	Units	Channel bandwidth				
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	6	6	7	9	10
RX parameter	Units	Channel bandwidth				
		30 MHz	40 MHz	50 MHz	60 MHz	80 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	11	12	13	14	15
RX parameter	Units	Channel bandwidth				
		90 MHz	100 MHz			
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	15.5	16			
NOTE 1: The transmitter shall be set to 4 dB below $P_{CMAX_L,f,c}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{CMAX_L,f,c}$ defined in clause 6.2.4.						

Table 7.7.1-1a: Spurious response parameters for NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

RX parameter	Units	Channel bandwidth				
		10 MHz	15 MHz	20 MHz	40 MHz	50 MHz
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	6	7	9	9	9
RX parameter	Units	Channel bandwidth				
		60 MHz	80 MHz	90 MHz	100 MHz	
Power in transmission bandwidth configuration	dBm	REFSENS + channel bandwidth specific value below				
	dB	9	9	9	9	
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.						

Table 7.7-2: Spurious response

Parameter	Unit	Level
$P_{\text{Interferer}}$ (CW)	dBm	-44
$F_{\text{Interferer}}$	MHz	Spurious response frequencies

7.7A Spurious response for CA

7.7A.1 Spurious response for Intra-band contiguous CA

Table 7.7A-1: Spurious response parameters for intra-band contiguous CA

RX parameter	Units	NR CA bandwidth class		
		C		
Power in transmission bandwidth configuration	dBm	REFSENS + CA bandwidth class specific value below		
	dB	9		
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.				

Table 7.7A-2: Spurious response for CA

Parameter	Unit	Level
$P_{\text{Interferer}} \text{ (CW)}$	dBm	-44
$F_{\text{Interferer}}$	MHz	Spurious response frequencies

7.7A.2 Void

7.7A.3 Spurious response for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the spurious response are defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.7 for each component carrier while all downlink carriers are active.

For the UE which supports inter-band CA configuration in Table 7.3A.3.2.1-1, $P_{\text{interferer}}$ power defined in Table 7.7-2 is increased by the amount given by $\Delta R_{\text{IB,c}}$ in Table 7.3A.3.2.1-1.

The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

7.7D Spurious response for UL MIMO

For UE with two transmitter antenna connectors in closed-loop spatial multiplexing scheme, the minimum requirements specified in clause 7.7 shall be met with the UL MIMO configurations described in clause 6.2D.1. For UL MIMO, the parameter P_{CMAX_L} is defined as the total transmitter power over the two transmit antenna connectors.

7.8 Intermodulation characteristics

7.8.1 General

Intermodulation response rejection is a measure of the capability of the receiver to receive a wanted signal on its assigned channel frequency in the presence of two or more interfering signals which have a specific frequency relationship to the wanted signal

7.8.2 Wide band Intermodulation

The wide band intermodulation requirement is defined using a CW carrier and modulated NR signal as interferer 1 and interferer 2 respectively.

The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2 and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1) with parameters specified in Table 7.8.2-1 for NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz and Table 7.8.2-2 for NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz. The relative throughput requirement shall be met for any SCS specified for the channel bandwidth of the wanted signal. For operating bands with an unpaired DL part (as noted in Table 5.2-1), the requirements only apply for carriers assigned in the paired part.

Table 7.8.2-1: Wide band intermodulation parameters for NR bands with $F_{DL\ high} < 2700\text{ MHz}$ and $F_{UL\ high} < 2700\text{ MHz}$

Rx parameter	Units	Channel bandwidth											
		5 MHz	10 MHz	15 MHz	20 MHz	25 MHz	30 MHz	40 MHz	50 MHz	60 MHz	80 MHz	90 MHz	100 MHz
P _w in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + channel bandwidth specific value below											
		6	6	7	9	10	11	12	13	14	15	15	16
P _{Interferer 1} (CW)	dBm	-46											
P _{Interferer 2} (Modulated)	dBm	-46											
BW _{Interferer 2}	MHz	5											
F _{Interferer 1} (Offset)	MHz	$-BW/2 - 7.5$ $/$ $+BW/2 + 7.5$											
F _{Interferer 2} (Offset)	MHz	2*F _{Interferer 1}											

NOTE 1: The transmitter shall be set to 4 dB below P_{C_{MAX}_L_{f,c}} at the minimum UL configuration specified in Table 7.3.2-3 with P_{C_{MAX}_L_{f,c}} defined in clause 6.2.4.

NOTE 2:	Reference measurement channel is specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1)
NOTE 3:	The modulated interferer consists of the Reference measurement channel specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1 and 15 kHz SCS.
NOTE 4:	The $F_{\text{interferer } 1}$ (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the CW interferer and $F_{\text{interferer } 2}$ (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the modulated interferer.

Table 7.8.2-2: Wide band intermodulation parameters for NR bands with $F_{DL\ low} \geq 3300$ MHz and $F_{UL\ low} \geq 3300$ MHz

[illegible]

7.8A Intermodulation characteristics for CA

7.8A.1 General

7.8A.2 Wide band intermodulation for CA

7.8A.2.1 Wide band intermodulation for Intra-band contiguous CA

Table 7.8A.2.1-1: Wide band intermodulation parameters for intra-band contiguous CA with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz

Rx parameter	Units	NR CA bandwidth class		
		C		
P_w in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + 6		
$P_{\text{Interferer 1 (CW)}}$	dBm	-46		
$P_{\text{Interferer 2 (Modulated)}}$	dBm	-46		
$BW_{\text{Interferer 2}}$	MHz	$BW_{\text{Channel_CA}}$		
$F_{\text{Interferer 1 (Offset)}}$	MHz	$-2BW_{\text{Channel_CA}}$ / $+2BW_{\text{Channel_CA}}$		
$F_{\text{Interferer 2 (Offset)}}$	MHz	$2 \cdot F_{\text{Interferer 1}}$		
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.				
NOTE 2: Reference measurement channel is specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).				
NOTE 3: The modulated interferer consists of the Reference measurement channel specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1 and the same SCS as the closest carrier.				
NOTE 4: The $F_{\text{interferer 1}}$ (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the CW interferer and $F_{\text{interferer 2}}$ (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the modulated interferer.				

Table 7.8A.2.1-2: Wide band intermodulation parameters for intra-band contiguous CA with $F_{DL_low} < 2700$ MHz and $F_{UL_low} < 2700$ MHz

Rx parameter	Units	NR CA bandwidth class
		C
P_w in Transmission Bandwidth Configuration, per CC	dBm	REFSENS + 22
$P_{\text{Interferer 1 (CW)}}$	dBm	-46
$P_{\text{Interferer 2 (Modulated)}}$	dBm	-46
$BW_{\text{Interferer 2}}$	MHz	5
$F_{\text{Interferer 1 (Offset)}}$	MHz	$-F_{\text{offset}} - 7.5$ / $F_{\text{offset}} + 7.5$
$F_{\text{Interferer 2 (Offset)}}$	MHz	$2 * F_{\text{Interferer 1}}$
NOTE 1: The transmitter shall be set to 4 dB below $P_{\text{CMAX_L,f,c}}$ at the minimum UL configuration specified in Table 7.3.2-3 with $P_{\text{CMAX_L,f,c}}$ defined in clause 6.2.4.		
NOTE 2: Reference measurement channel is specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).		
NOTE 3: The modulated interferer consists of the Reference measurement channel specified in Annexes A.3.2.2 and A.3.3.2 with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1 and 15 kHz SCS.		
NOTE 4: The $F_{\text{Interferer 1}}$ (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the CW interferer and $F_{\text{Interferer 2}}$ (offset) is the frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the modulated interferer.		

7.8A.2.2 Void

7.8A.2.3 Wide band intermodulation for Inter-band CA

For inter-band carrier aggregation with one component carrier per operating band and the uplink assigned to one NR band, the wide band intermodulation requirements are defined with the uplink active on the band(s) other than the band whose downlink is being tested. The UE shall meet the requirements specified in clause 7.8 for each component carrier while all downlink carriers are active.

For the UE which supports inter-band CA configuration in Table 7.3A.3.2.1-1, $P_{\text{interferer}}$ power defined in Table 7.8.2-1 and 7.8.2-2 is increased by the amount given by $\Delta R_{\text{IB,c}}$ in Table 7.3A.3.2.1-1.

The throughput of each carrier shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2, A.3.2, and A.3.3 (with one sided dynamic OCNG Pattern OP.1 FDD/TDD for the DL-signal as described in Annex A.5.1.1/A.5.2.1).

7.8D Intermodulation characteristics for UL MIMO

For UE(s) with two transmitter antenna connectors in closed-loop spatial multiplexing scheme, the minimum requirements in clause 7.8 shall be met with the UL MIMO configurations described in clause 6.2D.1. For UL MIMO, the parameter P_{CMAX_L} is defined as the total transmitter power over the two transmit antenna connectors.

7.9 Spurious emissions

The spurious emissions power is the power of emissions generated or amplified in a receiver that appear at the UE antenna connector.

The power of any narrow band CW spurious emission shall not exceed the maximum level specified in Table 7.9-1

Table 7.9-1: General receiver spurious emission requirements

Frequency range	Measurement bandwidth	Maximum level	NOTE
30 MHz $f < 1$ GHz	100 kHz	-57 dBm	
1 GHz $f \leq 12.75$ GHz	1 MHz	-47 dBm	
12.75 GHz $f \leq 5^{\text{th}}$ harmonic of the upper frequency edge of the DL operating band in GHz	1 MHz	-47 dBm	2
12.75 GHz – 26 GHz	1 MHz	-47 dBm	3
NOTE 1: Unused PDCCH resources are padded with resource element groups with power level given by PDCCH as defined in Annex C.3.1.			
NOTE 2: Applies for Band that the upper frequency edge of the DL Band more than 2.69 GHz.			
NOTE 3: Applies for Band that the upper frequency edge of the DL Band more than 5.2 GHz.			

7.9A Spurious emissions for CA

7.9A.1 Void

7.9A.2 Void

7.9A.3 Spurious emissions for Inter-band CA

For inter-band carrier aggregation including an operating band without uplink band, the UE shall meet the Rx spurious emissions requirements specified in clause 7.9 for each component carrier while all downlink carriers are active.